



CARDIOLOGY

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Contents	1
ACUTE RHEUMATIC FEVER	10
Acute Rheumatic Fever	10
History Background.....	10
Pathophysiology.....	10
Epidemiology.....	11
Clinical Presentation	12
Untreated rheumatic fever/chronic rheumatic fever	16
Differential Diagnoses.....	16
Diagnosis	17
Jones criteria, 2015 revision, high-risk populations (Oceania, Africa, South Asia, other lower - income areas).....	19
TREATMENT	20
INFECTIVE ENDOCARDITIS	22
Infective Endocarditis.....	22
Definition	22
Types of infective endocarditis	22
Etiology	23
Pathology	23
Clinical picture.....	25
Diagnosis	27
The Duke diagnostic criteria were developed by Durack and colleagues	29
Complication	31
Differential diagnosis	31
Prophylaxis.....	31
Treatment	33
Valvular Heart Disease	23
Aortic Valve Diseases	34
AORTIC STENOSIS.....	34
Causes:.....	34
Effects:.....	34
Clinical features.....	34



Investigations:.....	35
Complications:.....	37
D.D :.....	37
Treatment:.....	38
AORTIC REGURGITATION	40
Etiology:.....	40
Effects:.....	41
Clinical picture:.....	42
Investigations:.....	43
Complications:.....	44
D.D.....	44
Treatment	44
Mitral Valve Diseases	45
MITRAL STENOSIS	45
Proper closure of the MV depends on:.....	45
Etiology	47
Effects:.....	47
Clinical picture:.....	47
Investigations:.....	50
Complications:.....	53
Management:.....	54
MITRAL REGURGE	57
Proper closure of the MV depends on:.....	57
Etiology of MR:.....	57
Clinical presentations of mitral regurge	58
Investigation to MR:.....	61
Differential Diagnosis:.....	64
Complications:.....	64
Treatment of chronic MR:.....	64
Treatment of acute MR.....	65
Treatment of MV prolapse.....	65
Tricuspid Valve Disease	65



TRICUSPID STENOSIS	66
Etiology:	66
Physiology:	66
Clinical features:.....	66
Investigations:	67
Treatment:	67
TRICUSPID REGURGITATION	68
Etiology	68
Clinical features:.....	68
Investigations:	68
Treatment:	69
Pulmonary Valve Disease	69
PULMONARY STENOSIS.....	70
Etiology	70
The hemodynamic changes:	70
Clinical features:.....	70
Investigations:	71
Treatment:	71
PULMONARY REGURGITATION	72
Etiology	72
Clinical features:.....	72
Investigations	72
Treatment:	72
HEART FAILURE.....	73
HEART FAILURE	72
Definition	73
Epidemiology.....	73
Signs and symptoms	73
LEFT-SIDED FAILURE.....	74
• Common respiratory signs	74
• Other signs	74



RIGHT-SIDED FAILURE	75
Causes	76
Classification	77
Staging.....	78
Diagnosis	78
Management.....	80
Monitoring	81
Lifestyle	82
Medication	82
Other medications	84
Implanted devices	85
Surgical therapies.....	86
ISCHEMIC HEART DISEASE	87
Causes:	87
ATHEROMA OR ATHEROSCLEROSIS	87
Sequence:.....	88
Risk factors	88
CORONARY ARTERY DISEASES.....	89
Definitions.....	89
MYOCARDIAL ISCHEMIA; ANGINA	89
Angina	89
Diagnosis based on clinical history;	90
Differential diagnosis	91
Investigations	91
Angina Pectoris Management.....	92
ACUTE CORONARY SYNDROME	94
Definition:	94
ACUTE MYOCARDIAL INFARCTION.....	94
SYMPTOMS :	94
Signs	95
Diffrential diagnosis	95
Diagnosis	95



Complications.....	96
Management.....	97
CARDIOMYOPATHY.....	99
CARDIOMYOPATHY	99
•Cardiomyopathies are classified according to the predominant clinical presentation into:	99
1. DILATED CARDIOMYOPATHY	102
Definition:	102
Causes of DCM	102
Clinical presentation:	103
Prognosis.....	103
Investigations:.....	104
Treatment	107
2. HYPERTROPHIC CARDIOMYOPATHY	109
Definition	109
Patho-physiology:	109
Clinical presentation:	111
Investigations:.....	111
Treatment of HOCM:	113
3-RESTRICTIVE CARDIOMYOPATHY.....	114
Definition	114
Classification:	114
Clinical presentation of restrictive cardiomyopathy.....	115
Diagnostic testing:.....	116
Treatment:	116
4-ARRHYTHMOGENIC RIGHT VENTRICULAR DYSPLASIA.....	118
Definition	118
Histopathology:.....	118
Clinical presentation:	119
Diagnosis:	119
Treatment:	119
MYOCARDITIS.....	120
Definition:	120



Causes:	120
Clinical Features:	120
Investigations:	121
Treatment:	121
Prognosis:	122
PERIPARTUM CARDIOMYOPATHY	123
The severity of symptoms in patients with PPCM can be classified by the New York Heart Association (NYHA) system as follows:	124
Etiopathophysiology	124
Epidemiology.....	125
Presentation.....	125
The differential diagnosis includes the following:	126
Echocardiography	126
General Treatment Approach	127
Pericardial diseases.....	131
NORMAL PERICARDIUM.....	131
FUNCTION OF THE PERICARDIUM.....	132
PERICARDIAL SYNDROMES.....	133
Pericardial syndromes include:	133
PERICARDITIS	133
Types of pericarditis:.....	133
Acute pericarditis	134
Approach.....	134
Treatment	135
PERICARDIAL EFFUSION	137
Etiology of pericardial effusion.	137
Clinical presentation	137
CARDIAC TAMPONADE	139
Causes:	139
Clinical picture;.....	139
Investigations of pericardial effusion and tamponade	139
Treatment of pericardial effusion and tamponade:	141



CONSTRUCTIVE PERICARDITIS	141
Aetiology	142
Clinical presentation	142
Diagnosis	142
Treatment	143
Cardiac Arrhythmias	144
Impulse conduction.....	131
What are the parts of the cardiac conduction system?.....	131
Cardiac Action Potential.....	133
CARDIAC ARRHYTHMIAS	136
Altered Normal Automaticity.....	137
Triggered Activity	137
Re-entry.....	138
Mechanisms of bradycardias:	139
Atrial Arrhythmias.....	153
SUPRAVENTRICULAR TACHYCARDIAS	141
Causes of SVT	142
Sinus tachycardia	142
Premature supraventricular contractions or premature atrial contractions (PAC).....	144
Paroxysmal Supraventricular tachycardia [HR 160-250/min].....	144
Atrioventricular reciprocating tachycardia (AVRT)	146
PSVT	146
Acute Management	146
Long-term management	146
Wolf Parkinson White Syndrome (WPW)	147
► ECG shows:.....	160
Atrial flutter (HR200-350/min).....	148
Atrial flutter (TREATMENT)	148
Atrial fibrillation (AF).....	149
Management.....	149
VENTRICULAR TACHYARRHYTHMIAS	151
Ventricular tachycardia (VT)	151



Ventricular fibrillation (VF)	152
Torsades De Pointes:.....	153
Premature ventricular contractions (PVCs).....	155
BRADYCARDIAS	156
Sinus Bradycardia.....	156
Sick sinus syndrome	157
Atrioventricular (AV) Block	157
First degree A-V Block.....	157
Second degree A-V Block	158
Two types:.....	158
Third degree A-V Block.....	160
Bundle Branch Block (BBB):	161
Hemiblock	161
Bifascicular block.....	161
MANAGEMENT OF ARRHYTHMIAS	162
Classification of Antiarrhythmic Drugs based on Drug Action.....	163
HYPERTENSION	177
HIGH BLOOD PRESSURE (HYPERTENSION).....	164
What is the blood pressure (BP)?	164
Types of hypertension.....	164
Definition of hypertension:.....	164
Risk factors	165
Blood Pressure measurement.....	166
Postural hypotension (orthostatic hypotension).....	170
Complications of hypertension :	174
Management of hypertension	175
Treatment of HT in pregnancy	177
Blood pressure targets (Goals) :	177
Combination Therapy :	177
Resistant Hypertension	178
Secondary Causes of HT	179
Hypertensive Crisis.....	180



PULMONARY EMBOLISM.....	183
PULMONARY EMBOLISM	183
Causes	183
Risk factors.....	184
Symptoms:	185
Complications.....	185
Diagnosis	186
Treatment:	188
SHOCK.....	191
Shock.....	191
Definition:	191
Pathophysiology of shock:	191
Etiology	191
Clinical picture:.....	192
Management of Shock	194
Chronic arterial occlusion.....	218
Aortic diseases.....	221
Aortic dissection.....	221
Aortic aneurysm.....	224
Aortic coarctation.....	227

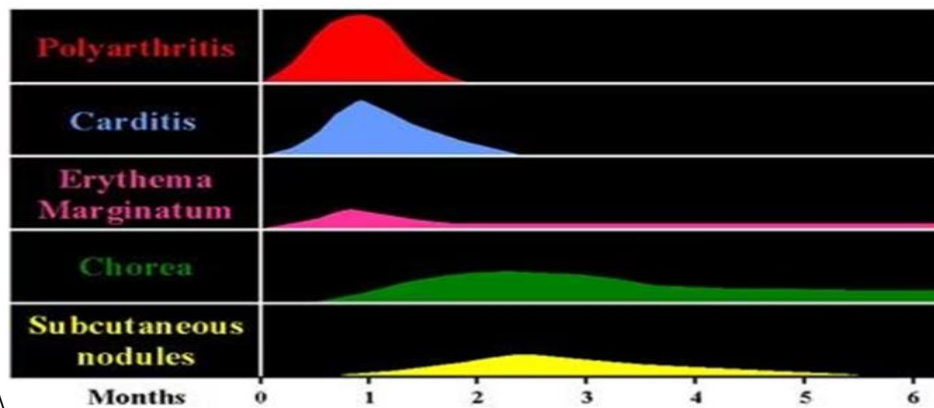
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Acute Rheumatic Fever

History Background

- Acute rheumatic fever (ARF) is an autoimmune inflammatory process that develops as a complication of streptococcal infection.
- ARF has extremely variable manifestations, and remains a clinical syndrome for which no specific diagnostic test exists.
- Persons who have experienced an episode of ARF are predisposed to recurrence following subsequent group A streptococcal infections.
- The most significant complication of ARF is rheumatic heart disease, which usually occurs after repeated bouts of acute illness.

Clinical Manifestations of Acute Rheumatic Fever



Pathophysiology

- ARF is characterized by nonsuppurative inflammatory lesions of the joints, heart, subcutaneous tissue, and central nervous system.



- An extensive literature search has shown that, at least in developed countries, rheumatic fever follows pharyngeal infection with hematogenic group A streptococci.
- The risk of developing rheumatic fever after an episode of streptococcal pharyngitis has been estimated at 0.3-3%.
- In this process, the patient's immune responses (both B- and T-cell mediated) are unable to distinguish between the invading microbe and certain host tissues. T helper 1 and cytokine Th17 appear to be key mediators of rheumatic heart disease
- Two basic theories have been postulated to explain the development of ARF and its sequelae following group A streptococcal infection:
 - (1) A toxic effect produced by an extracellular toxin of group A streptococci on target organs such as the myocardium, valves, synovium, and brain and (2) an abnormal immune response to streptococcal components. Increasing and compelling evidence now strongly favors the autoimmune explanation. It seems clear that an exaggerated immune response in a susceptible individual leads to rheumatic fever. This probably occurs through molecular mimicry, in which the immune response fails to differentiate between epitopes of the streptococcal pathogen and certain host tissues.

Epidemiology

- In developing countries, the magnitude of ARF is enormous.
- Recent estimates suggest that 33.4 million people worldwide have rheumatic heart disease and that 300,000-500,000 new cases of rheumatic fever (approximately 60% of whom will develop rheumatic heart disease) occur



annually, with 230,000 deaths resulting from its complications. Almost this entire toll occurs in the developing world.

- Rheumatic fever in the 21st century appears to be largely a disease of crowding and poverty.

Sex

Rheumatic fever does not have a clear-cut sexual predilection, although certain clinical manifestations, such as mitral stenosis and Sydenham chorea, are more common in females who have gone through puberty.

Age

- ARF is most common among children aged 5-15 years.
- It is relatively rare in infants and uncommon in preschool-aged children.
- ARF occurs in young adults, but the incidence of first episodes of ARF falls steadily after adolescence and is rare after age 35 years.
- The lower rate of ARF in adults may represent a decreased risk of streptococcal infections in this cohort.
- Recurrent episodes, with their predisposition to cause or exacerbate valvular damage, occur until middle age.

Clinical Presentation

Sore throat

- Although estimates vary, only 35%-60% of patients with rheumatic fever recall having any upper respiratory symptoms in the preceding several weeks.



- Many symptomatic individuals do not seek medical attention, go undiagnosed, or do not take the prescribed antibiotic for acute rheumatic fever (ARF) prevention.
- If a course of penicillin or another appropriate antibiotic is taken at this time, the risk for ARF is reduced by approximately 80%

Arthritis

- Overall, arthritis occurs in approximately 75% of first attacks of ARF. The likelihood increases with the age of the patient, and arthritis is a major manifestation of ARF in 92% of adults.
- The arthritis is typically polyarticular, but monoarthritic may occur.
- The arthritis of ARF is usually symmetrical and involves large joints, such as the knees, ankles, elbows, and wrists. Tenosynovitis is common in adults and may be severe enough to suggest a diagnosis of disseminated gonococcal disease.
- Aseptic monoarticular arthritis is seen in a substantial minority of cases from higher-risk areas, especially South Asia and Oceania, and can occur when NSAIDs are used early in the course.
- The evolution of polyarthritis in individual joints tends to overlap, causing more of an additive than a migratory pattern.
- In most instances, the entire bout of arthritis subsides within 4-6 weeks without any permanent damage. If not, a different diagnosis should be entertained.



Carditis

- Of first attacks of ARF, carditis occurs in 30%-60% of cases. It is more common in younger children but does occur in adults.
- Severe inflammation can cause congestive heart failure (CHF).
- Patients with carditis may present with shortness of breath, dyspnea upon exertion, cough, paroxysmal nocturnal dyspnea, chest pain, and/or orthopnea.
- Carditis may also be asymptomatic and may be diagnosed solely via auscultation or echocardiography

Sydenham chorea

- This occurs in up to 25% of ARF cases in children but is very rare in adults. It is more common in girls. Sydenham chorea in ARF is likely due to molecular mimicry, with autoantibodies reacting with brain ganglioside.
- Sydenham chorea may occur with other symptoms or as an isolated finding. It typically presents 1-6 months after the precipitating streptococcal infection and usually has both neurologic and psychological features.
- Evidence of a preceding streptococcal infection may be lacking.
- Like the arthritis, Sydenham chorea usually resolves without permanent damage but occasionally lasts 2-3 years and be a major problem for the patient and her family.
- Emotional liability, personality change, muscular weakness, and uncoordinated, involuntary, purposeless movements.
- The classic weakness is characterized by the inability to sustain a tetanic contraction. Patients are unable to maintain a clenched fist when attempting



to grip the examiner's hand. Other findings include dysarthric speech, gait problems, and poor fine-motor skills.

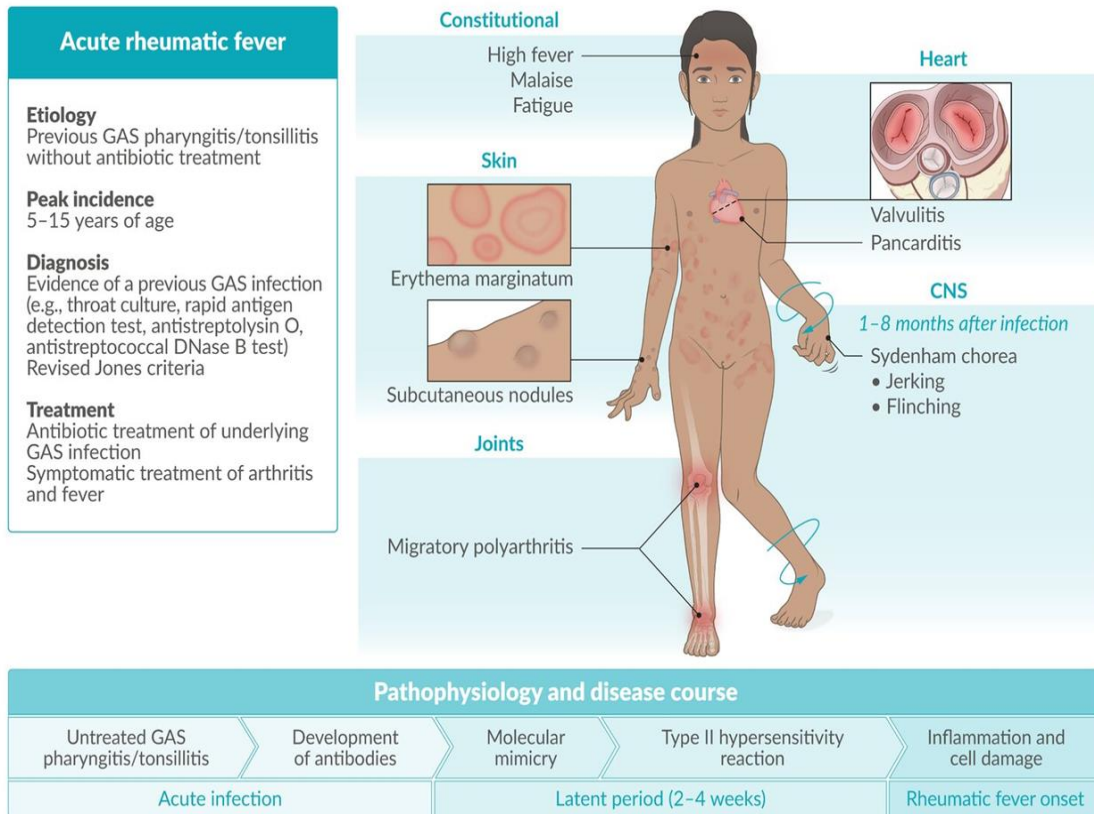
- The motor symptoms usually disappear during sleep and may be partially suppressed by sedation.
- They can involve the face, hands, and feet.

Erythema marginatum

- In first attacks of ARF in children, erythema marginatum occurs in approximately 10%. Like chorea, it is very rare in adults.
- Patients or parents may report a nonpruritic, painless, serpiginous, erythematous eruption on the trunk.
- The lesions may persist intermittently for weeks to months.
- They have a tendency to advance at the margins while clearing in the center.
- The lesions may be macular and can develop and disappear in minutes, appearing to change shape while being examined.
- They are found on the trunk and proximal aspects of the extremities and often go unnoticed by patients and parents, as they are usually covered by clothing.

Subcutaneous nodules

- Subcutaneous nodules are rarely noticed by the patient.
- Usually associated with severe carditis. They tend to occur several weeks after illness onset, are usually painless, and usually go unnoticed by the patient.
- They are found primarily over the bony surfaces or prominences and in tendon sheaths. The common sites include the elbows, knees, wrists, ankles, over the Achilles tendon, the back of the scalp, and the spinous process of the vertebrae



Untreated rheumatic fever/chronic rheumatic fever

- The average duration of an untreated ARF attack is 3 months. Chronic rheumatic fever, generally defined as disease persisting for longer than 6 months, occurs in less than 5% of cases.

Differential Diagnoses

1. Gonococcal Arthritis
2. Gout and Pseudogout
3. Infective Endocarditis
4. Juvenile Idiopathic Arthritis
5. Kawasaki Disease



6. Mixed Connective-Tissue Disease (MCTD)
7. Physical Medicine and Rehabilitation for Systemic Lupus Erythematosus
8. Reactive Arthritis
9. Rheumatoid Arthritis (RA)
10. Sarcoidosis
11. Septic Arthritis
12. Sickle Cell Disease (SCD)
13. Systemic Lupus Erythematosus (SLE)
14. Viral arthropathy

Diagnosis

Throat culture

- Throat culture remains the criterion standard for confirmation of group A streptococcal infection. Rapid antigen detection tests are not as sensitive.

Antibody titer tests

- Antibody titer tests used include ASO test, ant streptococcal DNase B (ADB) test, and the ant streptococcal hyaluronidase (AH) test.
- **ASO** is a test used to detect streptococcal antibodies directed against streptococcal lysine O. An elevated titer is proof of a previous streptococcal infection.
- These tests are of greater utility in areas with lower prevalence (e.g., in most Western countries).



Erythrocyte sedimentation rate, and C-reactive protein

- Acute-phase reactants, the erythrocyte sedimentation rate (ESR), and C-reactive protein levels (CRP) are usually elevated at the onset of ARF and serve as a minor manifestation in the Jones criteria. These tests are nonspecific, but they may be useful in monitoring disease activity.

Blood cultures

- Blood cultures are obtained to help rule out infective endocarditis, bacteremia, and disseminated gonococcal infection.

Imaging Studies

- Chest radiography
 - Chest radiography can reveal cardiomegaly and CHF in patients with carditis.
- Echocardiography
 - Echocardiography may demonstrate valvular regurgitant lesions in patients with ARF who do not have overt clinical manifestations of carditis. This is now considered an integral part of the evaluation of proven or suspected ARF everywhere. Patients with echocardiographic ally diagnosed subclinical carditis cases should receive the same long-term penicillin prophylaxis as those with the more classic clinical carditis, since they are also at risk for poor outcomes due to recurrent rheumatic heart disease.
 - Valvular stenotic lesions, especially of the mitral valve, can be observed in rheumatic heart disease. Pathological mitral and/or aortic regurgitation are also frequently encountered, though isolated echocardiographic disease of the aortic valve is uncommon with rheumatic heart disease.



Electrocardiography

- The most common finding on electrocardiography is a prolongation of the PR interval, which is a nonspecific finding, but counts as a minor manifestation in the Jones diagnostic criteria. It does not count as proof of carditis. On rare occasions, second- or third-degree heart block is present. In patients with chronic rheumatic heart disease, electrocardiography may show left atrial enlargement secondary to mitral stenosis.

Jones criteria, 2015 revision, high-risk populations (Oceania, Africa, South Asia, other lower-income areas)

Major criteria are as follows:

1. Carditis (clinical or echocardiographic diagnosis)
2. Polyarthritis or monoarthritis: Polyarthralgias can also be considered but only after careful consideration of the differential diagnoses.
3. Chorea (rare in adults)
4. Erythema marginatum (uncommon; rare in adults)
5. Subcutaneous nodules (uncommon; rare in adults)

Minor criteria are as follows:

1. Polyarthralgia (cannot count arthritis as a major criterion and arthralgia as a minor criterion)
2. Fever exceeding 38°C (note lower cutoff)
3. Elevated ESR (>30 mm/hr.; note lower ESR standard) or CRP level (>3 mg/L)
4. Prolonged PR interval



- If supported by evidence of preceding group A streptococcal infection, the presence of two major manifestations or one major and two minor manifestations indicates a high probability of ARF.
- Failure to fulfill the Jones criteria makes the diagnosis unlikely but not impossible. Clinical judgment is required.
- Recurrent ARF can be diagnosed based on 2 major, 1 major plus 2 minor, or 3 minor criteria.

TREATMENT

FOUR STAGES

Treatment of the group A streptococcal infection that led to the disease

- Although never proven to improve the one-year outcome, this is a standard practice. It may at least serve to reduce the spread of causative strain.

General treatment of the acute episode

- Anti-inflammatory agents are used to control the arthritis, fever, and other acute symptoms. Salicylates 8 Ws are the preferred agents, although other nonsteroidal agents are probably equally efficacious. And maybe preferred in children.
- Steroids are also effective but should probably be reserved for patients in whom salicylates fail, since there is a risk of rebound when they are withdrawn. 3 Ws
- Bed rest is a traditional part of ARF therapy and is especially important in those with carditis.



- Chorea is usually managed conservatively in a quiet no stimulatory environment; valproic acid is the preferred agent if sedation is needed.
- Some promising work suggests a possible role for hydroxychloroquine in the treatment of ARF, although no clinical data are yet available to recommend its use.

Cardiac management

- Bed rest is essential in patients with cardiac involvement.
- Carditis resulting in heart failure is treated with conventional measures; some use corticosteroids for severe carditis, although data to support this are scant.
- Diuretics and vasodilators are the mainstays of therapy.
- Monitor for development of arrhythmias in patients with active myocarditis.
- Atrial fibrillation requires aggressive management to reduce the risk of stroke.

Prophylaxis

- Children given penicillin G benzathine at a dose of 1.2 million U IM q4wk experienced a recurrence rate of 0.4 cases per 100 patient-years of observation. ARF recurrence rates have been found to be even lower if penicillin is administered q3wk instead of q4wk.
- This regimen may be appropriate in patients with severe rheumatic heart disease. Weigh the benefits of a 3-week regimen against patient compliance and cost; compliance is often poor to start with, at least partially due to the pain of the injections.
- Long-term administration of oral penicillin may be used in lieu of the intramuscular route. Erythromycin or sulfadiazine may be used in patients who are allergic to penicillin.



2

Infective Endocarditis

Definition

It is an infection of the endocardium and the cardiac valves.

Infective Endocarditis (IE) can occur in 2 ways:

1-Subacute bacterial endocarditis;

When the heart valves and the endocardium are damaged (RHD, congenital HD or artificial valve). Here a low virulence MO as streptococci can invade them and produce a slowly progressive infection.

2-Acute bacterial endocarditis:

Here normal valves and endocardium can be invaded by MO of high pathogenicity as staph, pneumo and gonococci in the course of fulminant septicemia originating in other organs

Types of infective endocarditis

1. Native valve endocarditis (NVE), acute and subacute
2. Prosthetic valve endocarditis (PVE), early and late
3. Intravenous drug abuse (IVDA) endocarditis



Etiology

- IE most commonly complicates: MR, AS, AR, calcific or sclerotic AV, VSD, PS, PDA, bicuspid AV, or artificial valve.
- Less common in cases of Fallot's tetralogy, and MS
- It is very rare in ASD.
- TV affection occurs among IV drug abusers who inject drug under septic conditions.

Organisms: (Bacteria, fungi, and rickettsia)

Fungal endocarditis is caused by candida or aspergillus in pts. receiving high doses of antibiotics, steroids, or cytotoxic drugs.

Also common in drug abusers, and in infection on prosthetic materials placed in the circulation as prosthetic valve, catheters.

Pathology

1-Platelets aggregates on the diseased endocardium and form sterile platelet vegetations stabilized by fibrin. Two factors are responsible for the formation of these thrombi:

- Damaged and deformed endocardium.
- Strong jet of blood.

2- MO in the blood then settles in these platelet thrombi start infection and form vegetations. The MO become embedded in the fibrin platelet mesh and become protected from phagocytosis and other body defenses.



Acute NVE

frequently involves normal valves and usually has an aggressive course. It is a rapidly progressive illness in healthy and debilitated persons alike. Virulent organisms, such as *S aureus* and group B streptococci, typically are the causative agents of this type of endocarditis. Underlying structural valve disease may be absent.

Sub-acute

Primarily affects abnormal valves. Its course, even in untreated patients, usually is more indolent than that of the acute form and may extend over many months. Alpha-hemolytic streptococci or enterococci, usually in the setting of underlying structural valve disease, typically are the causative agents for this type of endocarditis.

PVI

Early PVE occurs within 60 days of valve implantation. Traditionally, coagulase-negative staphylococci (CoNS), gram-negative bacilli, and *Candida* species have been the common infecting organisms. Late PVE occurs 60 days or more after valve implantation. Staphylococci, alpha-hemolytic streptococci, and enterococci are the common causative organisms. Data suggest that *S aureus* now may be the most common infecting organism in both early and late PVE.



IVDA

In 75% of cases of IVDA IE, no underlying valvular abnormalities are noted, and 50% of these infections involve the tricuspid valve.

Staphylococcus aureus is the most common causative organism

The effects of infective endocarditis are produced by:

- 1-Continuous release of the MO and their toxins into the blood stream leading to bacteremia, and toxemia.
- 2-Separation of parts of the vegetations in the form of infected emboli that cause metastatic infection, vascular obstruction or mycotic aneurysms in distant organs.
- 3-The infection destroy further the already damaged valves, endocardium... of the heart. Healing causes scar formation and deformities of the valve.
- 4-Ag-Ab reactions and formation of immune complexes

Clinical picture

1-Insidious onset of fever (low grade, but varies), sweating, arthralgia, malaise, toxic look, and pallor.

2-The heart may show:

- Always there is evidence of pre-existing heart dis.
- Change or increase in pre-existing murmurs or development of a new one due to destruction of the cardiac valves by infection

-In advanced cases, heart failure results from toxic myocarditis and the effects of cardiac defects.

3-Moderately enlarged, tender spleen.



4-Finger clubbing after 5-6 weeks.

5-Kidney affection may result in:

- Hematuria (macro, or microscopic), proteinurea.
- Immune complex acute GN.
- Renal failure may complicate advanced cases.
- Renal infarction (loin pain, and hematuria).

6-Embolism:

May involve any organ (spleen, kidney, limbs, brain, retina, mesentry,...) producing infarction of the affected organ. Pulmonary embolism may complicate the cases involving the Rt. Chambers. Retinal emboli or immune complexes cause areas of Hemorrhage with pale center “Roth spots”.

7-Neurological manifestations include:

- Cerebral embolism, septic infarction may result in cerebral abscess.
- Rupture of mycotic aneurysm may result in cerebral or sub-arachnoid Hemorrhage.

-A picture simulating meningitis or encephalitis may occur.

8-Skin lesions:

- Petechial Hemorrhage.
- Splinter Hemorrhage. (streaks of He under the nail), Janeway lesions (flat erythematous maculaes in the palms and the soles) are rarely seen.
- Osler nodules (tender intracutaneous nodules found in the pulps of the fingers and in the thenar and hypothenar eminences)



Clinical Findings in Infective Endocarditis Patients	(%)
Fever	75
Weight loss	52
Skin lesions	51
New murmur	33
Splenomegaly	20

Diagnosis

1-Bacterial endocarditis should always be suspected in any pt. with a murmur, or a known heart dis. Who develops an unexplained fever.

2-Blood culture:

-3 blood samples taken from 3 different sites must be tested on a period of 24 hours before any antibiotic is given.

-Culture for aerobic, anaerobic bacteria and fungi.

Culture must be incubated for up to 3 weeks to allow slowly growing MO to emerge.

- Culture sensitivity for the isolated MO.

- If the initial cultures are negative, another set of 3 samples must be taken after 48 hours.

Causes of –ve cultures are:

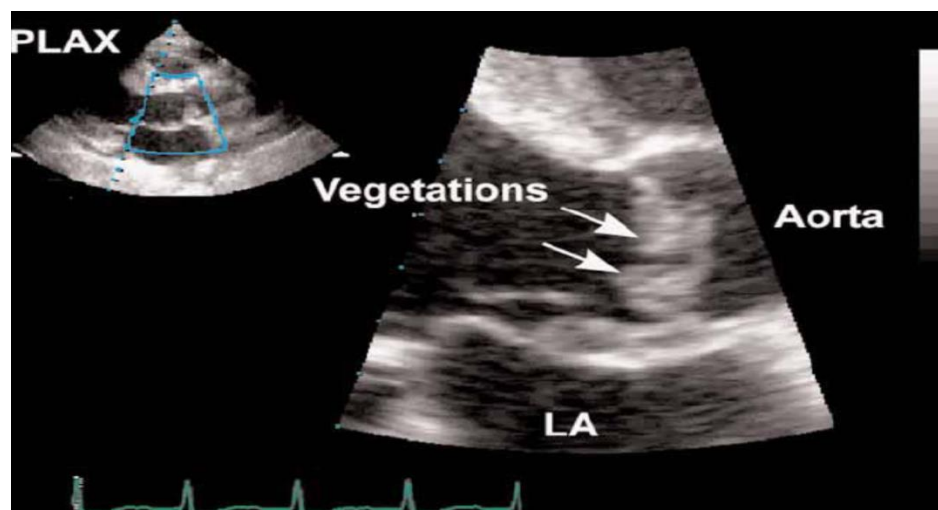
- Slowly progressing MO
- Fungal endocarditis,
- Prior antibiotic therapy

3- Echocardiography:

Is very essential, it may show:

- vegetation
- abscess
- leakage

TEE is more sensitive than TTE.





Echocardiography is useful for predicting the potential complications of IE, especially those that are embolic in nature.

Echocardiographic predictors of systemic embolization in patients with IE are the following:

- a. Large valvular vegetations (>10 mm in diameter)
- b. Multiple vegetations
- c. Mobile pedunculated vegetations
- d. Noncalcified vegetations
- e. Vegetations that are increasing in size
- f. Prolapsing vegetations

4- Urine examination: Show micro, or macroscopic hematuria. RBCs casts and heavy proteinuria indicate the presence of immune complex GN.

5- Blood examination may show elevated ESR, anemia, and sometimes leukocytosis.

The duke diagnostic criteria were developed by Durack and colleagues

Major blood culture criteria

- Two blood cultures positive for organisms typically found in patients with IE
- Blood cultures persistently positive for 1 of these organisms, from cultures drawn more than 12 hours apart
- Three or more separate blood cultures drawn at least 1 hour apart

Major echocardiographic

- Echocardiogram positive for IE, documented by an oscillating intracardiac mass on a valve or on supporting structures, in the path of regurgitant jets, or on implanted material, in the absence of an alternative anatomic explanation
- Myocardial abscess
- Development of partial dehiscence of a prosthetic valve
- New-onset valvular regurgitation

Minor criteria

- Predisposing heart condition or intravenous drug use (IVDA)
- Fever of 38°C (100.4°F) or higher
- Vascular phenomena, including major arterial emboli, septic pulmonary infarcts, mycotic aneurysm, intracranial hemorrhage, conjunctival hemorrhage, or Janeway lesions
- Immunologic phenomena such as glomerulonephritis, Osler nodes, Roth spots, or rheumatoid factor
- Positive blood culture results not meeting major criteria or serologic evidence of active infection with an organism consistent with IE
- Echocardiogram results consistent with IE but not meeting major echocardiographic criteria

definitive clinical diagnosis

Two major criteria

One major criterion and 3 minor criteria

Five minor criteria

Complication

	(%)
Stroke	15
Encephalopathy	9
Retinal emboli	3
Headache	4
Mycotic aneurysm	3
Abscess	1
Menigitis	1
Seizures	1
Total	37

Differential diagnosis

- 1-Rheumatic activity
- 2-Intercurrent infections and fevers.

Prophylaxis

- 1-Amoxicillin 2gm should be given orally to all patients with RHD, or cong. HD one hour before all procedure that may result in bacteremia.
- 2- Need to maintain good oral health
- 3- In cases of penicillin sensitivity, clindamycin, erythromycin, or vancomycin can be used.



4-Patients who are at high risk for developing endocarditis as:

-Artificial valve or

-Previous IE.

-GIT, genitourinary surgery, more intensive prophylaxis is required

(Ampicillin 2gm IV or IM 30 minutes before the procedure together with gentamycin 1.5 mg/kg. This should be repeated 6 hours later).

Prophylactic Regimens for Dental

Situation	Agent	Dosage Regimen *
Oral	Amoxicillin	2g
Unable to take oral	Ampicillin <i>or</i> ceftriaxone	2g IM or IV 1g IM or IV
Penicillin allergy, oral	Cephalexin, <i>or</i> clindamycin, <i>or</i> azithromycin	2g 600 mg 500 mg
Unable to take oral	Cefazolin, <i>or</i> ceftriaxone, <i>or</i> clindamycin	1g IM or IV 1g IM or IV 600 IM or IV

3-Early treatment for upper respiratory tract infections and other septic foci.

All carious teeth should be removed.

4-Surgical correction of underlying HD may be indicated only to prevent IE as closure of PDA.



Treatment

- The aim of treatment is to achieve bactericidal concentration of antibiotic in contact with the infective MO which is localized in avascular tissue or vegetation. That is why we should use High doses of parenteral bactericidal antibiotics.

-After culture sensitivity, The antibiotic must be given for at least 4-6 weeks.

The commonest causative MO is usually treated as follows:

*Strept viridans: 3 million units of cryst. Penicillin IV every 4 hours for 4 weeks together with gentamycin 3 mg/kg daily IM for the first 2 weeks.

Surgical Treatment

Surgery during acute infection does not increase mortality; in fact, restoration of a failing pump improves function and outcome.

Valve failure causing moderate to severe congestive heart failure (New York Heart Association Class III or IV) is a strong indication for urgent surgery.

An endocardial abscess, which can involve the aortic root, valve ring, or ventricular septum, is another indication for urgent intervention.

Other conditions favoring surgery include vegetations larger than 1 cm in diameter, a major embolic event, and failure or relapse of medical therapy

The contribution of the following factors to infective endocarditis (IE):

1. Antibiotic resistance
2. The increasing implantation of intravascular devices
3. COVID-19
4. Opiate use disorders
5. The growing number of immunosuppressed patients

3

Aortic Valve Diseases

AORTIC STENOSIS

Causes

- 1- Congenital (bicusped). 2- Rheumatic. 3- Sclerotic of old age.

Effect

- 1- Slightly narrowed valve (sclerotic) usually produces the murmur without significant hemodynamic changes.
- 2- Significant obstruction with the stenosis results in increased LV systolic pressure to become more than that in the aorta
- 3- Pressure overload lead to LVH, followed by dilatation.
- 4- Low COP, myocardial ischemia.

Clinical features

Symptoms

- Chest pain.
- Syncope (orthostatic-exertional-arrhythmia).
- Heart failure (left sided).
- Hemolytic anemia (calcification of the valve).
- GIT bleeding (Angiodysplasia).



-Infective endocarditis.

Signs

1-Pulse of low volume, plateau' rises slowly to its maximum with a notch on its ascending limb'.

2-Low systolic BP, small pulse pressure.

3-LVH result in sustained apex beat.

4-A systolic thrill over the aortic area propagated to the neck.

5-A harsh ejection systolic murmur heard over the aortic area and is propagating to the neck.

6-Gallavardin phenomenon:

- Dissociation between the harsh noisy and musical characters of systolic murmur Harsh noisy over the aortic area. Musical high frequency systolic murmur over the apex.

7-Aortic 2nd sound is soft.

Investigations

1-X-ray:

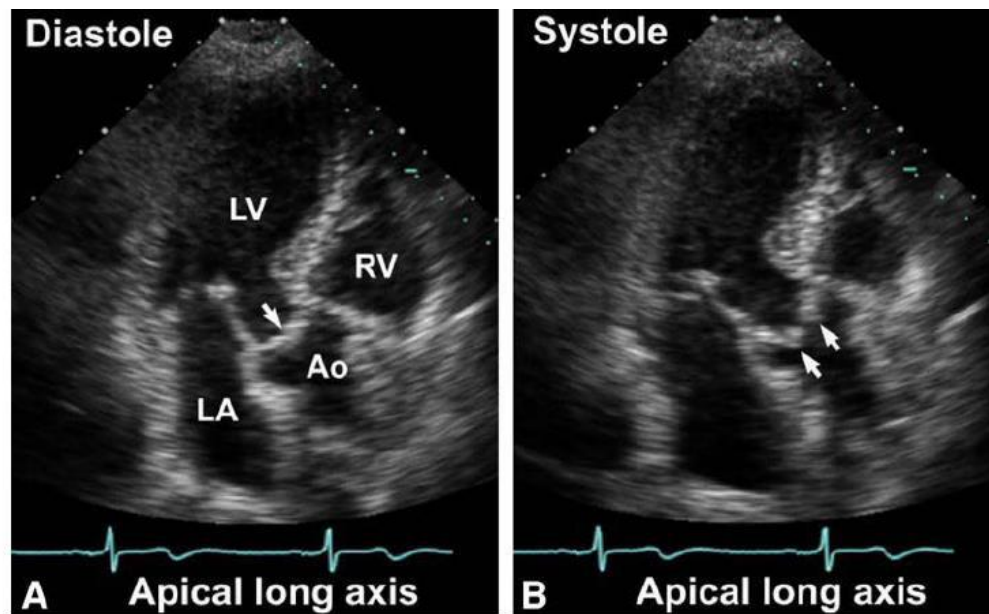
a-LV dilatation b-AV calcification c-Post stenotic dilatation of the aorta.

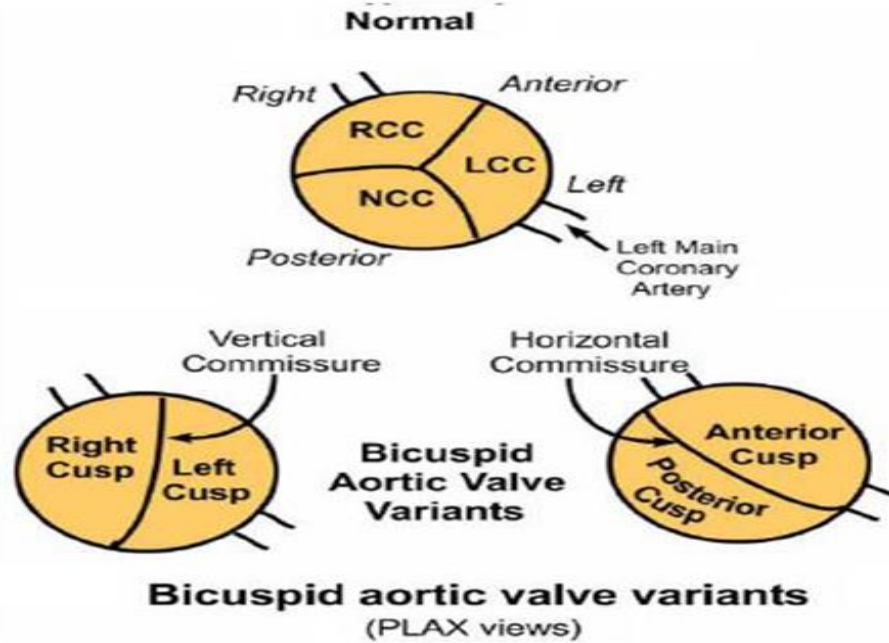
2-ECG: LVH and strain.

3-Echocardiography:

- Thickness and calcification of the valve.

- Pressure gradient across the valve:
 1. Mild AS if <50 mmHg.
 2. moderate AS 50-75 mmHg.
 3. severe AS if > 75 mmHg. (measured by echo).
- Cross sectional area of AV:
 - Normal size of the aortic orifice is 2.5 to 3cm².
 - Mild stenosis 0.75 to 2.5cm².
 - Moderate stenosis less than 0.5 to 0.75cm².
 - Severe stenosis less than 0.5cm².





4- Cardiac catheterization:

To detect the cause of myocardial ischemia.

Complications

- 1-infective endocarditis
- 2-Rheumatic fever activity
- 3-L.V. failure
- 4-Sudden death.

D.D

- 1-Supra and infra valvular lesions.
- 2-Hypertrophic cardiomyopathy.
- 3-Mitral regurge (gallavardin phenomenon).



4-V.S.D 5-Pulmonary stenosis.

Treatment

1- Rheumatic AS should receive prophylaxis against RF.

2- Prevention of IE. 3-Restrict physical activity, and pregnancy should be avoided in severe cases. 4-Surgery is indicated in:

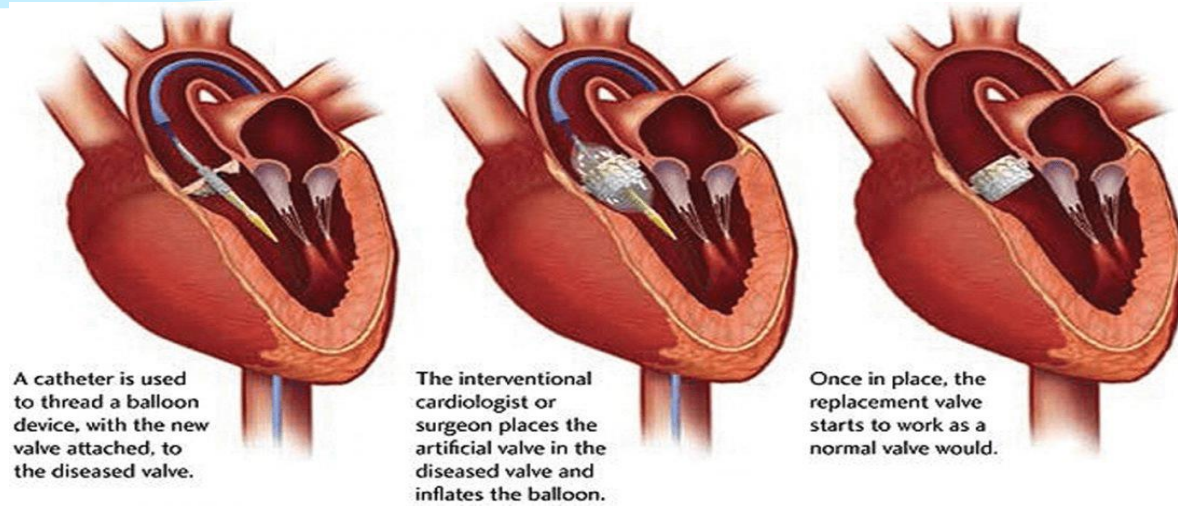
a-All symptomatic cases

b-All severe cases even asymptomatic. (for fear of sudden death). Either by replacement, or by commissurotomy (in children)

5- Ballon dilatation 'valvoplasty'.

6- Surgical treatment (valve replacement).

7- TAVI: During this minimally invasive procedure a new valve is inserted without removing the old, damaged valve. The new valve is placed inside the diseased valve. The surgery may be called a transcatheter aortic valve replacement (TAVR) or transcatheter aortic valve implantation (TAVI).

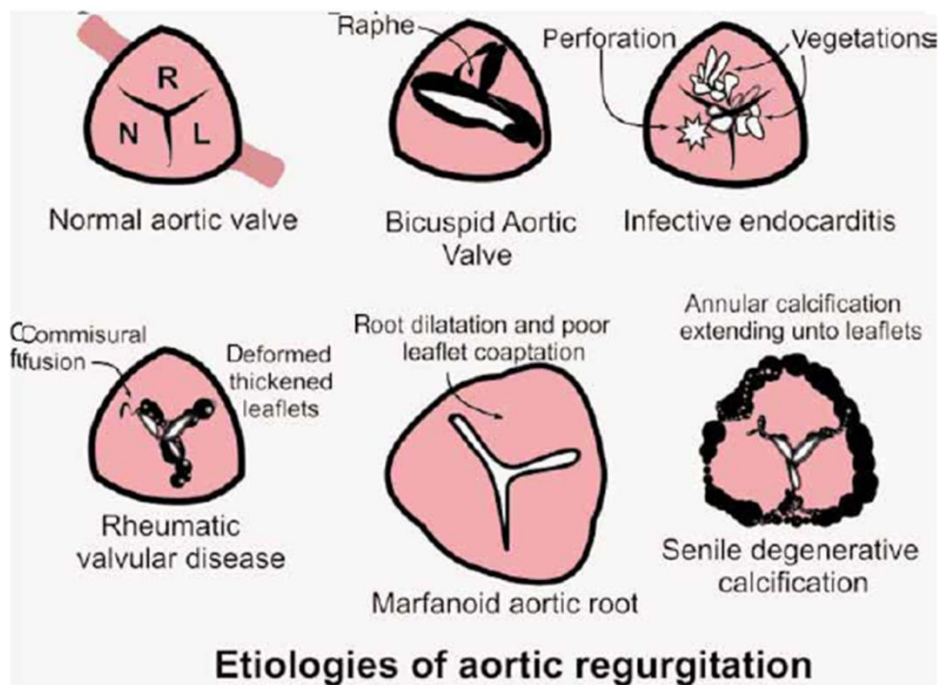
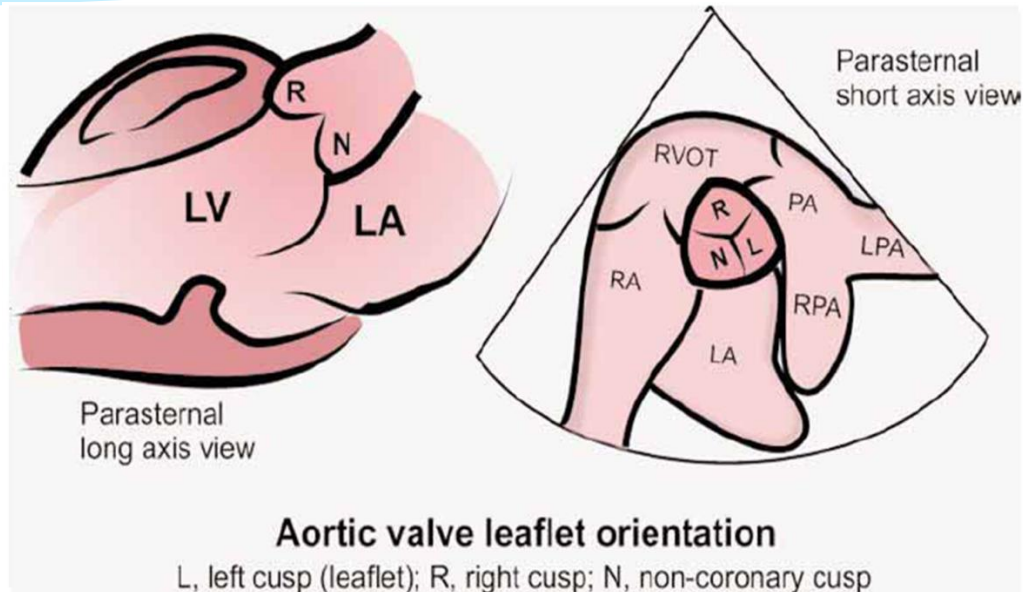




AORTIC REGURGITATION

Etiology

- 1-Rheumatic fever.
- 2-Myxomatous degeneration.
- 3-Senile calcifications.
- 4-congenital heart disease.
- 5-Syphilis.
- 6-Connective tissue disease.
- 7-Infective endocarditis.
- 8-Dissecting aneurysm of the aorta.
- 9-Traumatic.
- 10-Severe HTN.
- 11-Ankylosing spondylitis.
- 12-Marfan's syndrome: -
 - Hereditary disease of the connective tissue affecting the skeleton, ligaments and the blood vessels.
 - The pt. is long and thin, with arachnodactyly (long fingers), High arched palate, dislocated patella, subluxation of the lens.
 - The CVS affection include: aortic aneurysms, dissection, or aortic regurgitation, MVP, and long chordae tendinae (which may rupture.)



Effects

1-During diastole blood regurgitate from the aorta into LV causing sudden and marked drop in the diastolic pressure in the aorta and this result in decrease in



coronary blood flow “myocardial ischemia”.

2-LV hypertrophy and dilatation and high systolic pressure.

Clinical picture:

A-Symptoms:

- 1-Asymptomatic
- 2-LV failure manifestations.
- 3-Angina.

B-Signs:

-High systolic BP then rapid drop of diastolic BP.

-“Hill’s sign” popliteal systolic blood pressure exceeding brachial systolic blood pressure by > 60 mmHg (normal >40 mmHg).

-The peripheral pulse has the following characteristics:

- a. High volume.
- b. Collapsing pulse
- c. Pulsations in the neck “**Corrigan’s sign**”.
- d. Sudden flow of blood in the arteries during systole produces a sound “pistol shot” heard on big arteries.
- e. **Duroziez’s sign**: audible diastolic murmur over femoral artery when compressed by bell of stethoscope.
- f. **Traube’s sign**: systolic and diastolic murmur heard over femoral artery when it’s gradually compressed.

-Generalized vasodilation result in:

- a. Warm hands and feet.



- b. Capillary pulsations (wide pulse pressure may be transmitted across the dilated arterioles to the capillaries)
- c. **Quineke's sign** visualization of capillary pulsation upon light compression applied by tip of finger nail bed.
- d. **Müller sign**: bobbing of uvula occurs during systole
- e. **Becker's sign**: presence of visible pulsation of retinal artery by ophthalmoscope.
- f. Hyperdynamic apex.

C. Auscultation

-Loud aortic component of 2nd HS (A2).

-High pitched early diastolic murmur best heard on the 2nd aortic area.

-A functional ejection systolic murmur may be heard at the aortic area and is due to increased volume of blood pumped through the aortic valve during systole. (DD organic AS).

- "Austin Flint murmur" A short mid diastolic murmur 'The regurgitant stream from the AV pushing back the anterolateral leaflet of the MV in diastole causing partial closure and vibrations of the MV and turbulence of the stream of blood coming from the LA'.

Investigations:

1-X-ray:

- LV dilatation
- Thickness of LV wall.

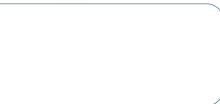
- Prominent aortic knob.
- Degree of LV and aortic dialation.

2-ECG: signs of LVH.

3-Echocardiography

Complications:

1. Infective endocarditis.
2. Rheumatic fever.
3. Heart failiure.



1-Pulmonary regurgitation

Graham Steel murmur

2-V.S. D

Treatment

- 1-Prevention of endocarditis.
- 2-Treatment of H.F.
- 3-Surgical treatment (valve replacement).

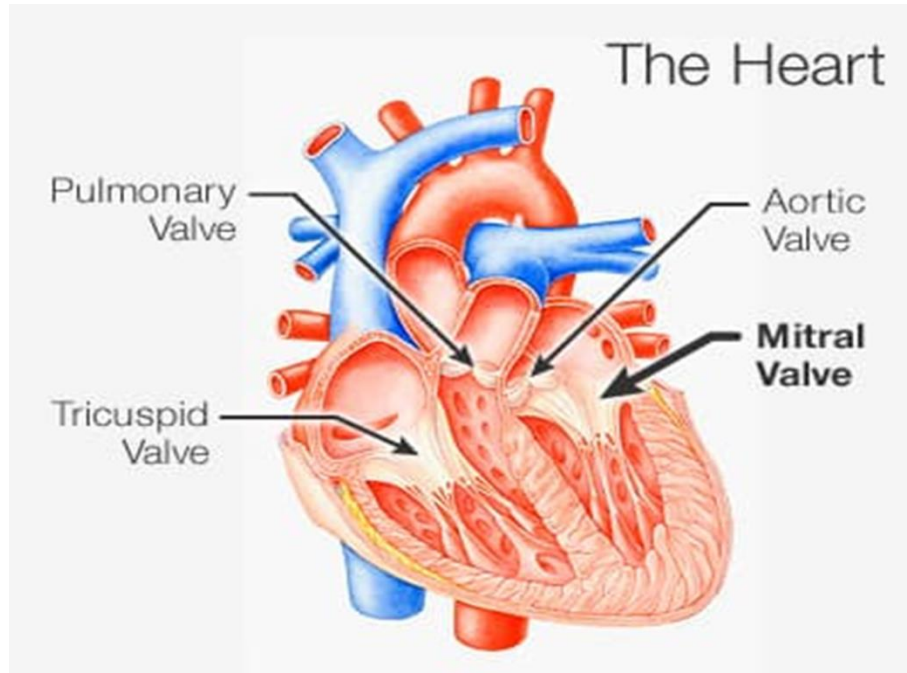
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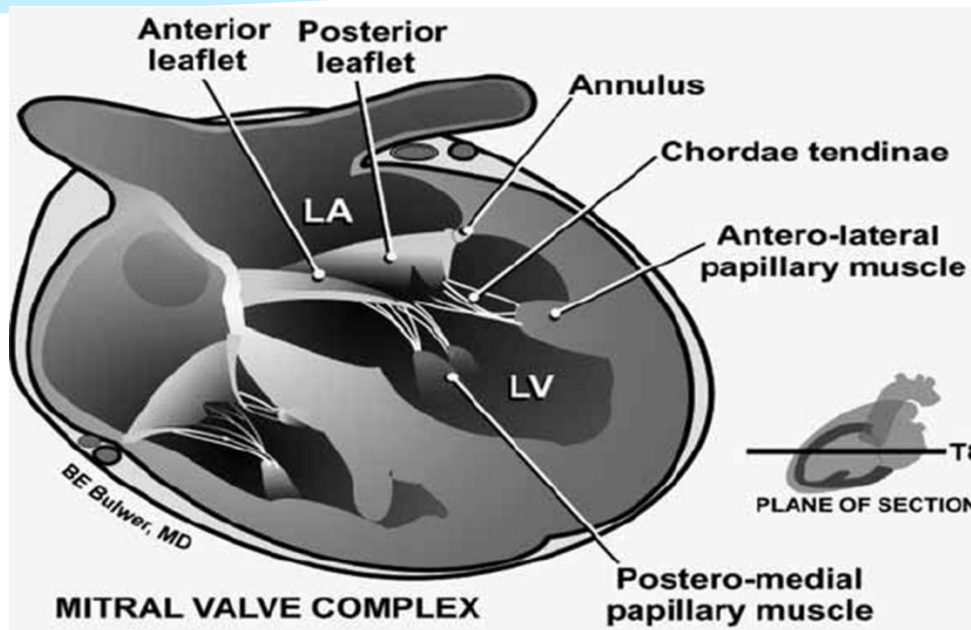
MITRAL STENOSIS

The normal MV consists of two leaflets, CT, and PMs. Stenosis results from fusion and rigidity of the two leaflets

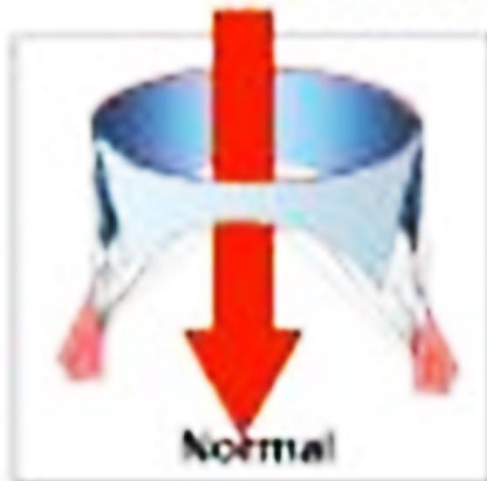
Proper closure of the MV

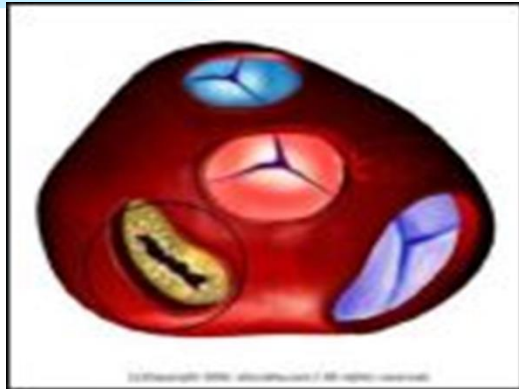
- Normal leaflets,
- Normal CT to keep the leaflet in place,
- Normal contractile papillary muscles to pull down the valve cusps during systole and prevent their displacement into the atrium
- Normal size of the MV ring.





Mitral 'Stenosis'





Etiology:

- 1- Organic MS always results from rheumatic carditis & congenital
- 2- Relative MS occurs when excessive blood flow across normal MV.

Effects:

- 1- Pressure rises in the LA resulting in LA dilatation.
- 2- Pulmonary congestion which may lead to pulmonary edema
- 3- pulmonary arteriolar vasoconstriction which leads to pulmonary hypertension, RVH, and congestive HF.
- 4- Stretching and fibrosis of the LA may result in AF.

Clinical picture:

- * Over 75% of pts. with MS are females under the age of 45.
- * CP consists of the following depending on the stage and the severity of the disease:
 - Signs of MS alone.
 - Signs and symptoms of pulmonary congestion
 - Development of pulmonary HTN.
 - Development of CHF.



Symptoms:

1-Mild cases are asymptomatic

2-Moderate and severe cases present with symptoms of pulmonary congestion (Dyspnea, orthopnea, PND, cough and haemoptysis). Pulmonary oedema may occur.

3- Palpitation when AF complicates the disease.

4- When PH occurs dyspnea is ameliorated and is replaced by low COP symptoms

Signs

1-Apex beat is normal in position

2-A long diastolic thrill may be palpable at the apex.

3-1st HS is accentuated because the valve is kept open to the end of diastole to allow the flow of blood, and the MV is closed from a widely open position.

4- The opening of the thickened MV by the high LA pressure produces a loud high pitched sound” opening snap”. It follows the 2nd HS by a short interval which is the isometric phase of relaxation.

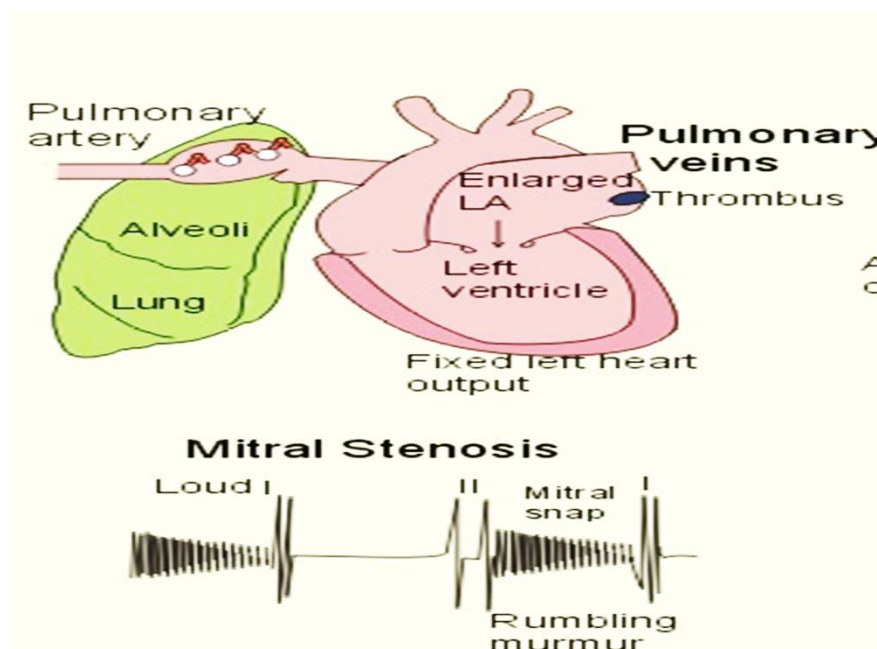
- In tight MS, the LA pressure is very high and opens the MV very shortly after the 2nd HS and the opening snap is used to judge the severity of stenosis” the shorter this interval the severer the stenosis.

5- Mid diastolic murmur is heard just inside the apex, It becomes louder just before the 1st HS” presystolic accentuation”, because atrial contraction

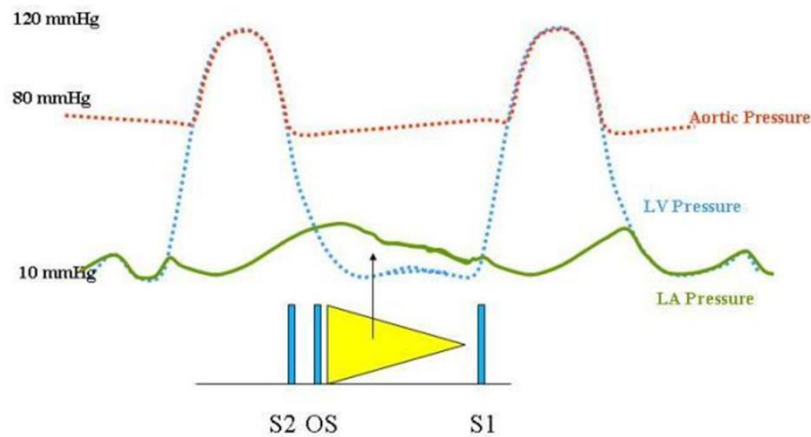
increases blood flow across the MV. When AF occurs the presystolic part of the murmur is lost.

6-When the valve becomes very thick, immobile, and calcified, the opening and closing movement of the valve becomes poor. The sound of its opening” opening snap” and of its closure”1st HS” becomes soft or inaudible.

7-Low COP in severe cases causes small pulse volume and peripheral cyanosis, malar flush.



Mitral Stenosis



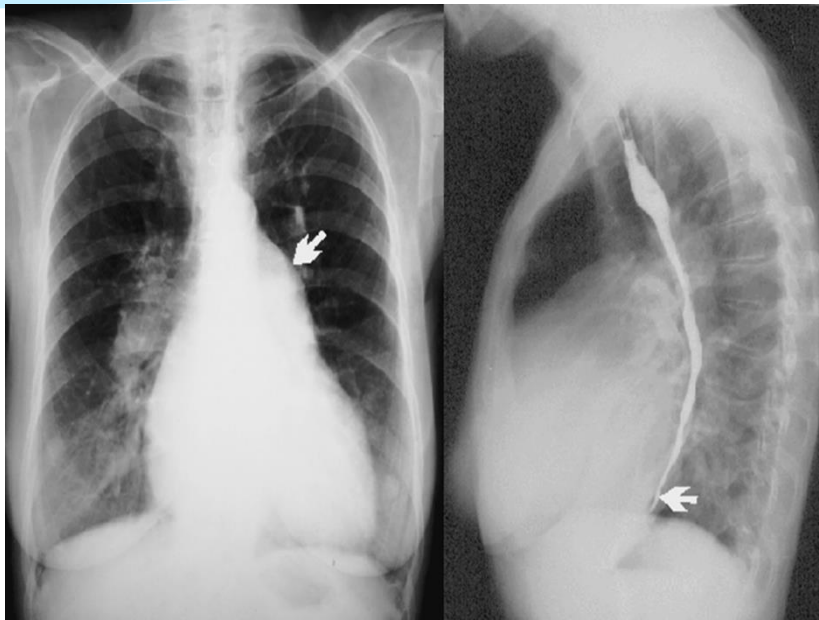
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(c) 2007, Munther K. Homoud, MD

Investigations:

1-X-Ray:

- 1-Dilatation of the LA (mitralization of the left border of the heart, double contour)
- 2-Signs of pulmonary congestion
- 3-Signs of pulmonary HTN.
- 4-Signs of CHF.

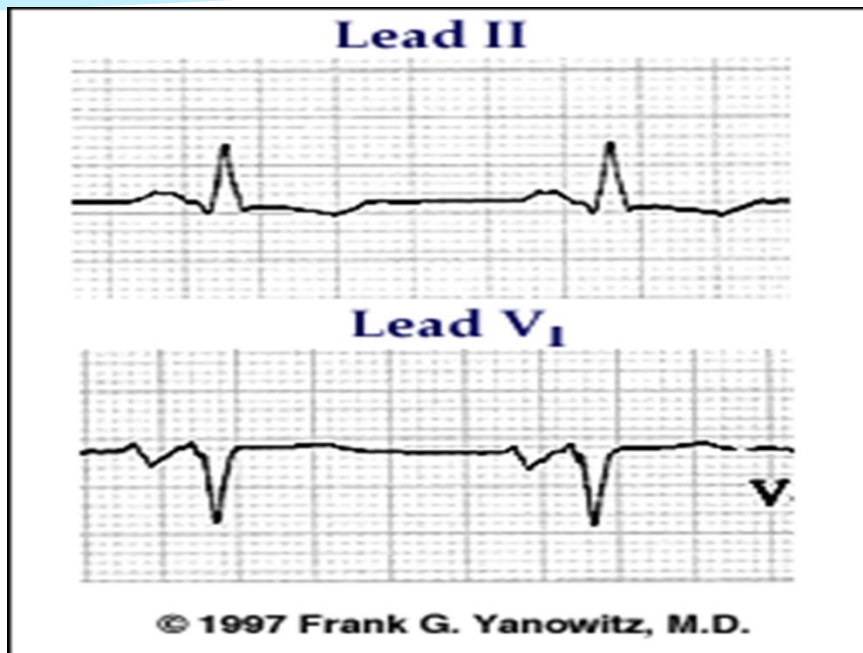


2.ECG:

1-LA hypertrophy (P mitrale)

2-RVH

3-AF



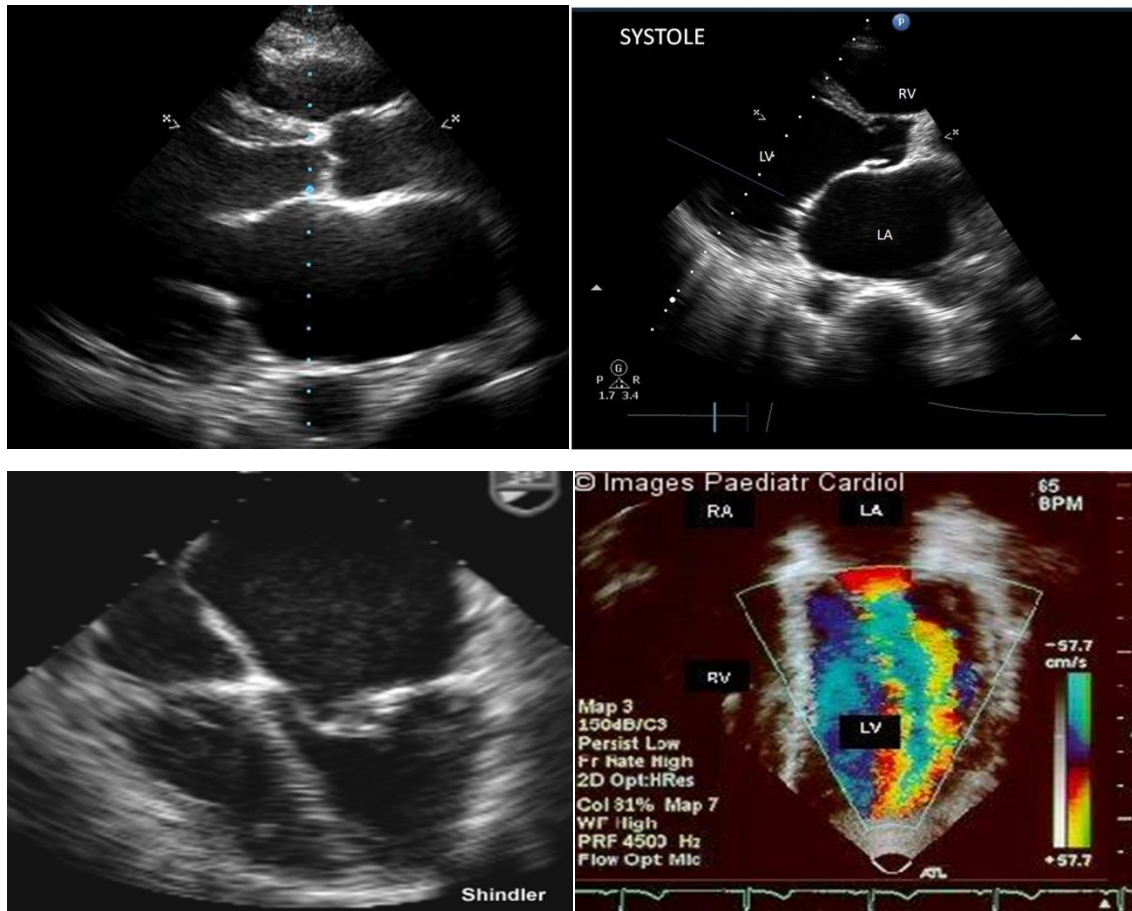
3-Echocardiography:

A.MITRAL VALVE AREA (MVA):

- Normal 5-4 cm²,
- Mild 4-1.75 cm²,
- Moderate 1.75-1.25 cm²
- Severe 1.25-1 cm²,

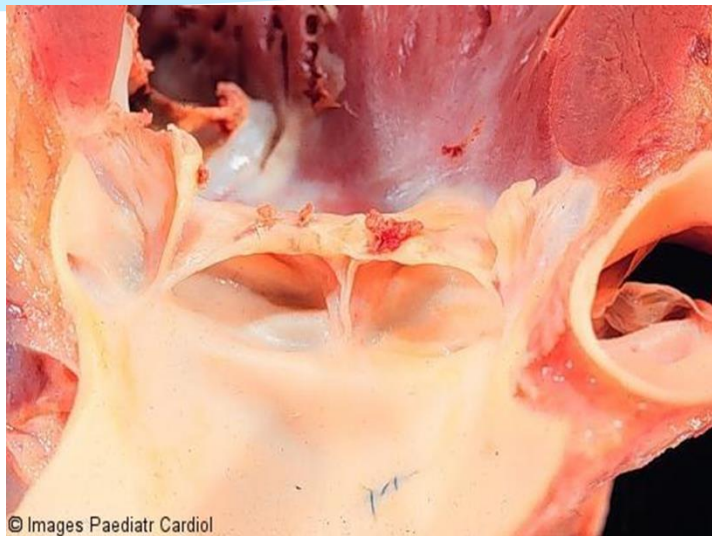
- Critically severe $<1 \text{ cm}^2$

B. THICKENING, CALCIFICATION AND MOBILITY of the cusps and involvement of the subvalvular apparatus.



Complications:

- 1-Pulmonary congestion, hypertension, and CHF
- 2-AF
- 3-Thrombi in the LA may lead to systemic embolization
- 4-Thrombosis in the RA may also occur and result in pulmonary embolism
- 5-Infective endocarditis
- 6-Calcification of the MV
- 7-Recurrence of rheumatic activity



Management:

- 1-Asymptomatic or minimally symptomatic pts. remains so far many years and should be put on medical treatment only.
- 2-Once significant symptoms develop, the disease progress relatively rapidly to total disability and death so relief of the stenosis by intervention should be advised.

I. Medical treatment:

- 1-Continuous penicillin prophylaxis to prevent recurrence of rheumatic activity.
- 2-Antibiotic prophylaxis for infective endocarditis.
- 3-If dyspnea developed restrict salt in diet and give diuretic to reduce pulmonary congestion
- 4-Prevent and treat all factors ppt deterioration (arrhythmias, infection,)
- 5-Treat complications as HF, embolism,



II. Intervention

Splitting of the commissures:

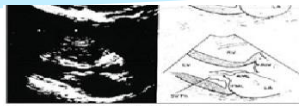
- **Percutaneous balloon mitral valvuloplasty (PBMV).** is now the method of choice when the valve is not severely fibrosed or calcified and no thrombi in the LA.

Indications:.

- 1-Symptomatic moderate to severe MS
- 2-Evidence of PH, RVH or failure.
- 3-History of embolic manifestations.

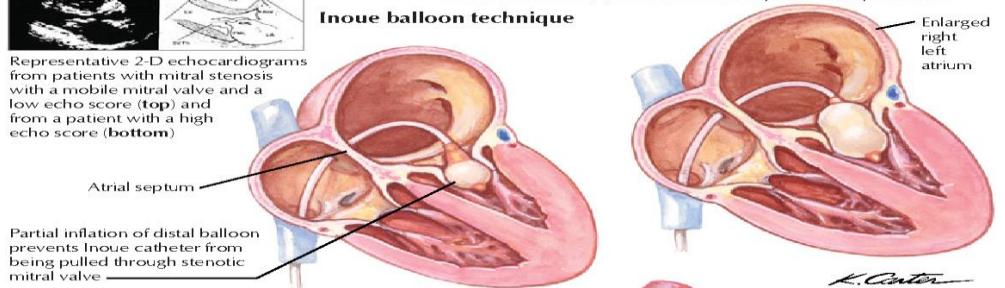
Contraindications:.

- 1- presence of rheumatic activity
- 2-Presence of infective endocarditis.
- 3-Associated significant MR (MV replacement is the treatment of choice here)

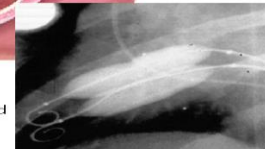
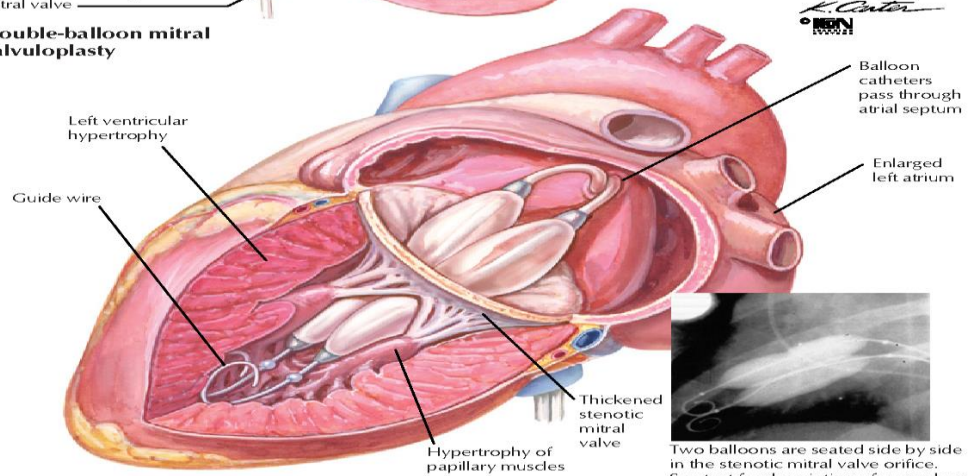


Representative 2-D echocardiograms from patients with mitral stenosis with a mobile mitral valve and a low echo score (top) and from a patient with a high echo score (bottom)

Inoue balloon technique



Double-balloon mitral valvuloplasty



Two balloons are seated side by side in the stenotic mitral valve orifice. See text for description of procedure.

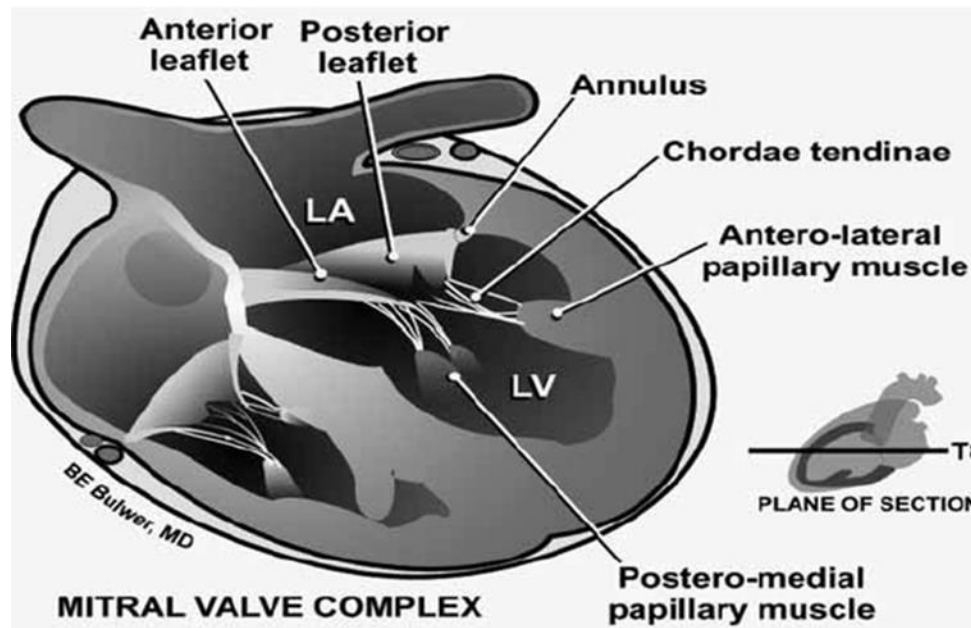
iii. Surgical treatment:

- 1-Surgical MV valvotomy.
- 2-MV replacement.

4

MITRAL REGURGE

- Normal leaflets,
- Normal CT to keep the leaflet in place,
- Normal contractile papillary muscles to pull down the valve cusps during systole and prevent their displacement into the atrium
- Normal size of the MV ring.



Etiology of MR:

MR can result from diseases involving any of the different components of the MV as:

1-Mitral valve prolapses



2-Rheumatic fever

3-Dilatation of the MV ring as in HF.

4-Dysfunction of the MV leaflets due to ischemia, infarction,

5-Connective tissue disease

6-Annular calcifications

7-Congenital cardiac conditions

8-Acute mitral regurgitation (infective endocarditis – myxomatous degeneration – truma – acute myocardial infarction)

9-Weight loss agents as: dexfenfluramine – fenfluramine & fenteramine

Clinical presentations of mitral

1.Chronic MR

Effects:.

1-LA is dilated and its pressure rises due to regurgitation of blood from LV into LA each systole.

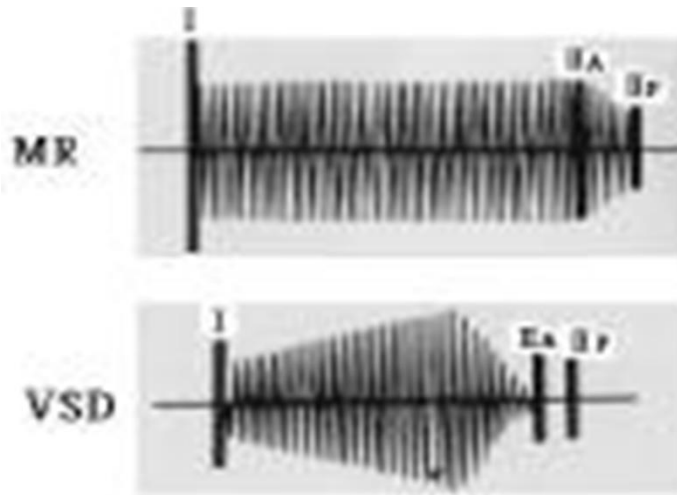
2-During diastole the large amount of blood in the LA flows to the LV across a no stenotic MV producing turbulence of blood flow and relative MS.

3-LV is dilated due to volume overload.

4-LV failure lead to pulmonary congestion, HTN

5-AF may occur

Clinical Picture:



Symptoms:

- 1-Mild to moderate MR is asymptomatic.
- 2-When LV failure occurs, symptoms of pulmonary congestion appear.
- 3-Symptoms of pulmonary HTN.

Signs:

- 1-LV volume overload causes: -Hyperdynamic apex beat displaced outward and downward.
- 2-Regurgitation through MV causes:
 - Systolic thrill at the apex.
 - Pan systolic murmur, propagating to the axilla.



-Faint 1st HS due to in diastole, the MV floats on the top of blood filling the LV. When systole starts the valve is already semi closed

3-Rapid flow of blood across the MV in diastole causes:

-3rd HS

-Mid diastolic murmur.

2.Acute mitral regurge:.

Mechanism:

- 1-Papillary muscle dysfunction, or rupture complicating AMI.
- 2-MV perforation in infective endocarditis.
- 3-Rupture CT in MVP (myxomatous degeneration).

The degree and symptoms of pulmonary congestion are much greater than in chronic MR. This is because LV contraction forces the regurgitant blood into non-compliant LA still of normal size. Acute pulmonary edema may develop, and shock may develop. These patients may need urgent surgical correction.

3.Mitral valve prolapse:

- Prolapse of the redundant MV leaflets into the LA may occur in 10% of normal population.

Causes:

- 1-Unknown
- 2-Marfan syndrome

3-Congenital anomaly either isolated MVP, or in association with ASD.

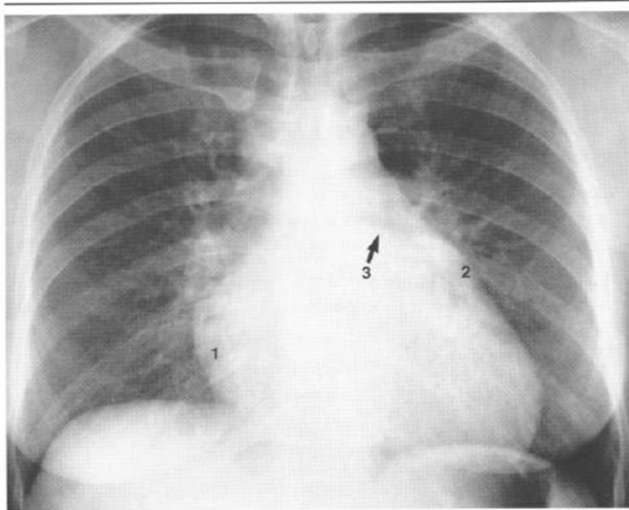
4-Myxomatous degeneration of the MV.

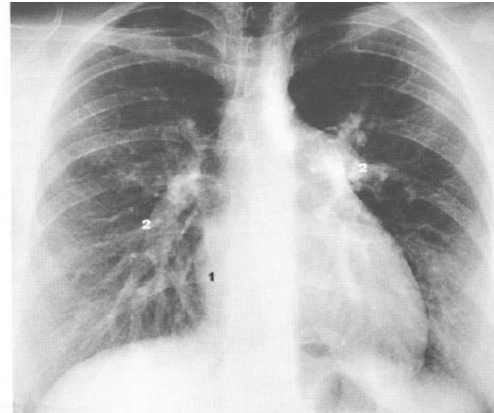
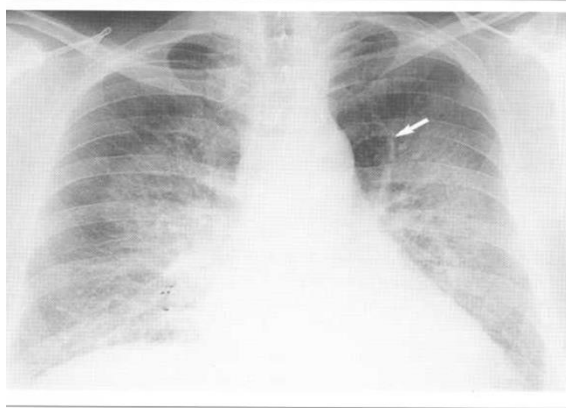
- MR may develop.
- Non-specific chest pain and palpitation due to recurrent arrhythmia (usually respond to Beta blockers).
- By auscultation: mid systolic click and may be late systolic murmur due to MR (click murmur syndrome).

Investigation to MR:

1-X-ray:

- LA enlargement
- LV enlargement
- Pulmonary congestion & HTN





2-ECG:

- a) Normal
- b) LA dilatation
- c) LV hypertrophy – dilatation
- d) RV hypertrophy – dilatation
- e) AF

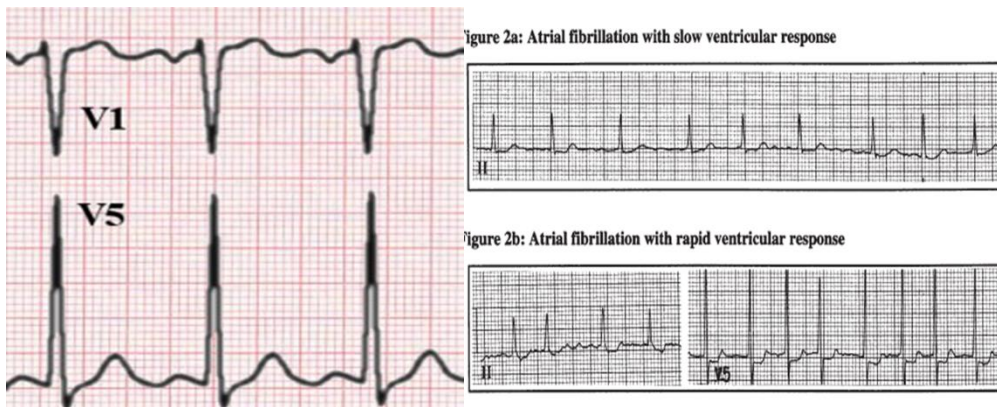
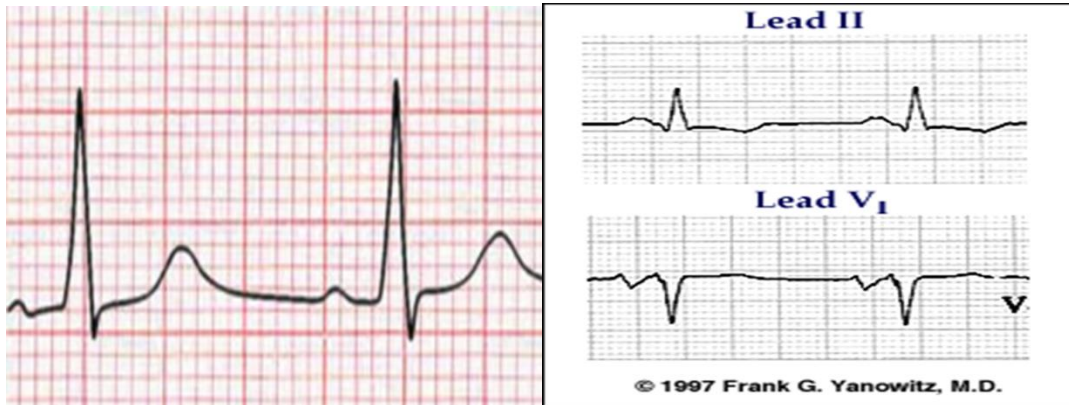


Figure 2b: Atrial fibrillation with rapid ventricular response



3-Echocardiography:

- a) Degree of LV, and LA dilation
- b) Degree of MR
- c) PH

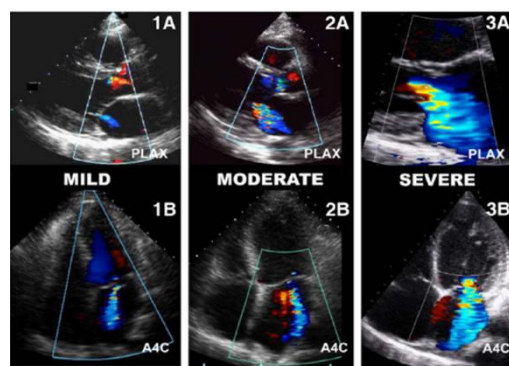
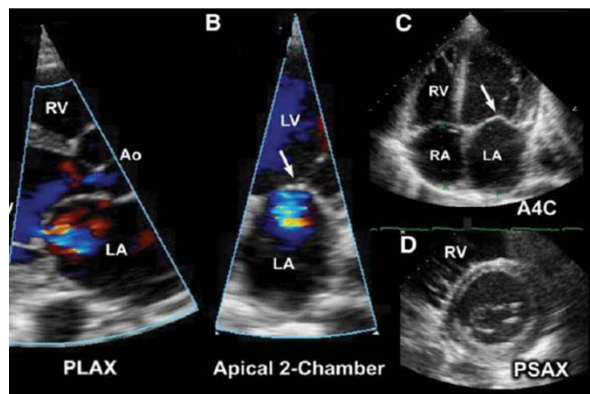
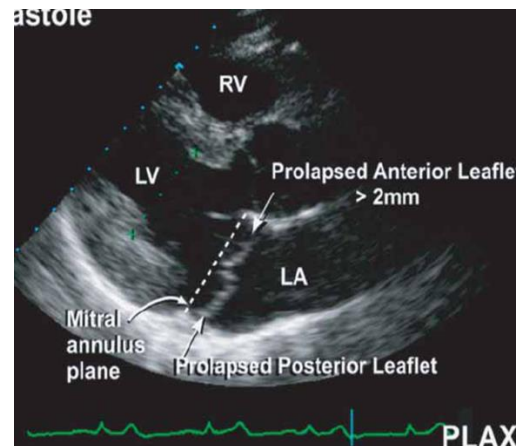
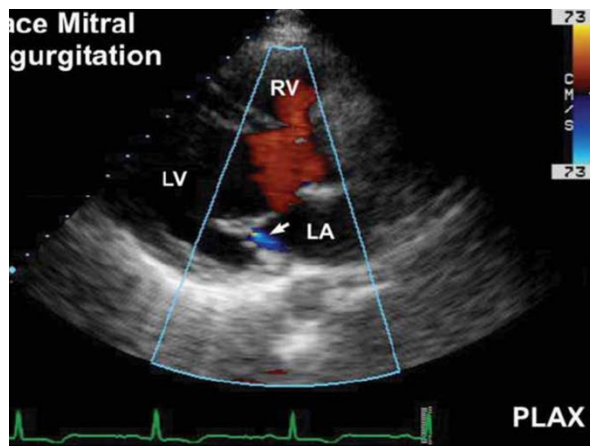
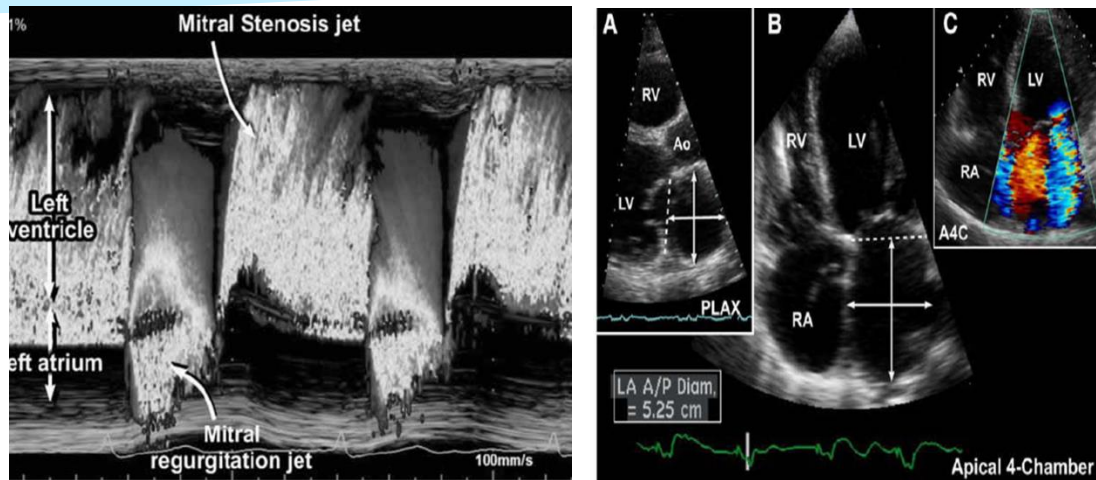


Fig. 19. Color flow jet areas. (Please see companion DVD for corresponding video.)



Differential Diagnosis:

1-TR

2-VSD

Complications:

1-LV failure, PH and RV failure.

2-AF.

3-Thrombi in the LA may lead to systemic embolization

4-Thrombosis in the RA may also occur and result in pulmonary embolism

5-Infective endocarditis

6-Calcification of the MV

7-Recurrence of rheumatic activity

Treatment of chronic MR:

1-Prophylaxis against Rh-F

2-Prophylaxis against Infective Endocarditis.



3-Treat complications as HF, arrhythmias.

4-Afterload reducing agent as: Nitrates & vasodilators

5- ACE inhibitors (decrease the LV size)

6-Regular follow up of all patients, clinically and by Echo.

*Asymptomatic patients with stable LV dimensions can be treated medically.

*Symptomatic patients, severe MR, and evidence of progressive LV dilatation
Surgery must be advised before impairment of LV function.

* MV repair

* MV replacement.

Treatment of acute MR

ACUTE MR is a medical emergency

Acute left ventricular failure needs urgent CC U admission and prepare for **urgent** surgical mitral valve **replacement**.

- Assurance
- Beta blocker
- Follow up as cases of mild chronic MR



5

TRICUSPID STENOSIS

Etiology:

- 1- Rh. Fever.
- 2- Carcinoid tumor.
- 3- Endocardial fibrosis.
- 4- Rt. Atrial myxoma.

Physiology

Tricuspid stenosis causes high Rt. Atrial pressure and elevated systemic venous pressure.

Clinical features:

Symptoms:

- Dyspnea.
- Fatigue.

Signs:

- Pulsation of the neck veins (giant A wave).
- L.L. edema. - Hepatomegally.
- Accentuated 1st heart sound.
- Mid diastolic murmur along the Lt. lower sternal border (may be presystolic).
- Opening snap.



Investigations:

1 - chest x - ray:

- Large Rt. Atrium.
- No prominence of the pulmonary arteries.

2 - ECG:

- P wave tall and tented.
- No Rt. Ventricular hypertrophy.

3 - Echocardiography:

- Cross sectional area of the valve.
- Echo Doppler: detect and assess the gradient across the tricuspid valve.

4 - Cardiac catheterization:

Treatment:

- Medical: - Surgical:

TRICUSPID REGURGITATION

Etiology:

- 1- 2ry to Rt. Ventricular dilation.
- 2- Infective endocarditis.
- 3- Trauma.
- 4- Prolapse.
- 5- Congenital.

Clinical features:

Symptoms

Sign

- Peripheral edema.
- Large C & V waves.
- Muffled 1st heart sound.
- Holosystolic murmur.

Investigations:

- 1 - Chest x - ray: - Rt. Atrium and Rt. Ventricle.
- 2 - ECG: - Atrial fibrillation.
- 3 - Echocardiography:



Treatment:

1 - Medical: - Antibiotic prophylaxis. - Treatment of Rt. Ventricular failure. 2 - Surgical; - Annuloplasty. - Valve replacement.



5

PULMONARY STENOSIS

Etiology:

- 1- Congenital (isolated or part of other syndromes like fallot's tetralogy)
- 2- Carcinoid syndrome.
- 3- Rheumatic (rare).

The hemodynamic changes:

- The arterial oxygen saturation decreases causing cyanosis in sever degree.
- Rt. Ventricular end diastolic pressure increases
- Increase of the Rt. Atrial pressure result in Rt. to Lt. shunting.

Clinical features:

Symptoms:

- No Symptoms.
- Effort dyspnea, chest pain and fainting.

Sign

- Prominent A wave.
- Decreased P2. - Systolic murmur (mid systolic peaking).
- Pulmonic ejection click.

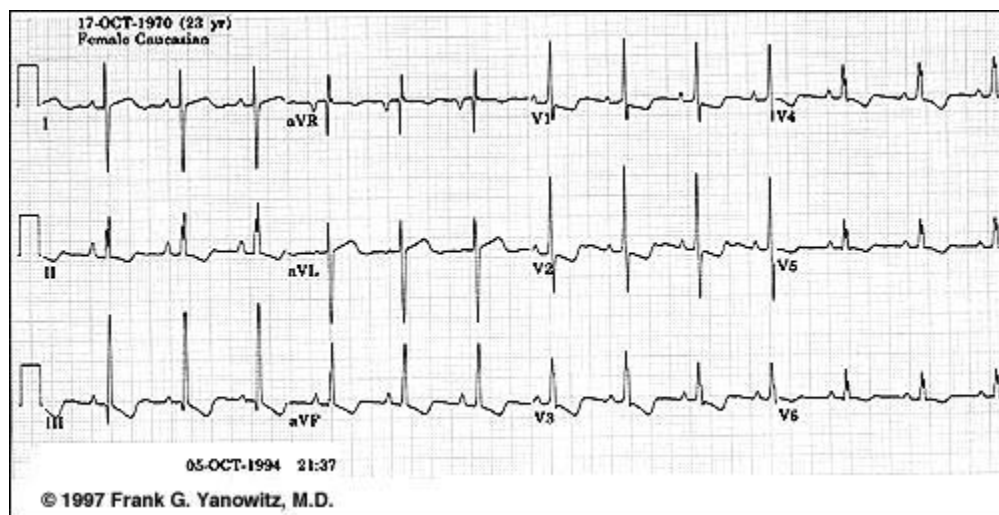
Investigations:

1 - Chest x - ray:

- Lung oligemia.
- Rt. Atrial enlargement.
- Rt. Ventricular enlargement.

2 - ECG:

- Rt. Atrial enlargement.
- Rt. Ventricular hypertrophy.



3 - Echocardiography: - Estimate the severity.

Treatment:

- 1- Balloon valvotomy.
- 2- Surgical treatment.



PULMONARY REGURGITATION

Etiology:

- 1- Pulmonary hypertension.
- 2- Infective endocarditis.
- 3- Traumatic.

Clinical features:

- Manifestation of right-side HF.
- Graham Steel murmur.

Investigations:

- 1 - **Chest x - ray:** - Prominent pulmonary artery.
- 2 - **ECG:**
- 3 - **Echocardiography:**

Treatment:

- 1- Medical:
- 2- Surgical:



Definition

- Heart failure is not a disease but a syndrome – a combination of signs and symptoms – caused by the failure of the heart to pump blood to support the circulatory system at rest or during activity.
- It develops when the heart fails to properly fill with blood during diastole, resulting in a decrease in intracardiac pressures or in ejection during systole, reducing cardiac output to the rest of the body.
- The filling failure and high intracardiac pressure can lead to fluid accumulation in the veins and tissue. This manifests as water retention and swelling due to fluid accumulation (edema) called congestion. Impaired ejection can lead to inadequate blood flow to the body tissues, resulting in ischemia

Epidemiology

In 2022, heart failure affected about 64 million people globally. Overall, around 2% of adults have heart failure. In those over the age of 75, rates are greater than 10%

Signs and symptoms

The term "**congestive heart failure**" is often used because one of the most common symptoms is congestion or fluid accumulation in the tissues and veins of the lungs or other parts of a person's body. Congestion manifests itself particularly in the form of fluid accumulation and swelling (edema), in the form of peripheral edema (causing swollen limbs and feet) and pulmonary edema (causing difficulty breathing) and ascites .



Symptoms of heart failure are traditionally divided into left-sided and right-sided because the left and right ventricles supply different parts of the circulation. In biventricular heart failure, both sides of the heart are affected. Left-sided heart failure is the more common

LEFT-SIDED FAILURE

- The left side of the heart takes oxygen-rich blood from the lungs and pumps it to the rest of the circulatory system in the body, except for the pulmonary circulation).
- Failure of the left side of the heart causes blood to back up into the lungs, causing breathing difficulties and fatigue due to an insufficient supply of oxygenated blood.
- **Common respiratory signs** include increased respiratory rate and labored breathing (nonspecific signs of shortness of breath). Rales or crackles heard initially in the lung bases and when severe in all lung fields indicate the development of pulmonary edema (fluid in the alveoli). Cyanosis, indicates deficiency of oxygen in the blood, is a late sign of extremely severe pulmonary edema.
- **Other signs** of left ventricular failure include a laterally displaced apex beat (which occurs when the heart is enlarged) and a gallop rhythm (additional heart sounds), which may be heard as a sign of increased blood flow or increased intracardiac pressure. Heart murmurs may indicate the presence of valvular heart disease, either as a cause (e.g., aortic stenosis) or as a consequence (e.g., mitral regurgitation) of heart failure.



- Reverse insufficiency of the left ventricle causes congestion in the blood vessels of the lungs, so that symptoms are predominantly respiratory. Reverse insufficiency can be divided into the failure of the left atrium, the left ventricle, or both within the left circuit.
- Patients will experience shortness of breath (dyspnea) on exertion and, in severe cases, dyspnea at rest. Increasing breathlessness while lying down, called orthopnea, also occurs. It can be measured by the number of pillows required to lie comfortably, with extreme cases of orthopnea forcing the patient to sleep sitting up.
- Another symptom of heart failure is paroxysmal nocturnal dyspnea: a sudden nocturnal attack of severe shortness of breath, usually occurring several hours after falling asleep.
- There may be "cardiac asthma" or wheezing. Impaired left ventricular forward function can lead to symptoms of poor systemic perfusion such as dizziness, confusion, and cool extremities at rest.

RIGHT-SIDED FAILURE

- Right-sided heart failure is often caused by pulmonary heart disease (cor pulmonale), which is typically caused by issues with pulmonary circulation such as pulmonary hypertension or pulmonic stenosis.
- Physical examination may reveal pitting peripheral edema, ascites, liver enlargement, and spleen enlargement. Jugular venous pressure is frequently assessed as a marker of fluid status, which can be accentuated by



testing hepatojugular reflux. If the right ventricular pressure is increased, a parasternal heave

- Backward failure of the right ventricle leads to congestion of systemic capillaries. This causes swelling under the skin (peripheral edema or anasarca) and usually affects the dependent parts of the body first, causing foot and ankle swelling in people who are standing up and sacral edema in people who are predominantly lying down.
- Nocturia may occur when fluid from the legs is returned to the bloodstream while lying down at night. In progressively severe cases, ascites (fluid accumulation in the abdominal cavity causing swelling) and liver enlargement may develop. Significant liver congestion may result in impaired liver function (congestive hepatopathy), jaundice, and coagulopathy

Causes

Common causes of heart failure include coronary artery disease, including a previous myocardial infarction (heart attack), high blood pressure, atrial fibrillation, valvular heart disease, excess alcohol use, infection, and cardiomyopathy of an unknown cause. In addition, viral infections of the heart can lead to inflammation of the muscular layer of the heart and subsequently contribute to the development of heart failure. Genetic predisposition plays an important role. If more than one cause is present, progression is more likely and prognosis is worse.

Classification

- One historical method of categorizing heart failure is by the side of the heart involved (left heart failure versus right heart failure). However, mixed presentations are common and left heart failure is a common cause of right heart failure.
- More accurate classification of heart failure type is made by measuring ejection fraction, or the proportion of blood pumped out of the heart during a single contraction. Ejection fraction is given as a percentage with the normal range being between 50 and 75%. The types are:
 - 1. Heart failure with reduced ejection fraction (HFrEF):** Synonyms no longer recommended are "heart failure due to left ventricular systolic dysfunction" and "systolic heart failure". HFrEF is associated with an ejection fraction less than 40%.
 - 2. Heart failure with mildly reduced ejection fraction (HFmrEF),** previously called "heart failure with mid-range ejection fraction", is defined by an ejection fraction of 41–49%.
 - 3. Heart failure with preserved ejection fraction (HFpEF):** Synonyms no longer recommended include "diastolic heart failure" and "heart failure with normal ejection fraction." HFpEF occurs when the left ventricle contracts normally during systole, but the ventricle is stiff and does not relax normally during diastole, which impairs filling

Staging

- Heart failure is commonly stratified by the degree of functional impairment conferred by the severity of the heart failure, as reflected in the New York Heart Association (NYHA) functional classification.
- The NYHA functional classes (I–IV) begin with class I, which is defined as a person who experiences no limitation in any activities and has no symptoms from ordinary activities.
- People with NYHA class II heart failure have slight, mild limitations with everyday activities; the person is comfortable at rest or with mild exertion. With NYHA class III heart failure, a marked limitation occurs with any activity; the person is comfortable only at rest.
- A person with NYHA class IV heart failure is symptomatic at rest and becomes quite uncomfortable with any physical activity. This score documents the severity of symptoms and can be used to assess response to treatment

Diagnosis

1. Ultrasound

An echocardiogram (ultrasound of the heart) is commonly used to support a clinical diagnosis of heart failure

2. Chest X-ray



Chest X-rays are frequently used to aid in the diagnosis of CHF. In a person who is compensated, this may show cardiomegaly

3. *Electrophysiology*

An electrocardiogram (ECG or EKG) may be used to identify arrhythmias, ischemic heart disease, right and left ventricular hypertrophy, and presence of conduction delay or abnormalities (e.g. left bundle branch block). Although these findings are not specific to the diagnosis of heart failure, a normal ECG virtually excludes left ventricular systolic dysfunction

4. **Blood tests**

- N-terminal pro-BNP (NT-proBNP) is the favoured biomarker for the diagnosis of heart failure, according to guidelines published 2018 in the UK. Brain natriuretic peptide 32 (BNP) is another biomarker commonly tested for heart failure. An elevated NT-proBNP or BNP is a specific test indicative of heart failure. Additionally, NT-proBNP or BNP can be used to differentiate between causes of dyspnea due to heart failure from other causes of dyspnea. If myocardial infarction is suspected, various cardiac markers may be used.
- Blood tests routinely performed include electrolytes (sodium, potassium), measures of kidney function, liver function tests, thyroid function tests, a complete blood count, and often C-reactive protein if infection is suspected

5. **Angiography**

Angiography is the X-ray imaging of blood vessels, which is done by injecting contrast agents into the bloodstream through a thin plastic tube (catheter), which is placed directly in the blood vessel. X-ray images are called angiograms.^[74] Heart



failure may be the result of coronary artery disease, and its prognosis depends in part on the ability of the coronary arteries to supply blood to the myocardium (heart muscle). As a result, coronary catheterization may be used to identify possibilities for revascularisation through percutaneous coronary intervention or bypass surgery.

Management

Treatment focuses on improving the symptoms and preventing the progression of the disease. Reversible causes of heart failure also need to be addressed (e.g. infection, alcohol ingestion, anemia, thyrotoxicosis, arrhythmia, and hypertension). Treatments include lifestyle and pharmacological modalities, and occasionally various forms of device therapy. Rarely, cardiac transplantation is used as an effective treatment when heart failure has reached the end stage

Acute decompensation:

In acute decompensated heart failure, the immediate goal is to re-establish adequate perfusion and oxygen delivery to end organs. This entails ensuring that airway, breathing, and circulation are adequate. Immediate treatments usually involve some combination of vasodilators such as nitroglycerin, diuretics such as furosemide, and possibly noninvasive positive pressure ventilation. Supplemental oxygen is indicated in those with oxygen saturation levels below 90%, but is not recommended in those with normal oxygen levels in normal atmosphere

Chronic management

The goals of the treatment for people with chronic heart failure are the prolongation of life, prevention of acute decompensation, and reduction of symptoms, allowing for greater activity.



- Heart failure can result from a variety of conditions. In considering therapeutic options, excluding reversible causes is of primary importance, including thyroid disease, anemia, chronic tachycardia, alcohol use disorder, hypertension, and dysfunction of one or more heart valves.
- Treatment of the underlying cause is usually the first approach to treating heart failure. In the majority of cases, though, either no primary cause is found or treatment of the primary cause does not restore normal heart function.
- In these cases, behavioral, medical and device treatment strategies exist that can provide a significant improvement in outcomes, including the relief of symptoms, exercise tolerance, and a decrease in the likelihood of hospitalization or death.
- Breathlessness rehabilitation for chronic obstructive pulmonary disease and heart failure has been proposed with exercise training as a core component.
- Rehabilitation should also include other interventions to address shortness of breath including psychological and educational needs of people and needs of caregivers.
- Iron supplementation appears to reduce hospitalization but not all-cause mortality in patients with iron deficiency and heart failure.

Monitoring

The various measures often used to assess the progress of people being treated for heart failure include fluid balance (calculation of fluid intake and excretion) and monitoring body weight (which in the shorter term reflects fluid shifts). Remote monitoring can be effective to reduce complications for people with heart failure.

Lifestyle

Behavior modification is a primary consideration in chronic heart failure management program, with dietary guidelines regarding fluid and salt intake. Fluid restriction is important to reduce fluid retention in the body and to correct the hyponatremic status of the body. The evidence of benefit of reducing salt, however, is poor as of 2018.

Exercise and physical activity

Exercise should be encouraged and tailored to suit individual's capabilities. A meta-analysis found that centre-based group interventions delivered by a physiotherapist are helpful in promoting physical activity in HF. The inclusion of regular physical conditioning as part of a cardiac rehabilitation program can significantly improve quality of life and reduce the risk of hospital admission for worsening symptoms, but no evidence shows a reduction in mortality rates as a result of exercise.

Home visits and regular monitoring at heart-failure clinics reduce the need for hospitalization and improve life expectancy.

Medication

- Quadruple medical therapy using a combination of angiotensin receptor-neprilysin inhibitors (ARNI), beta blockers, mineralocorticoid receptor antagonists (MRA), and sodium/glucose cotransporter 2 inhibitors (SGLT2 inhibitors) is the standard of care as of 2021 for heart failure with reduced ejection fraction (HFrEF).



- There is no convincing evidence for pharmacological treatment of heart failure with preserved ejection fraction (HFpEF). Medication for HFpEF is symptomatic treatment with diuretics to treat congestion. Managing risk factors and comorbidities such as hypertension is recommended in HFpEF.
1. **Inhibitors of the renin–angiotensin system (RAS)** are recommended in heart failure. The angiotensin receptor-neprilysin inhibitors (ARNI) sacubitril/valsartan is recommended as first choice of RAS inhibitors in American guidelines published by AHA/ACC in 2022. Use of angiotensin-converting enzyme (ACE) inhibitors (ACE-I), or angiotensin receptor blockers (ARB) if the person develops a long-term cough as a side effect of the ACE-I, is associated with improved survival, fewer hospitalizations for heart failure exacerbations, and improved quality of life in people with heart failure. European guidelines published by ESC in 2021 recommends that ARNI should be used in those who still have symptoms while on an ACE-I or ARB, beta blocker, and a mineralocorticoid receptor antagonist. Use of the combination agent ARNI requires the cessation of ACE-I or ARB therapy at least 36 hours before its initiation.
 2. **Beta-adrenergic blocking agents (beta blockers)** add to the improvement in symptoms and mortality provided by ACE-I/ARB. The mortality benefits of beta blockers in people with systolic dysfunction who also have atrial fibrillation is more limited than in those who do not have it. If the ejection fraction is not diminished (HFpEF), the benefits of beta blockers are more modest; a decrease in mortality has been observed, but reduction in hospital admission for uncontrolled symptoms has not been observed.
 3. In people who are intolerant of ACE-I and ARB or who have significant kidney dysfunction, the use of **combined hydralazine and a long-acting nitrate**, such



as isosorbide dinitrate, is an effective alternate strategy. This regimen has been shown to reduce mortality in people with moderate heart failure. It is especially beneficial in the black population.

4. . Use of a **mineralocorticoid antagonist**, such as spironolactone or eplerenone, in addition to beta blockers and ACE-I, can improve symptoms and reduce mortality in people with symptomatic heart failure with reduced ejection fraction (HFrEF). SGLT2 inhibitors are used for heart failure

Other medications

- **Second-line medications** for CHF do not confer a mortality benefit.

1. **Digoxin** is one such medication. Its narrow therapeutic window, a high degree of toxicity, and the failure of multiple trials to show a mortality benefit have reduced its role in clinical practice. It is now used in only a small number of people with refractory symptoms, who are in atrial fibrillation, and/or who have chronic hypotension.

2. **Diuretics** have been a mainstay of treatment against symptoms of fluid accumulation, and include diuretics classes such as loop diuretics (such as furosemide), thiazide-like diuretics, and potassium-sparing diuretics. Although widely used, evidence on their efficacy and safety is limited, with the exception of mineralocorticoid antagonists such as spironolactone.

4. Anemia is an independent factor in mortality in people with chronic heart failure. **Treatment of anemia** significantly improves quality of life for those with heart failure, often with a reduction in severity of the NYHA classification, and also improves mortality rates. European Society of Cardiology recommends



screening for iron deficiency and treating with intravenous iron if deficiency is found.

5. The decision to **anticoagulate** people with HF, typically with left ventricular ejection fractions $<35\%$ is debated, but generally, people with coexisting atrial fibrillation, a prior embolic event, or conditions that increase the risk of an embolic event such as amyloidosis, left ventricular noncompaction, familial dilated cardiomyopathy, or a thromboembolic event in a first-degree relative.

6. **Vasopressin receptor antagonists** can also be used to treat heart failure. Conivaptan is the first medication approved by US Food and Drug Administration for the treatment of euvolemic hyponatremia in those with heart failure. In rare cases hypertonic 3% saline together with diuretics may be used to correct hyponatremia.

7. **Ivabradine** is recommended for people with symptomatic heart failure with reduced left ventricular ejection fraction who are receiving optimized guideline-directed therapy (as above) including the maximum tolerated dose of beta-blocker, have a normal heart rhythm and continue to have a resting heart rate above 70 beats per minute. Ivabradine has been found to reduce the risk of hospitalization for heart failure exacerbations in this subgroup of people with heart failure.

Implanted devices

In people with severe cardiomyopathy (left ventricular ejection fraction below 35%), or in those with recurrent VT or malignant arrhythmias, treatment with an automatic implantable cardioverter-defibrillator (AICD) is indicated to reduce the risk of severe life-threatening arrhythmias. The AICD does not improve symptoms or



reduce the incidence of malignant arrhythmias but does reduce mortality from those arrhythmias, often in conjunction with antiarrhythmic medications.

Cardiac contractility modulation (CCM) is a treatment for people with moderate to severe left ventricular systolic heart failure (NYHA class II–IV), which enhances both the strength of ventricular contraction and the heart's pumping capacity.

About one-third of people with LVEF below 35% have markedly altered conduction to the ventricles, resulting in dyssynchronous depolarization of the right and left ventricles. This is especially problematic in people with left bundle branch block . Using a special pacing algorithm, biventricular cardiac resynchronization therapy (CRT) can initiate a normal sequence of ventricular depolarization

Surgical therapies

- People with the most severe heart failure may be candidates for ventricular assist devices, which have commonly been used as a bridge to heart transplantation, but have been used more recently as a destination treatment for advanced heart failure.
- In select cases, heart transplantation can be considered. While this may resolve the problems associated with heart failure, the person must generally remain on an immunosuppressive regimen to prevent rejection, which has its own significant downsides.



ISCHEMIC HEART DISEASE

- Most common cause of death in developed countries.

Occur when there is imbalance between Oxygen supply and myocardial demand

Causes:

- Usually due to decreased blood flow:
 1. Most common is atheroma and its complications.
 2. Coronary spasm
 3. Coronary embolism
 4. Congenital anomalies e.g. coronary ostial stenosis.
 5. Coronary arteritis as PAN, SLE
 6. Aortic valve disease
- Decreased oxygenated blood as in anemia, methemoglobinemia.
- Increased demand as in thyrotoxicosis and myocardial hypertrophy

ATHEROMA OR ATHEROSCLEROSIS

- It is a patchy focal disease of the arterial wall.



Sequence

1. Various risk factors contribute to initial endothelial injury causing accumulation of oxidized lipoproteins inside the macrophages in the intima; Fatty streaks
2. Release of fat and appearance of extracellular lipid pool; Transitional plaque
3. Release of cytokines causing proliferation of cells over lipid layer; Fibrinoid plaque
4. Growth into the lumen or thrombus formation; Complicated plaque

Risk factors

A. Fixed: non-modifiable

Age, gender, Family history, Genetic factors.

B. Modifiable:

1. Smoking.
2. Hypertension
3. Dyslipidemia
4. DIABETES
5. Diet
6. Lack of exercise
7. OBESITY
8. Alcohol
9. Coagulation factors; s. fibrinogen, factor VII
10. Homocysteine Mia

CORONARY ARTERY DISEASES

Definitions

- Stable, classical or exertional angina Ischemia due to fixed atheromatous stenosis of one or more coronary arteries. Usually provoked by physical exertion, aggravated by stress, after meals or in cold weather, usually fades quickly.
- Unstable or crescendo angina or acute coronary syndrome
- Dynamic obstruction due to plaque rupture with superimposed thrombosis or spasm. Occur at rest, prolonged and rapidly worsening angina
- Variant or Prinzmetal's angina

Angina occurs without provocation, usually at rest as a result of coronary artery spasm

- Myocardial infarction

Acute occlusion of a coronary artery due to plaque rupture and thrombosis resulting in myocardial necrosis.

MYOCARDIAL ISCHEMIA; ANGINA

Chronic coronary syndromes (CCS);

Angina; a clinical syndrome used to describe the discomfort due to transient myocardial ischemia.

CCS are a range of clinical presentations that arise due to structural and or functional changes in epicardial coronary arteries and or coronary microcirculation.



This term describes CAD during stable periods and includes the following presentations;

- 1- Stress induced angina or equivalent with obstructive CAD.
- 2- Stabilized phase after ACS, PCI or CABG.
- 3- LV dysfunction or HF of ischemic etiology.
- 4- Angina or equivalent without obstructive CAD.
- 5- Asymptomatic patients with abnormal coronary anatomical or functional tests.

Diagnosis based on clinical history

A. Chest pain:

described as heaviness, tightness or gripping.

- Typically, central or retrosternal, occasionally epigastric or interscapular
- May radiate to Jaw, arms or hands

A. Angina pectoris

- Typical chest pain
- Physical examination is frequently negative.
- Carefully search for evidence of:
 1. Risk factors
 2. Contributory disease as obesity, myxedema, hypertension,
 3. Signs of left ventricular dysfunction
 4. Aortic valve disease

Differential diagnosis

1. Musculoskeletal pain
2. Pericarditis
3. Eosophagitis
4. Eosophageal spasm

Investigation

1. Resting ECG:

Reversible changes during pain
Normal between attacks, sometimes evidence of previous infarction, left ventricular hypertrophy or left BBB.

2. Exercise ECG

Useful in assessing the presence and severity of coronary disease. Specificity 80% and sensitivity 70%.

3. Cardiac Scintigraphy (isotope scanning):

Helpful in an equivocal or uninterpretable exercise test. IV radioactive isotope as Thallium201 is used & the test is done at rest and after physical or pharmacological stress.

4. Echocardiography

To assess ventricular function and ventricular wall involvement.

5. Coronary angiography

In patients considered for revascularization



Angina Pectoris Management

A. General management:

- Correct risk factors.
- Correct underlying problem as anemia or thyroid disease.
- Management of coexistent conditions as diabetes or hypertension.

B. Medical management:

- **Prognostic therapies:**

1. **Aspirin:**

Reduces the risk of coronary events, should be prescribed for all patients unless contraindicated (75-150 mg daily).

2. **Lipid-lowering therapy: statins**

- **Symptomatic treatment:**

3. **Nitrates:**

Act directly on vascular smooth muscle to produce venous and arteriolar dilatation, it reduce myocardial O₂ demand and increase myocardial O₂ supply.

- **Glyceryl trinitrate:** used sublingually (300-500 μ g)
- **Long acting nitrates:** useful in patients who gain relief from sublingual nitrates.
- **Intravenous nitrates.**

4. **Beta-blockers:**

- It reduce myocardial O₂ demand by reducing heart rate and the force of ventricular contraction.
- Drug of choice in patients with previous myocardial infarction for their proven benefit in secondary prevention.



- Atenolol 50-100mg/day, Metoprolol, 25-50 twice/day, bisoprolol 5-10mg/day

5. Ca-channel blockers:

- Block Ca flux into the cells, so they relax coronaries, cause peripheral vasodilatation and reduce the force of ventricular contraction.
- Non-dihydropyridine derivatives (verapamil and diltiazem) also reduce heart rate.
- Short acting dihydropyridine as nifedipine and amlodipine cause reflex tachycardia.

6. K-channel activator:

- Nicorandil, used when others are contraindicated

C. Invasive management: PCI

Coronary Angioplasty:

- Percutaneous transluminal coronary angioplasty; PTCA refers to the technique of dilating coronary atheromatous obstructions by inflating a balloon within the obstruction.
- It may improve symptoms but confer no significant prognostic benefit.
- Complications include acute coronary occlusion & restenosis.
- PTCA + stent insertion has significantly better results

Coronary Artery Bypass

- Two principal indications: Symptom control & Improved survival.
- Internal mammary artery or reverse saphenous vein are used to bypass major coronary artery stenosis



ACUTE CORONARY SYNDROME

- 1) UNSTABLE ANGINA
 - 2) NON-STEMI
 - 3) STEMI
- Pathophysiological changes are the same: dynamic CA obstruction provoked by plaque rupture followed by thrombus formation

Definition

The spectrum of acute ischemia related syndromes ranging from UA to MI with or without ST elevation that are secondary to acute plaque rupture or plaque erosion.

ACUTE MYOCARDIAL INFARCTION

- MI is irreversible necrosis of heart muscle secondary to prolonged ischemia
- AMI is a part of ACS
- AMI could be STEMI or NON-STEMI and Q-wave or non-Q-wave

SYMPTOMS

1-chest pain:

- severe substernal; compressing;squeezing;burning and even sharp pain
- prolonged pain >30 minutes
- radiates to left arm; neck;epigastrium
- associated with nausea; vomiting;excessive sweating;fatigue



2-dyspnea: angina equivalent or developing HF

3-atypical symptoms: in old age; diabetics; women

Signs

: variable

-may be normal or pt. may be markedly distressed

-low grade fever;hypotension or hypertension

-audible S4; systolic murmur(MR or VSD)

signs of HF: S3; basal crepitations raised JVP

Differential diagnosis

- Angina pectoris: stable & unstable
- Massive PE
- Myocarditis; pericarditis
- Dissecting aortic aneurysm
- Esophageal diseases:GERD;spasm
- Acute gastritis; peptic ulcer
- Tension pneumothorax

Diagnosis

1-ECG: +VE up to 80% of cases

ST elevation; pathologic Q-wave;T wave inversion

2-CARDIAC ENZYMES:

a- troponins: specific&sensitive: start3-12h; peak24-48h; last for two weeks



b- CK: MB isoenzyme: start 3-12h; peak 24 h; lasts 72h

3-MYOGLOBIN: highly sensitive but non-specific

4-BLOOD COUNT: leucocytosis

5-ELEVATED CRP & ESR

6-IMAGING STUDIES: X-RAY CHEST; ECHO; SCINTIGRAPHY

Complications

1-post MI angina

- in 1/3 of pts; treated as unstable; most pts undergo PCI or CABG

2-ARRHYTHMIAS:

a-sinus bradycardia: in inferior MI

b-supraventricular tachyarrhythmias

c-ventricular arrhythmias

d-conduction disturbances: AV block; bundle branch block

3-MYOCARDIAL DYSFUNCTION:

a-acute LV failure

b-hypotension & shock

c-late CHF: remodelling effect

4-RV INFARCTION

5-MECHANICAL COMPLICATIONS:

a-rupture IVS: VSD

b-rupture papillary muscle: acute MR

c-myocardial rupture: fatal



6-LV ANEURYSM

7-PERICARDITIS: early&late

8-THROMBOEMBOLISM: mural thrombus &DVT

Management

1) GENERAL MEASURES:

- admission to CCU as soon as possible
- O2 therapy
- continuous monitoring for cardiac rhythm and vital signs

2) POTENT ANALGESIA: iv opiate for pain

3) ANTIPLATELET THERAPY: initiated immediately

- aspirin :150-300mg/day forever
- clopidogrel :300mg initially then 75mg/day for 1 year

4) THROMBOLYTIC THERAPY:

- decrease mortality and size of infarction
- greatest benefit in the first 3 hours
- but can be used up to first 12 hours
- streptokinase:1.5m units in50cc dextrose 5% infused in 60 min. cheap but side effects can occur: hypotension; allergy;bleeding
- t-PA: high clot selectivity; no allergy but more expensive and should followed by iv heparin

5) PRIMARY PCI: *superior* to thrombolysis when done by experienced operator provided a door to balloon time less than 90 min.



PCI is the treatment of choice in pts with shock; contraindication to or failed thrombolysis

6) β -ADRENERGIC BLOCKERS: reduce rate of re-infarction and post MI angina. Should be given for life unless contraindicated

7) ACE INHIBITORS: should be given unless contraindicated; more beneficial in pts with LV dysfunction; ARBS could be alternative in case of side effects

8) NITRATES: only for symptomatic relief in post MI ANGINA and for preload reduction

9) MANAGEMENT OF COMPLICATIONS if developed



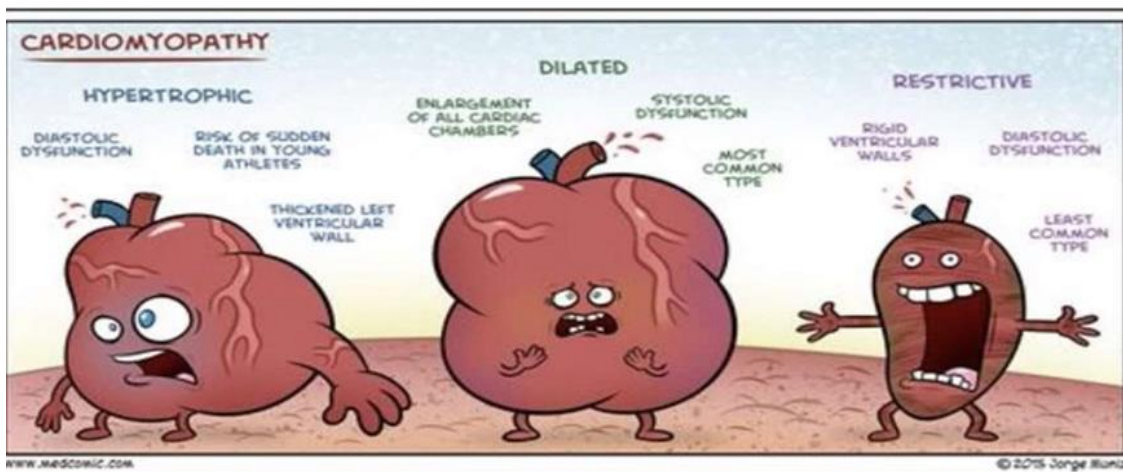
CARDIOMYOPATHY

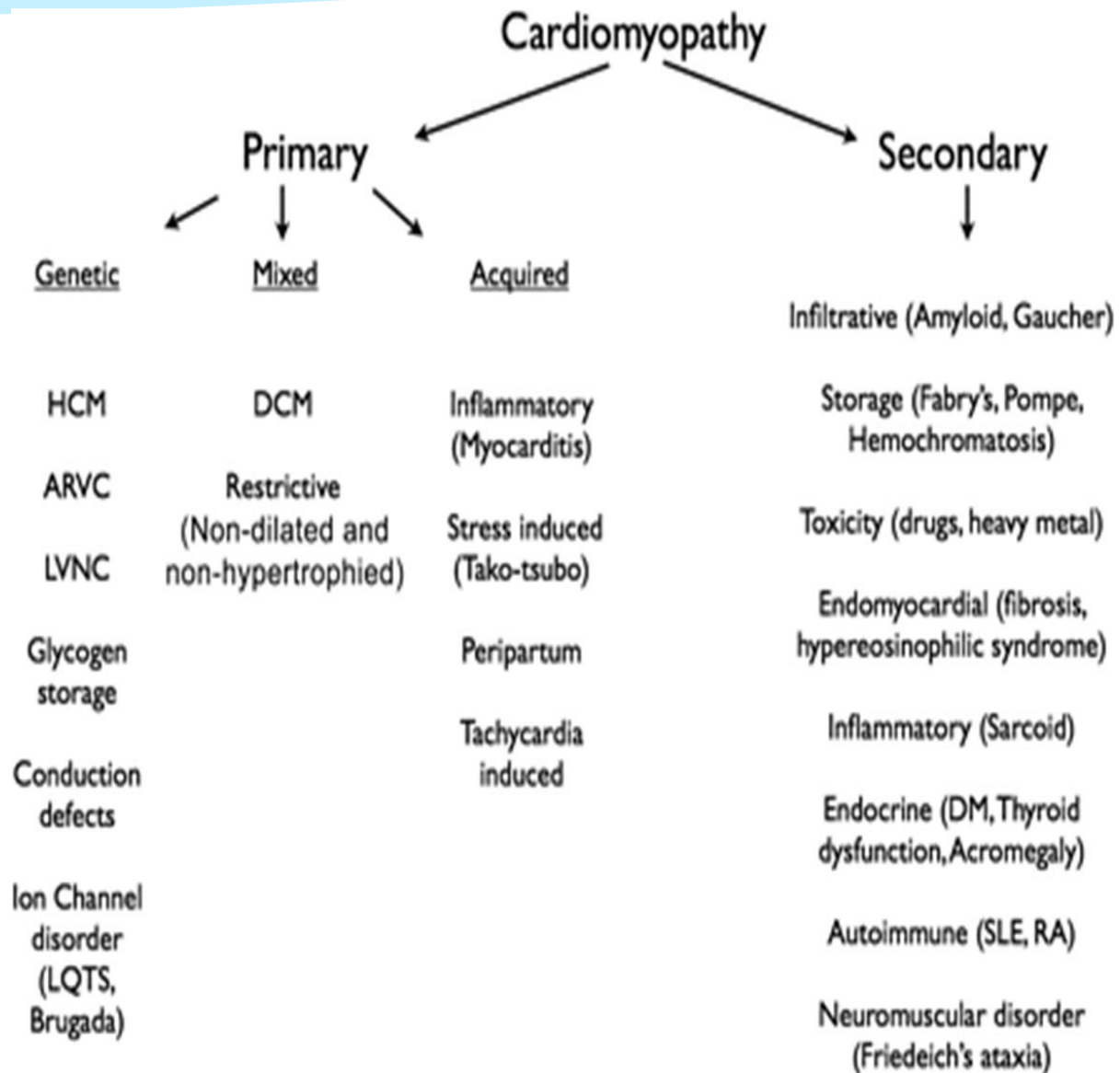
Cardiomyopathy is a general term indicating disease of the cardiac muscle.

• ***Cardiomyopathies are classified according to the predominant clinical presentation into:***

1. Dilated cardiomyopathy-DCM ventricular dilation.
2. Hypertrophic cardiomyopathy-HCM myocardial hypertrophy.
3. Restrictive cardiomyopathy- RCM impaired ventricular filling.
4. Arrhythmogenic right ventricular cardiomyo- pathy (ARVC)-right ventricular involvement is prominent with high frequency of ventricular arrhythmias.

Cardiomyopathy





DILATED CARDIOMYOPATHY

Definition

DCM is characterized by dilation and impaired systolic function of LV and/ or RV, in the absence of abnormal loading conditions (e.g. HTN, valve dis.). Dilated cardiomyopathy is a progressive disease of heart muscle that is characterized by ventricular chamber enlargement and contractile dysfunction. The right ventricle may also be dilated and dysfunctional. Dilated cardiomyopathy is the third most common cause of heart failure and the most frequent reason for heart transplantation

Causes of DCM

- 1- **Genetic:** - Autosomal dominant DCM.
X-linked DCM.
- 2- **Inflammatory:** - post infective, autoimmune, connective tissue dis. (SLE, scleroderma.)
- 3- **Metabolic:** e.g. glycogen storage dis.
- 4- **Nutritional:** e.g. Thiamine def.
- 5- **Endocrinal:** acromegaly, thyrotoxicosis, myxedema, DM
- 6- **Infiltrative:** Hereditary haemochromatosis.
- 7- **Neuromuscular:** muscle dystrophy, Friderichs ataxia.
- 8- **Toxic:** alcohol, cocaine, adriamycin.
- 9- **Haematological:** sickle cell anemia.



10- **Puerperal cardiomyopathy.**

11- **Ischemic cardiomyopathy.**

Clinical

- The reported incidence is 400,000-550,000 cases per year, with a prevalence of 4-5 million people
 - Cardiomyopathy is a complex disease process that can affect the heart of a person of any age, and clinical manifestations appear most commonly in the third or fourth decade.
- 1- Congestive heart failure: symptoms and signs of left and/or right heart failure.
 - 2- Syncope or sudden cardiac death due to ventricular arrhythmias, or conduction defect.
 - 3- Pulmonary or systemic embolism.
- Clinical evaluation should include careful family history and evaluation of relatives.

Prognosis

- Although some cases of dilated cardiomyopathy reverse with treatment of the underlying disease, many progress inexorably to heart failure. With continued decompensation, mechanical circulatory support or heart transplantation may be necessary.
- The prognosis for patients with heart failure depends on several factors, with the etiology of disease being the primary factor. Other factors play important roles in determining prognosis; for example, higher mortality rates are associated with



increased age, male sex, and severe congestive heart failure (CHF). Prognostic indices include the New York Heart Association (NYHA) functional classification.

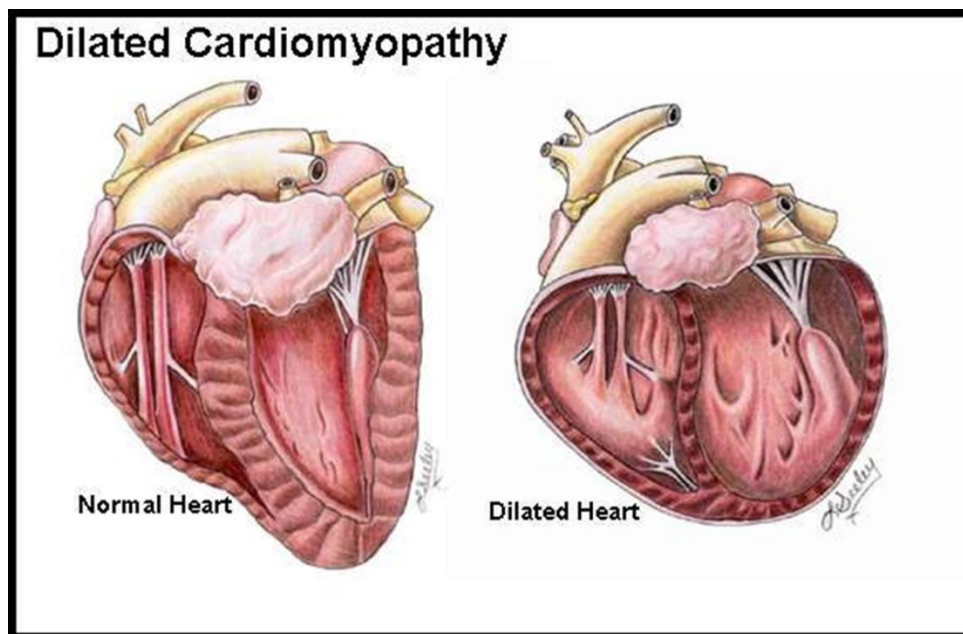
- The Framingham Heart Study found that approximately 50% of patients diagnosed with CHF died within 5 years.
- Individuals with a high likelihood of myocardial recovery following appropriate therapy include those with alcohol-induced cardiomyopathy, hypertensive cardiomyopathy, tachycardia-induced cardiomyopathy, Takotsubo cardiomyopathy, or ischemic cardiomyopathy after revascularization.

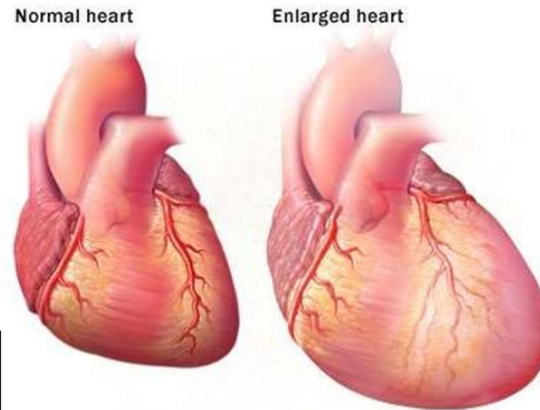
Investigations

- 1- Chest X-ray: shows cardiomegaly
- 2- ECG: sinus tachycardia, nonspecific ST, T changes conduction abnormalities QT prolongation and arrhythmias.
- 3- Echocardiography: dilation of LV and/ or RV with poor global contraction. Functional valve regurgitation.
- 4- Cardiac Biomarkers: Cardiac enzymes are useful for assessing acute or recent myocardial injury. Serum markers for myocardial necrosis (eg, troponin, creatine kinase, creatine kinase-MB) may be acutely elevated in persons with myocarditis. Levels are markedly elevated in persons with muscular dystrophy. Elevated biomarker levels may indicate acute coronary syndrome, which should be considered as a potential etiology for acute decompensation in a patient with a history of heart failure. Further, while the precise role of cardiac biomarkers is still being defined, there is evidence that patients who present with elevated markers experience more severe heart failure and higher mortality.

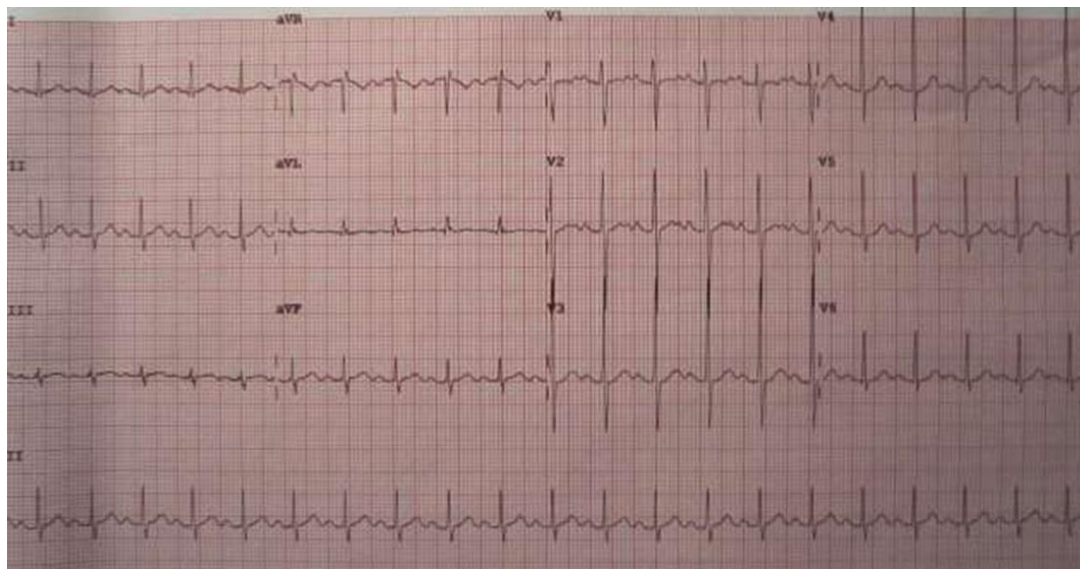
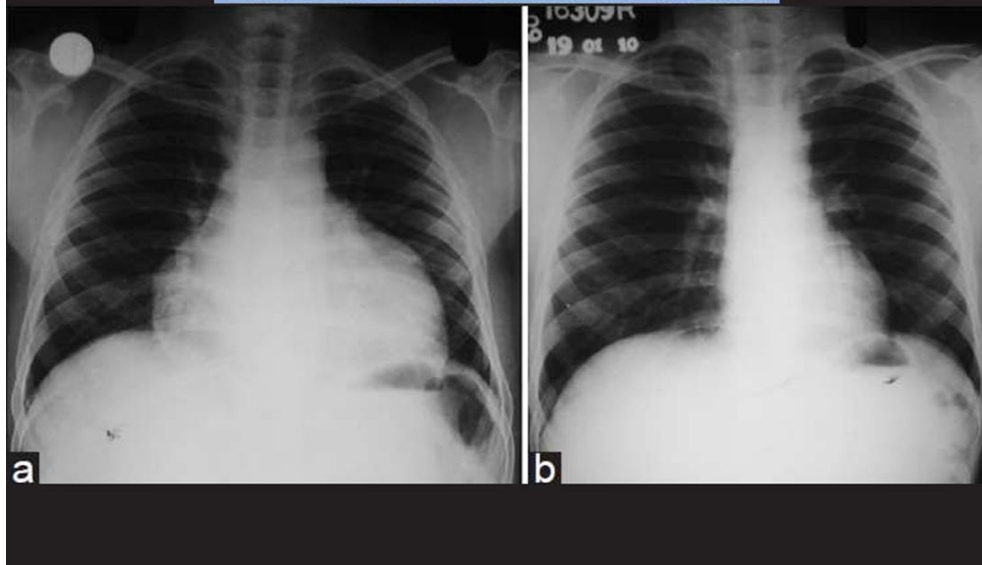
- 5- B-type natriuretic peptide (BNP) assays help monitor the presence and severity of fluid overload. Changes in BNP level can reflect response to treatment. A low level of BNP is helpful in ruling out the condition. A serum BNP below 100 pg/mL proved useful in excluding heart failure
- 6- Coronary angiography should be performed to exclude coronary artery disease in all patients at high risk.

Now replaced by CCTA





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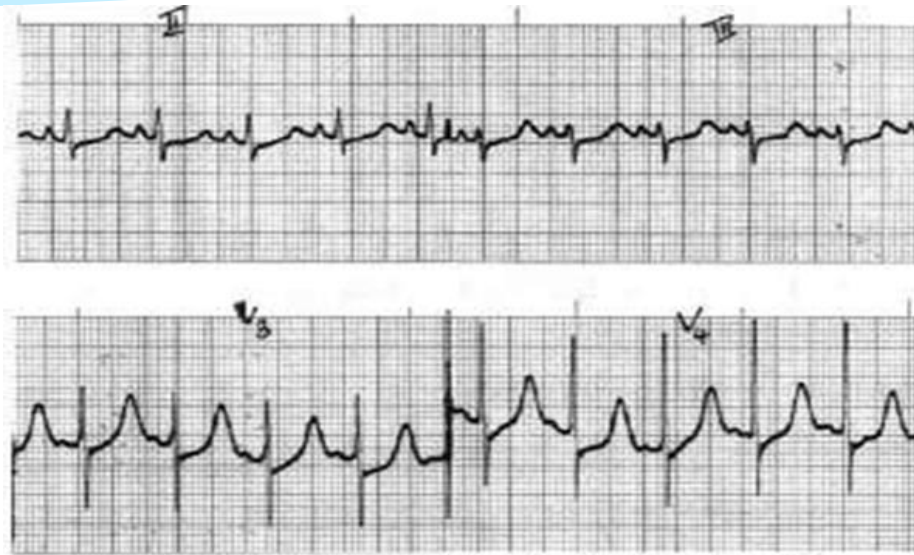
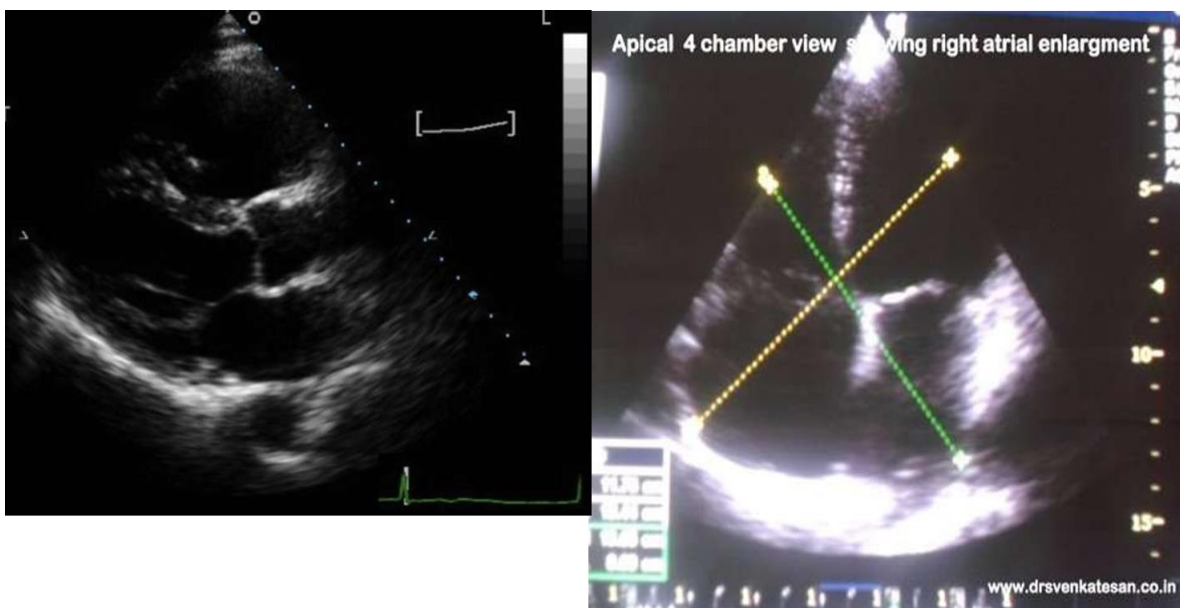


Fig. 1 : ECG at time of admission showing QTc prolongation.



Treatment

- **Goals of treatment:**

- 1- To relieve symptoms.
- 2- To retard disease progression.



3- To prevent complications.

• ***Lines of treatment:***

1- Conventional management of heart failure:

- . Diuretics .ACE-inhibitors (ARNI)
- . B-Blockers.MRAI . SGLT2I

2- Management of arrhythmias:

- . antiarrhythmic drugs . Permanent pacing .CRT
- . Implantable cardioverter-defibrillators.

3- Anticoagulation treatment:

Indicated in: -severe ventricular dilation and dysfunction, documented atrial fibrillation, or history of embolization. The patient is treated with oral anticoagulants maintaining (INR= 2-3).

4- Cardiac transplantation:

The principal option for refractory HF, and advanced disease.

5- Alternatives to cardiac transplantation:

- Ventricular assist devices (Extracorporeal devices, implantable pulsatile devices, total artificial heart).
- Cardiomyoplasty.

2

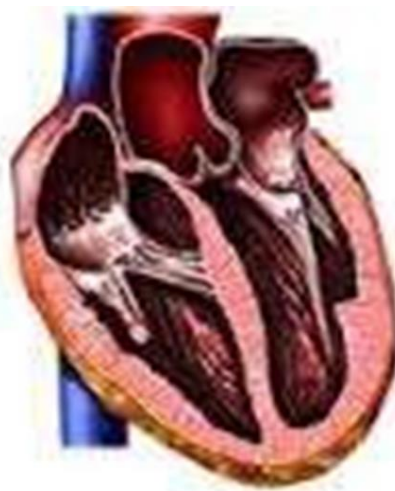
HYPERTROPHIC CARDIOMYOPATHY

Definition

Hypertrophic obstructive cardiomyopathy (HOCM) is defined as myocardial hypertrophy in the absence of known cause.

Patho-physiology

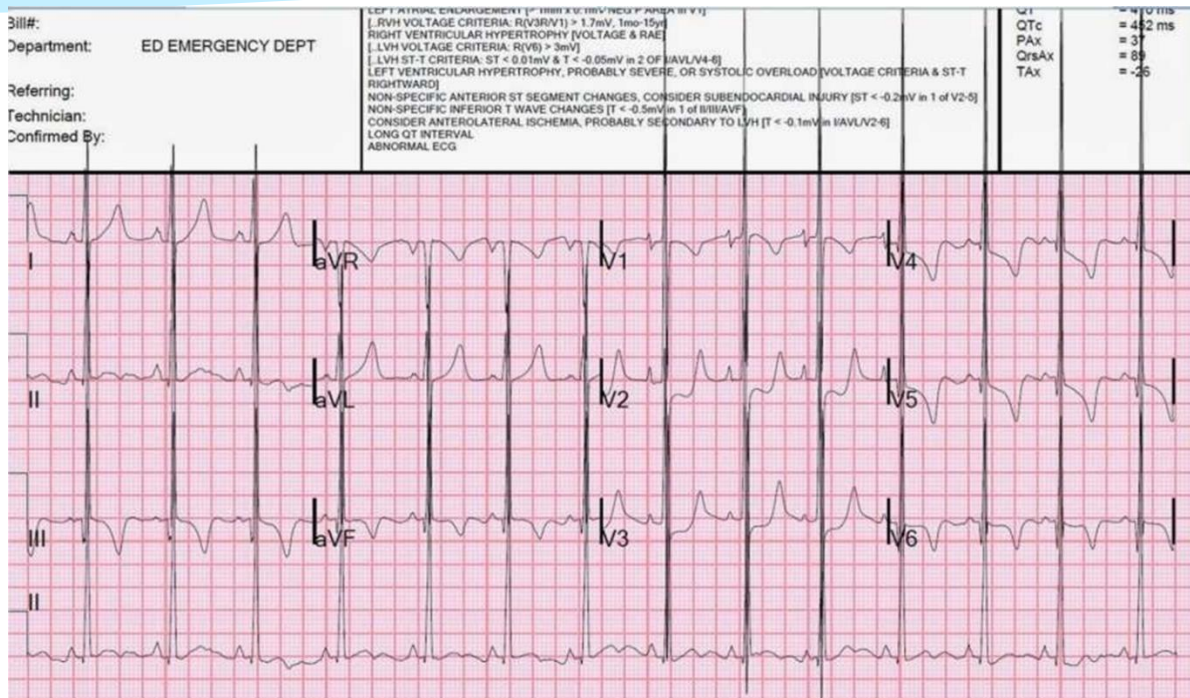
- There is disorder of cell-to-cell arrangement, disorganization of cellular architecture and fibrosis.
- The most common sites of ventricular involvement are the septum, apex and mid ventricle.



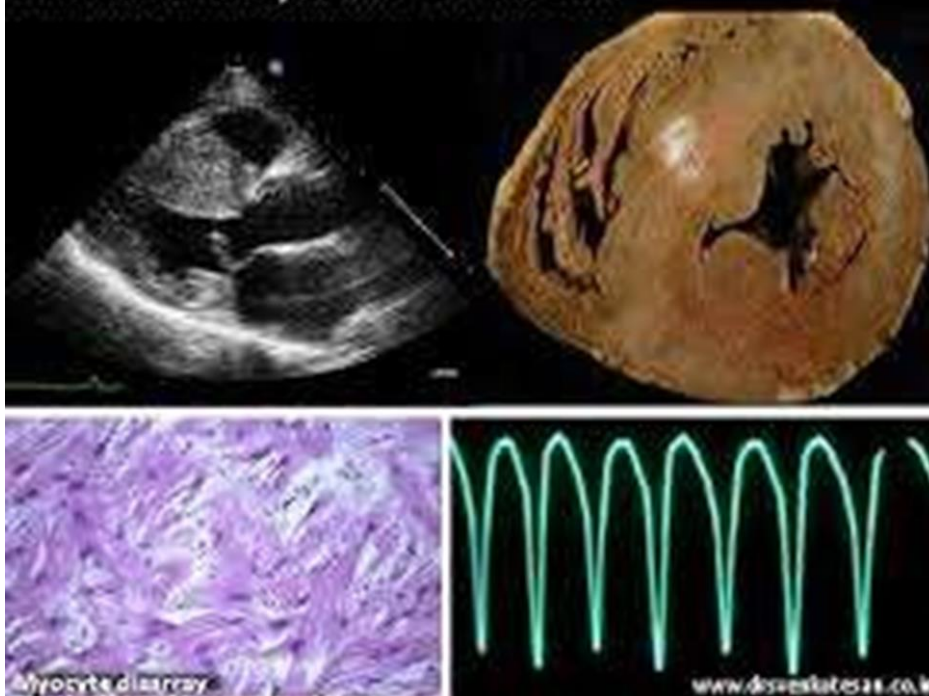
Normal heart
(cross section)



Hypertrophic
cardiomyopathy



How often LVOT is the arrhythmic focus in ventricular tachycardia of HOCM?



Clinical presentation

- The prevalence of HCM is about 1 in 500.

Signs and symptoms:

1. Heart failure: dyspnea, dyspnea on exertion, PND and fatigue (diastolic HF).
2. Myocardial ischemia
3. Syncope as a low COP symptom, and is associated with increased risk of sudden Death.
4. Sudden death: it is more common among children and young adults, and it occurs mainly during physical activity. Both arrhythmias and ischemia can be the underlying mechanisms.
5. Paradoxically, cardiovascular examination is often normal but, in patients with LV outflow tract obstruction (LVOTO), a number of typical features may be identified including a rapid up-and-down stroke to the arterial pulse and an ejection systolic murmur at the left sternal edge that radiates to the right upper sternal edge and apex. The intensity of the murmur is increased by manoeuvres that reduce ventricular preload or afterload, such as standing up from the squatting position and forceful attempted exhalation against a closed airway (Valsalva manoeuvre). Most patients with LVOTO also have signs of mitral regurgitation.

Investigations

1. **ECG:** LVH, RVH, and RA, and LA enlargement.
2. **Echocardiography:** RVH, LVH and may be obstruction, diastolic dysfunction. In an adult, HCM is defined by a wall thickness ≥ 15 mm in one or more LV myocardial segments—as measured by any imaging technique.

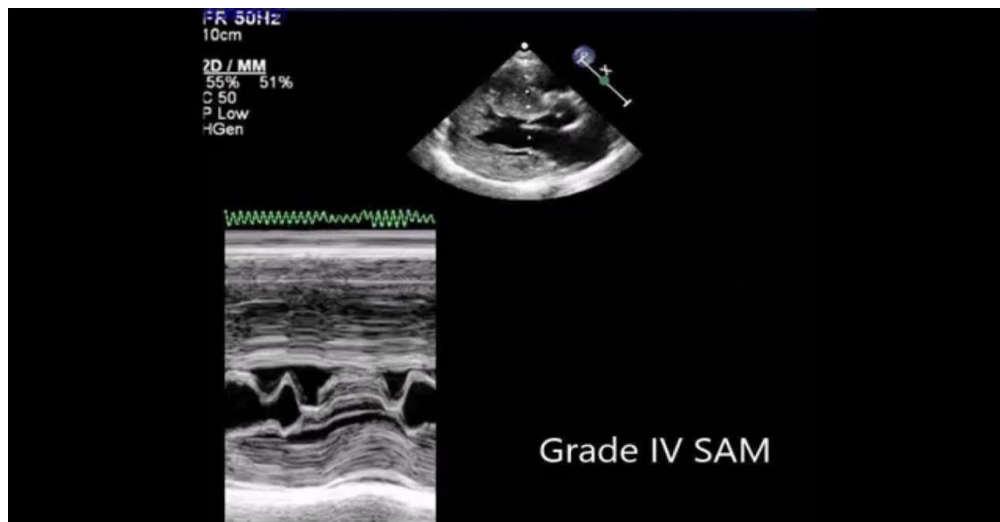
Systolic anterior motion (SAM) or elongation of the mitral valve leaflet(s) and abnormal papillary muscles], particularly in the presence of an abnormal ECG, increases the probability of disease. Approximately one-third of patients have resting SAM of the mitral valve leaflets that results in obstruction to the LV outflow tract, while another third have latent obstruction only during maneuvers that change loading conditions and LV contractility.

3. Cardiovascular magnetic resonance imaging

Cardiovascular magnetic resonance imaging embraces several modalities that provide detailed information on cardiac morphology, ventricular function and myocardial tissue characteristics.

4. Endomyocardial biopsy

Many of the genetic and non-genetic causes of HCM have characteristic histological appearances, but the diagnosis of HCM is clinical and relies on non-invasive testing in the first instance.





Treatment of HOCM

Aims of treatment:

- 1- Management of heart failure (diastolic and systolic HF).
- 2- Prevention of SD.

I-Medical therapy

1-B-blockers: are effective in relieving angina, dyspnea and syncope.

2-Calcium channel blockers.

3- diuretics

✓ **No digitalis no ACE**

II-non-pharmacological therapy:

1-Septal myomectomy

2-Alcohol septal ablation

3-ICD or amiodarone to prevent SCD.



3

RESTRICTIVE CARDIOMYOPATHY

Definition

- Restrictive cardiomyopathies are group of diseases characterized by (restricted filling function, increase atrial pressure and hence CHF), and normal LV contractility. When the disease advances, LV contraction is decreased.
- Restrictive cardiomyopathy (RCM) is a rare disease of the myocardium and is the least common of the three clinically recognized and described cardiomyopathies

Classification

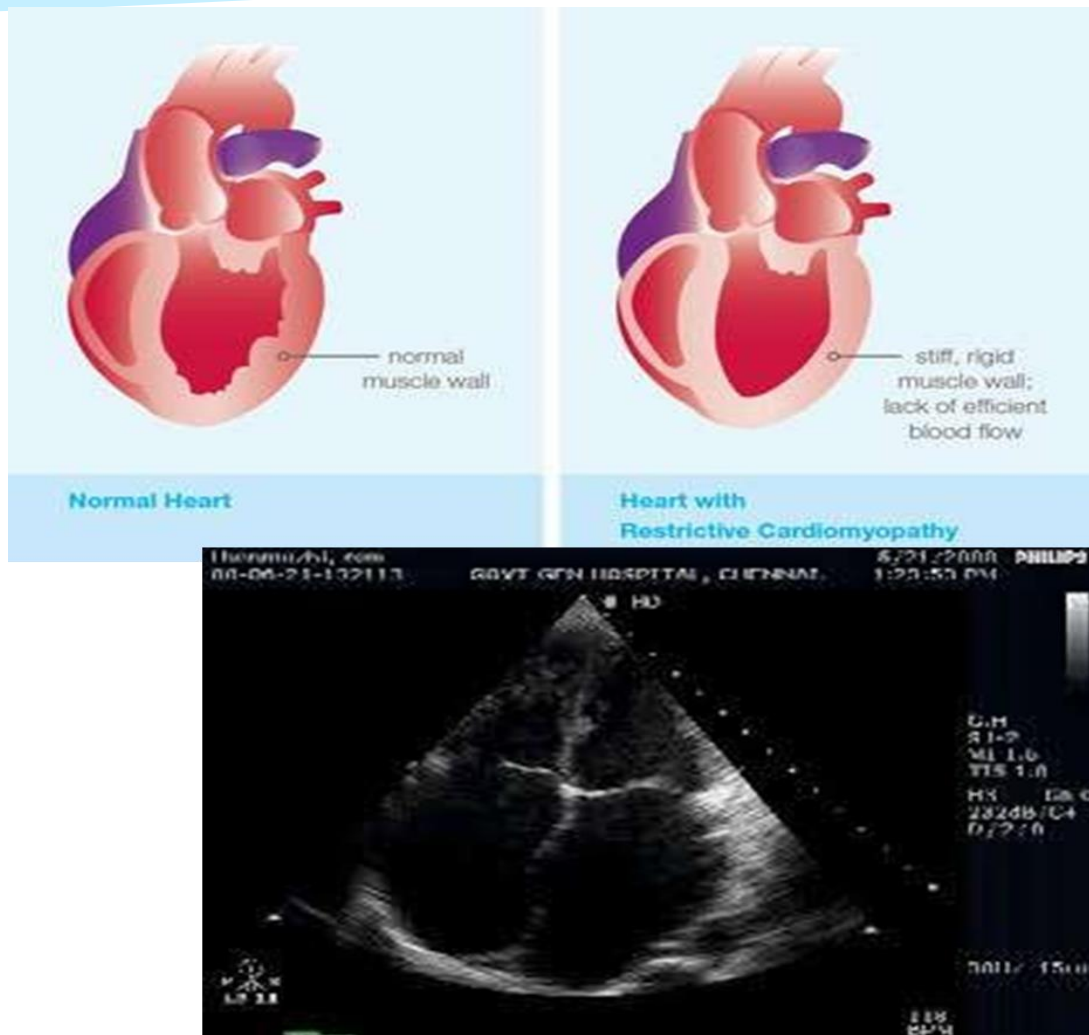
1. Primary restrictive cardiomyopathy:

- Idiopathic restrictive cardiomyopathy
- Endomyocardial fibrosis
- **Eosinophilic cardiomyopathy**

2-secondry restrictive cardiomyopathy

- Amyloidosis
- Hemochromatosis
- Sarcoidosis
- Storage diseases

-Others: radiation therapy may lead to myocardial fibrosis.....



Clinical presentation of restrictive cardiomyopathy

I- At the beginning pts.

1. Presented with manifestations of diastolic dysfunction as dyspnea on exertion, and decreased exercise tolerance.
2. Dyspnea at rest, PND, and orthopnea.
3. Congestive heart failure: oedema, rt. Upper abdomen pain, abdominal bloating, and increased JVP....

II- Then manifestations of systolic heart failure present as: Low COP



Diagnostic testing

- 1- **Echocardiography:** will reveal diminished diastolic function of the heart.
- 2- **Endomyocardial biopsy** may be needed to reveal the cause of myocardial restriction.

Treatment

RCM has no specific treatment. However, therapies directed at individual causes of RCM have been proven to be effective.

I- Pharmacologic therapy may include:

1. Beta blockers
2. Amiodarone
3. Cardioslective calcium channel blockers
4. Diuretics
5. Anticoagulants for patients with atrial fibrillation
6. Digoxin (use with caution)
7. Melphalan (for antiplasma cell therapy in systemic amyloidosis)
8. Chemotherapy (in amyloidosis)
9. Corticosteroids, cytotoxic agents (eg, hydroxyurea), and interferon (in Loeffler endocarditis)
10. Chelation therapy or therapeutic phlebotomy (in hemochromatosis)

II- Other treatments:

1. Pacemaker implantation
2. Endomyocardectomy
3. Cardiac transplantation



4. Novel therapies (eg, RNA interference or gene silencing molecules that target abnormal protein production in familial amyloidosis)

3

ARRHYTHMOGENIC RIGHT VENTRICULAR DYSPLASIA

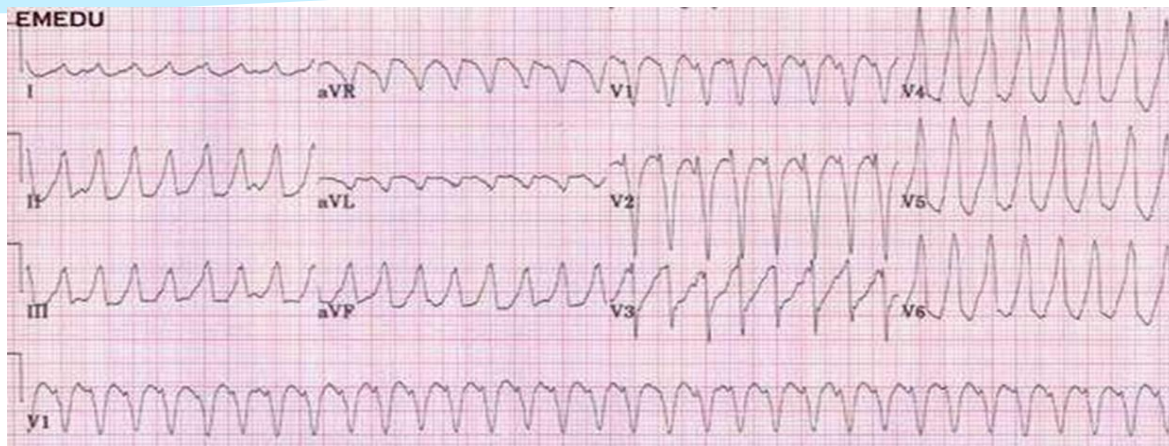
Definition:

It is a primary myocardial dis. That mainly affects the RV and later on both ventricles are affected.

Histopathology

There is an evidence of patchy programmed cell death (apoptosis) of cardiac myocytes, and myocytes are replaced by fat tissue





Clinical presentation

- ARVD usually present itself among young persons as syncope, sudden death (due to ventricular arrhythmias), Or more rarely as right sided heart failure.

Diagnosis

1. **Echocardiography:** may show areas of decreased RV contractility, and later on dilatation of the RV.
2. **MRI** helps to identify fatty infiltration of the RV.

Treatment

1. Antiarrhythmic drugs, ICD
2. Diuretics, digoxine, and sodium restriction

MYOCARDITIS

Definition

Myocarditis is an acute inflammation of the myocardium

Causes

1. Idiopathic

2. Infective:

- viral (cox.v, adeno v, HIV, hepatitis v.,polio,...COVID 19....)
- Bacterial (strept, diphtheria....)
- Parasitic: (toxoplasma, trypanosoma,)
- Spirochaetal: (Lyme dis., leptospirosis,)
- Fungal. -Rickettsial.

3. Toxic:

- drugs: methyldopa, penicillin, anti TB, adriamycin.
- bacterial toxins (diphtheria).

4.Radiation.

5. Autoimmune: rheumatic myocarditis, collagen diseases and hypersensitivity reaction.

Clinical Features

- A history of previous respiratory or febrile illness may be present.
- Fever, and may be chest pain.
- Cardiac failure: Low COP symptoms, dyspnea and or orthopnea.
- Physical exam:



- a. . Tachycardia out of proportion of fever
- b. . Soft heart sounds, gallop rhythm, systolic murmurs, may be pericardial friction rub.
- c. -Occasionally, presentation may mimic AMI, or be with syncope, or sudden cardiac death due to conduction system involvement or ventricular arrhythmias.

Investigations:

- 1- **Chest X-ray**.... may show some cardiac dilation.
- 2- **ECG:**ST, T wave abnormalities, may be heart block and ventricular arrhythmias.
- 3- **Cardiac enzymes** are elevated.
- 4- **Echocardiography:** may show ventricular dysfunction.
- 5- **MRI:** - areas of necrosis and fibrosis.
- 6- **Endomyocardial biopsy** may show acute inflammation but it is of limited value outside specialized units.

Treatment

- I. **General management:** bed rest, eradication of any acute infection, and stop toxic drugs.
- II. Management of cardiac failure.
- III. Management of arrhythmias.
- IV. Immunosuppressive therapy



V. Antiviral therapy.

Prognosis

Depends on the aetiology, and is usually good, although a chronic cardiomyopathy may be happened.

PERIPARTUM CARDIOMYOPATHY

Peripartum (postpartum) cardiomyopathy (PPCM) is the most common cardiomyopathy in pregnancy. PPCM is defined as an idiopathic cardiomyopathy that presents with heart failure secondary to left ventricular (LV) systolic dysfunction toward the end of pregnancy or in the months after delivery, in the absence of any other cause of heart failure. PPCM is a diagnosis of exclusion, and the majority are diagnosed postpartum. Although the LV may not be dilated, the ejection fraction is nearly always reduced below 45%.

This definition specifically excludes women who develop LV dysfunction early in their pregnancy and explicitly notes that not all cases of PPCM present with LV dilatation. Thus, it helps avoid misdiagnosis of other conditions that present with pulmonary edema in pregnancy, such as diastolic dysfunction from preeclampsia and other disorders (see the section on Differential Diagnosis).

PPCM is more common in women older than 30 years, black women, multiparous women, women with preeclampsia or hypertension, and those who smoke or are malnourished. The overlap between PPCM and preeclampsia is clinically important, as patients with preeclampsia can present with noncardiogenic edema and the coexistence of these two conditions highlights potential shared pathogenic mechanisms.



The severity of symptoms in patients with PPCM can be classified by the New York Heart Association (NYHA) system as follows:

Class I - Disease with no symptoms

Class II - Mild symptoms/effect on function or symptoms only with extreme exertion

Class III - Symptoms with minimal exertion

Class IV - Symptoms at rest

Etioopathophysiology

Although many potential mechanisms for peripartum (postpartum) cardiomyopathy (PPCM) exist, its exact cause remains unknown but is likely to be multifactorial.

The timing of presentation argues against the hemodynamic changes of pregnancy for causing PPCM, because such changes usually peak by 28 weeks' gestation. Furthermore, the timing highlights a potential hormonal pathogenesis given that hormonal changes, such as the production of prolactin, continue in late pregnancy. Relatively recent research has focused on the “vasculo-hormonal hypothesis” with soluble Fms-like tyrosine kinase 1 (sFlt1) and prolactin as molecules involved in the pathophysiology of PPCM.

- **Prolactin**

Prolactin is associated with endothelial and myocyte apoptosis.

- **sFLT1**

sFlt1 is a molecule secreted by the placenta in late gestation, and in higher levels in preeclampsia and twin gestations. Taken together, the increased production of



prolactin and placental secretion of sflt1 in late pregnancy could be toxic to both the vasculature and the cardiac myocytes.

- **Genetics**

Critics argue that the recovery of left ventricular ejection fraction and the lack of recurrence in all subsequent pregnancies challenge the notion that PPCM is a solely genetically-mediated condition.

- **Myocarditis**

The equal prevalence of myocardial inflammation.

- **Nutritional factors**

Lower levels of selenium or iron have been proposed as causative factors.

Epidemiology

The prevalence is reported to be 1 case per 6000 live births in Japan, 1 case per 1000 live births in South Africa, and 1 case per 350-400 live births in Haiti. A high prevalence in Nigeria has been attributed to the tradition of ingesting kanwa (dried lake salt) while lying on heated mud beds twice a day for 40 days postpartum.

Age- and race-related demographics

Although PPM can occur at any age, approximately 50% of cases occur in women older than 30 years.

Presentation

Many PPCM patients present with heart failure or a major adverse event (eg, stroke or respiratory failure) without any previous signs or symptoms to alert the clinician that a cardiomyopathy is going to develop; 19% of patients may present with the



syndrome before the last gestational month. Symptoms of PPCM are the same as those in nonpregnant patients with systolic dysfunction. New or rapid onset of the following symptoms requires prompt evaluation.

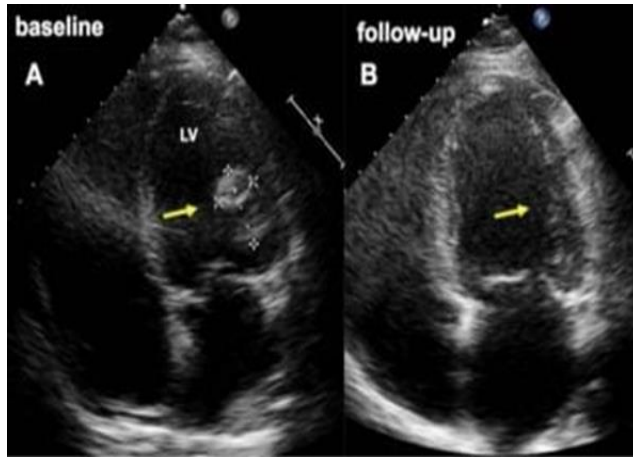
The differential diagnosis includes the following

- Aortic Stenosis
- Alcoholic Cardiomyopathy
- Cocaine-Related Cardiomyopathy
- Dilated Cardiomyopathy
- Hypertrophic Cardiomyopathy
- Restrictive Cardiomyopathy
- Cardiovascular Disease and Pregnancy
- Coronary Artery Atherosclerosis
- Hypertension
- Hypertension and Pregnancy
- Malignant Hypertension
- Mitral Stenosis
- Noncardiogenic pulmonary edema during pregnancy.

Echocardiography

All women in whom the diagnosis of peripartum (postpartum) cardiomyopathy (PPCM) is being considered should undergo echocardiographic evaluation to assess left ventricular function, valve structure, chamber size, and wall motion. Cardiac chambers do enlarge slightly during pregnancy, usually within normal limits.

Normal function suggests a pulmonary process or noncardiogenic pulmonary edema, and diastolic dysfunction can be observed in patients with severe preeclampsia.



Pericardial cardiomyopathy.

Baseline (A) and follow-up (B) four-chamber view echocardiograms in a woman who developed cardiomyopathy 4 weeks after an elective cesarean delivery. Image A demonstrates a large thrombus (arrow) attached to the lateral wall of the left ventricle (LV) at baseline (A), which completely resolved after two months (B). The patient's younger sister also experienced peripartum cardiomyopathy.

General Treatment Approach

An overarching principle to treating patients with peripartum (postpartum) cardiomyopathy (PPCM) is to remember that a healthy fetus depends on a healthy mother. Formulate the care plan and consider the consequences of not treating the mother before addressing the potential or theoretical effects of a test/treatment on the fetus. This approach will help clarify the best plan for clinicians who infrequently address medical issues during pregnancy.

Transfer

Consider rapid transfer to an intensive care unit (ICU) for monitoring of maternal status. If the mother is at less than 37 weeks' gestation, transfer her to a center with a neonatal ICU (NICU).



Nonpharmacologic

The patient should follow a low-sodium (2 g/day sodium chloride) diet. Strict bed rest may increase the risk of venous thromboembolism and is no longer recommended as a mainstay of therapy. Activity should be limited only by the patient's symptoms. In severe cases of true PPCM, bed rest may promote better uteroplacental perfusion.

Pharmacologic

- Medications should be used when there is clear benefit to the mother. The US Food and Drug Administration (FDA) has outlined a classification system for medications in pregnancy, with medications in category X clearly having more risks to the fetus than benefits. Any medication in class A through D may be used when the potential benefit justifies the potential risk. Organogenesis is completed by 13 weeks' gestation. Although some medications may have direct effects on the fetus, no risk of teratogenesis is present after the first trimester.
- Medical therapy for patients with systolic dysfunction during pregnancy is similar to that for the nonpregnant patient. The mainstays of medical therapy are digoxin, loop diuretics, and beta-adrenergic blockade with carvedilol or metoprolol succinate, as they have been shown to decrease all-cause mortality and hospitalization in those with systolic dysfunction. Afterload reduction is typically accomplished with hydralazine and nitrates, because angiotensin-converting enzyme inhibitors (ACEIs) and angiotensin-receptor blockers (ARBs) are contraindicated in pregnancy owing to fetal renal dysgenesis and death. When the ejection fraction (EF) is less than 35%, anticoagulation is recommended due to a high risk of venous and arterial thrombosis. Warfarin,



low-molecular weight heparin (LMWH), or unfractionated heparin (UFH) may be considered for anticoagulation, with determinants of choice of anticoagulant the gestational age, risk of fetal toxicity of warfarin, and renal function.

- After delivery, the contracted uterus squeezes 300-500 mL of blood into the systemic circulation. This autotransfusion, along with the release of inferior vena cava compression by the gravid uterus, leads to an increase in cardiac output as high as 10%-20% over predelivery levels. Therefore, worsening of heart failure is anticipated typically at 2-3 days post-partum due these fluid shifts, the loss of the low-resistance placental bed, and the decrease in the vascular compliance that was maintained by the hormonal changes of pregnancy.

Other considerations

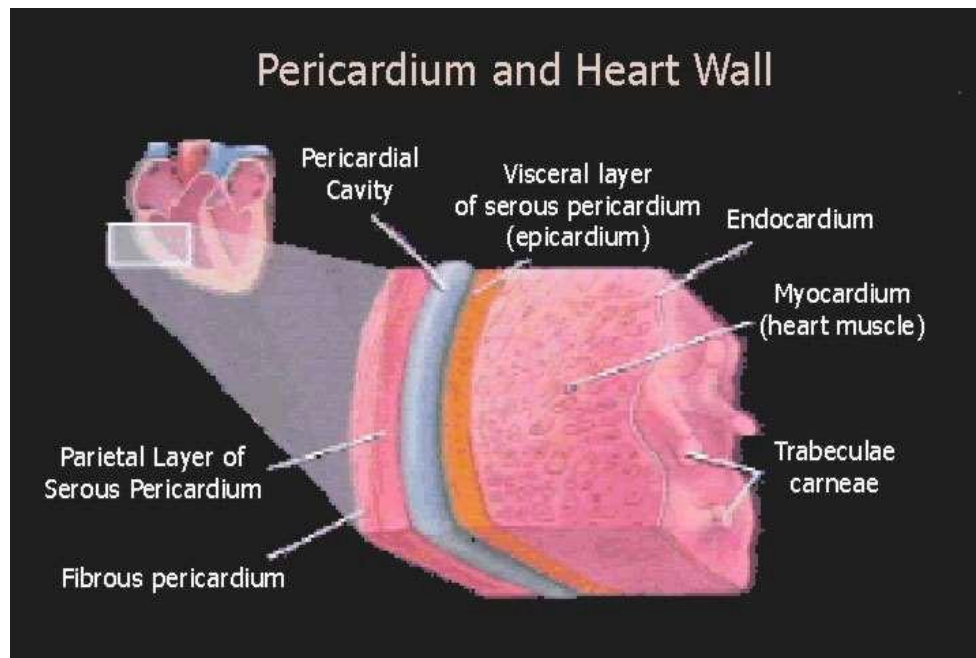
- After delivery, patients on hydralazine/nitrates should be placed on an ACEI, and the dose should be maximized. Captopril and enalapril are thought to be compatible with breastfeeding.
- Postdelivery breastfeeding is an important consideration. The risks and benefits of breastfeeding must be weighed. Breastfeeding increases metabolic demands. Prolactin remains elevated during breastfeeding, and this may contribute to the ongoing LV dysfunction. Furthermore, administration of bromocriptine inhibits prolactin secretion, effectively limiting breast milk production.



- At present, breastfeeding is discouraged in patients with severe LV dysfunction and New York Heart Association (NYHA) class 3-4 symptoms. Recommendations for patients with less severe LV dysfunction or NYHA class 1-2 status are less well defined.
- An echocardiogram should be ordered as indicated by new clinical findings or by a decline in function. If an echocardiogram reveals abnormal systolic function during pregnancy, a repeat study should be obtained approximately 2 months after delivery. If the results of that study show that systolic function has improved but has not returned to normal levels, another study should be obtained within the year to determine the patient's new baseline.
- Genetic testing for familial cardiomyopathy may be considered in patients with persistent LV dysfunction and a family history of cardiomyopathy.
- For women considering pregnancy or those who desire an evaluation to estimate the risk of a future pregnancy, recovery of systolic function is a prime concern. In women with a normalized LVEF, consideration could be made for weaning off one cardiac medication at a time with continued close clinical and echocardiographic surveillance throughout the discontinuation process. Women with persistent systolic dysfunction should be maintained on vasodilators, nitrates, and diuretics as tolerated and indicated.

NORMAL PERICARDIUM

Normal pericardium consists of an outer sac or fibrous pericardium and an inner double-layered sac called the *serous pericardium*. The 2 layers of serous pericardium include the visceral layer, or *epicardium*, which covers the heart and proximal great vessels. The visceral layer is reflected to form the parietal pericardium, which lines the fibrous pericardium. The visceral and parietal layers are separated by the pericardial cavity, which in healthy people contains 15 to 50 mL of a plasma ultrafiltrate.





FUNCTION OF THE PERICARDIUM

1. Stabilization of the heart within the thoracic cavity -ligamentous attachments
-- limiting the heart's motion.
2. Protection of the heart from mechanical trauma and infection from adjoining structures.
3. The pericardial fluid functions as a lubricant and decreases friction of cardiac surface during systole and diastole.
4. Prevention of excessive dilation of heart especially during sudden rise in intra-cardiac volume

PERICARDIAL SYNDROMES

Pericardial

1. Pericarditis
2. Pericardial effusion
3. Cardiac tamponade
4. Constrictive pericarditis

PERICARDITIS

Aetiological classification for pericardial diseases is to consider infectious and non-infectious causes. In developed countries, viruses are usually the most common aetiological agents of pericarditis. whereas tuberculosis (TB) is the most frequent cause of pericardial diseases in the world and developing countries, where TB is endemic. In this setting, TB is often associated with human immunodeficiency virus (HIV) infection, especially in sub-Saharan Africa. (table 1)

Types of pericarditis:

1. Acute pericarditis (less than 6 weeks)
2. Incessant pericarditis (lasting more than 6 weeks and less than 3 months without remission)
3. Recurrent pericarditis (recurrence after a documented first episode of acute pericarditis and symptom free interval of 4-6 weeks or longer)
4. Chronic pericarditis (lasting more than 3 months)



Acute

Acute pericarditis is an inflammatory pericardial syndrome with or without pericardial effusion. The clinical diagnosis can be made with two of the following criteria;

1. Chest pain (>85–90% of cases)—typically sharp and pleuritic, improved by sitting up and leaning forward.
2. pericardial friction rub ($\leq 33\%$ of cases)—a superficial high pitch (rasping, scratching, grating, walking on crunchy snow or dry leaves) sound best heard with the diaphragm of the stethoscope over the left sternal border at end of expiration and leaning forward.
3. Electrocardiogram (ECG) changes (up to 60% of cases)—with new widespread ST elevation or PR depression in the acute phase.
4. pericardial effusion (up to 60% of cases, generally mild)

Other manifestations:

additional signs and symptoms may be present according to the underlying aetiology or systemic disease (i.e. signs and symptoms of systemic infection such as fever and leucocytosis, or systemic inflammatory disease or cancer)

Major differential diagnoses include acute coronary syndromes and Pleuritis.

Approach

- It's a clinical diagnosis based on history and examination.
- ECG changes
- Elevation of markers of inflammation (CRP) and (ESR), as well as elevation of the white blood cell count is a common and supportive finding in patients

with acute pericarditis and may be helpful for monitoring the activity of the disease and efficacy of therapy. Others based on suspected aetiology.

- Patients with concomitant myocarditis may present with an elevation of markers of myocardial injury [i.e. creatine kinase (CK), troponin].
- A chest X-ray is generally normal in patients with acute pericarditis.

Treatment

1. Treatment of the underlying cause.
2. Restrict physical activity until resolution of symptoms and normalization of CRP.
3. Aspirin or NSAIDs are mainstays of therapy for acute pericarditis. Ibuprofen (600mg – 800mg TDS) Aspirin (2-4mg/d) with gradual withdrawal of the dose over 1-2 weeks
4. Colchicine is recommended at low, weight-adjusted doses to improve the response to medical therapy and prevent recurrences (0.5 mg once daily if body weight is <70 kg or 0.5 mg twice daily if it is ≥70 kg). Tapering of colchicine is not mandatory.
5. Corticosteroids should be considered as a second option in patients with contraindications and failure of aspirin or NSAIDs. in this case they are used with colchicine. low to moderate doses (i.e. prednisone 0.2–0.5 mg/kg/day or equivalent) should be recommended. The initial dose should be maintained until resolution of symptoms and normalization of CRP, then tapering should be considered.

Table 1

A. Infectious causes:
Viral (common): Enteroviruses (coxsackieviruses, echoviruses), herpesviruses (EBV, CMV, HHV-6), adenoviruses, parvovirus B19 (possible overlap with aetiological viral agents of myocarditis).
Bacterial: <i>Mycobacterium tuberculosis</i> (common, other bacterial rare), <i>Coxiella burnetii</i> , <i>Borrelia burgdorferi</i> , rarely: <i>Pneumococcus</i> spp, <i>Meningococcus</i> spp, <i>Gonococcus</i> spp, <i>Streptococcus</i> spp, <i>Staphylococcus</i> spp, <i>Haemophilus</i> spp, <i>Chlamydia</i> spp, <i>Mycoplasma</i> spp, <i>Legionella</i> spp, <i>Leptospira</i> spp, <i>Listeria</i> spp, <i>Providencia stuartii</i> .
Fungal (very rare): <i>Histoplasma</i> spp (more likely in immunocompetent patients), <i>Aspergillus</i> spp, <i>Blastomyces</i> spp, <i>Candida</i> spp (more likely in immunocompromised host).
Parasitic (very rare): <i>Echinococcus</i> spp, <i>Toxoplasma</i> spp
B. Non-infectious causes:
Autoimmune (common): Systemic autoimmune and auto-inflammatory diseases (systemic lupus erythematosus, Sjögren syndrome, rheumatoid arthritis, scleroderma), systemic vasculitides (i.e. eosinophilic granulomatosis with polyangiitis or allergic granulomatosis, previously named Churg-Strauss syndrome, Horton disease, Takayasu disease, Behçet syndrome), sarcoidosis, familial Mediterranean fever, inflammatory bowel diseases, Still disease.
Neoplastic: Primary tumours (rare, above all pericardial mesothelioma). Secondary metastatic tumours (common, above all lung and breast cancer, lymphoma).
Metabolic: Uraemia, myxoedema, anorexia nervosa, other rare.
Traumatic and Iatrogenic: Early onset (rare): • Direct injury (penetrating thoracic injury, aoesophageal perforation). • Indirect injury (non-penetrating thoracic injury, radiation injury). Delayed onset: Pericardial injury syndromes (common) such as postmyocardial infarction syndrome, postpericardiotomy syndrome, posttraumatic, including forms after iatrogenic trauma (e.g. coronary percutaneous intervention, pacemaker lead insertion and radiofrequency ablation).
Drug-related (rare): Lupus-like syndrome (procainamide, hydralazine, methyldopa, isoniazid, phenytoin); antineoplastic drugs (often associated with a cardiomyopathy, may cause a pericardiopathy): doxorubicin, daunorubicin, cytosine arabinoside, 5-fluorouracil, cyclophosphamide; penicillins as hypersensitivity pericarditis with eosinophilia; amiodarone, methysergide, mesalazine, clozapine, minoxidil, dantrolene, practolol, phenylbutazone, thiazides, streptomycin, thiouracils, streptokinase, p-aminosalicylic acid, sulfa-drugs, cyclosporine, bromocriptine, several vaccines, GM-CSF, anti-TNF agents.
Other (common): Amyloidosis, aortic dissection, pulmonary arterial hypertension and chronic heart failure.
Other (uncommon): congenital partial and complete absence of the pericardium.



PERICARDIAL EFFUSION

The normal pericardial sac contains 10–50 ml of pericardial fluid as a plasma ultrafiltrate.

Etiology of pericardial effusion.

1. **Inflammation:** infection, or immunologic process.
2. **Trauma:** bleeding in pericardial space.
3. **Non-infectious:**
 - a. increase in pulmonary hydrostatic pressure e.g., congestive heart failure.
 - b. increase in capillary permeability e.g., hypothyroidism.
 - c. decrease in plasma oncotic pressure e.g., cirrhosis.
4. **Decreased drainage** : obstruction of thoracic duct as in malignancy or damage during surgery.

Clinical presentation

It varies according to the speed of pericardial fluid accumulation. If pericardial fluid is rapidly accumulating, such as after wounds or iatrogenic perforations, even small amounts of blood may cause an increase in intrapericardial pressure within minutes and overt cardiac tamponade. On the other hand, a slow accumulation of pericardial fluid allows the collection of a large effusion in days to weeks before a significant increase in pericardial pressure causes symptoms and signs.

Small effusions; usually do not produce hemodynamic abnormalities.



Large effusions usually cause

1. Hemodynamic Instability
2. Compression of adjoining structures dysphagia (compression of esophagus) hoarseness (recurrent laryngeal nerve compression) hiccups (phrenic nerve compression)
3. Dyspnea and orthopnea (pleural inflammation/effusion), Chest pain, discomfort, Light-headedness, syncope and Palpitations

CARDIAC TAMPONADE

Cardiac tamponade is a life-threatening, slow or rapid compression of the heart due to the pericardial accumulation of fluid, pus, blood, clots or gas.

Causes:

_inflammation, trauma, rupture of the heart, aortic dissection, or malignancy.

Clinical picture;

Tachycardia, tachypnea, hepatojugular reflux, weakened peripheral pulses, raised JVP, muffled heart sounds, tubular breath sounds: left axilla or left base : bronchial compression.

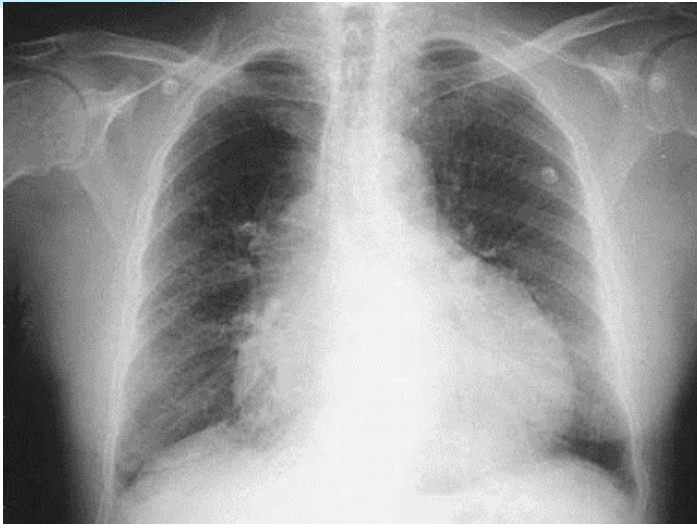
Ewart sign: dullness to percussion beneath the angle of left scapula and increased fremitus

Beck's triad: hypotension, soft or absent heart sounds, jugular venous distension with prominent x descent but an absent y descent

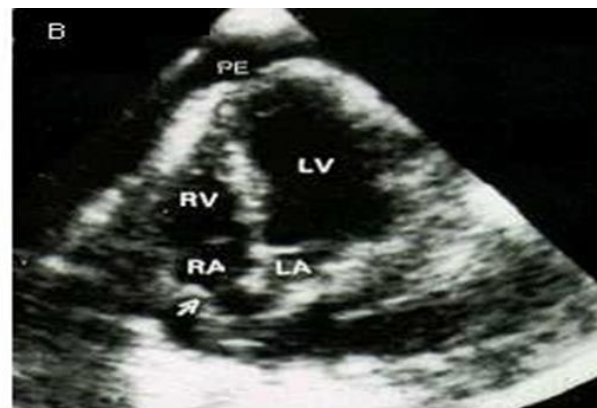
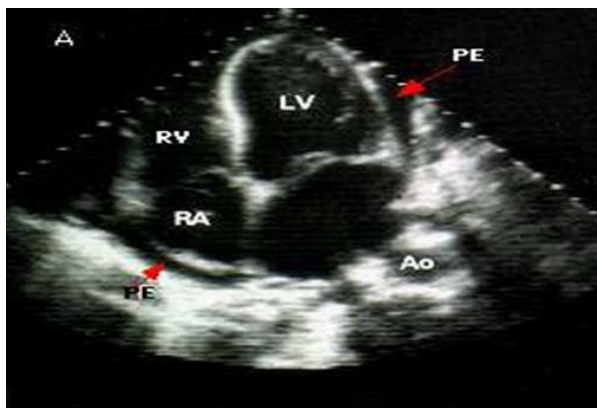
Pulsus paradoxus: A key diagnostic finding is defined as an inspiratory decrease in systolic arterial pressure of >10 mmHg during normal breathing.

Investigations of pericardial effusion and tamponade

- 1- **Echocardiography:** the most important investigation to detect the amount of effusion and its hemodynamic effect.
- 2- **Chest X- ray:** flask or bottle shape
- 3- **ECG:** low voltage, and electrical alternans
- 4- **CT and MRI**



Electrical alternans Sinus tachycardia with electrical alternans which is characterized by beat-to-beat alternation in the QRS appearance (best seen in leads V2 to V4). These findings are strongly suggestive of pericardial effusion, usually with tamponade. The alternating ECG pattern is related to back-and-forth swinging motion of the heart in the pericardial fluid. Courtesy of Ary Goldberger, MD.



Treatment of pericardial effusion and tamponade:

(Tamponade is an emergency and life-threatening condition)

- 1- Treatment of the cause
- 2- Surgical procedures
 - Pericardiocentesis with prolonged pericardial drainage
 - Pericardiectomy
 - Pericardial window placement
 - Pericardiotomy

Pericardiocentesis



CONSTRUCTIVE PERICARDITIS

Late complication of pericardial disease, ---- Fibrous scar formation, ---- Fusion of pericardial layers, ----- Calcification further stiffens pericardium.

Etiology

1. Any cause of pericarditis
2. Idiopathic
3. Post-surgery
4. Tuberculosis
5. Radiation
6. Neoplasm

Clinical presentation

The classic clinical picture is characterized by signs and symptoms of right heart failure with preserved right and left ventricular function in the absence of previous or concomitant myocardial disease. Patients complain about fatigue, peripheral oedema, breathlessness and abdominal swelling, Venous congestion, hepatomegaly, pleural effusion and ascites may occur. Haemodynamic impairment of the patient can be additionally aggravated by a systolic dysfunction due to myocardial fibrosis or atrophy in more advanced cases.

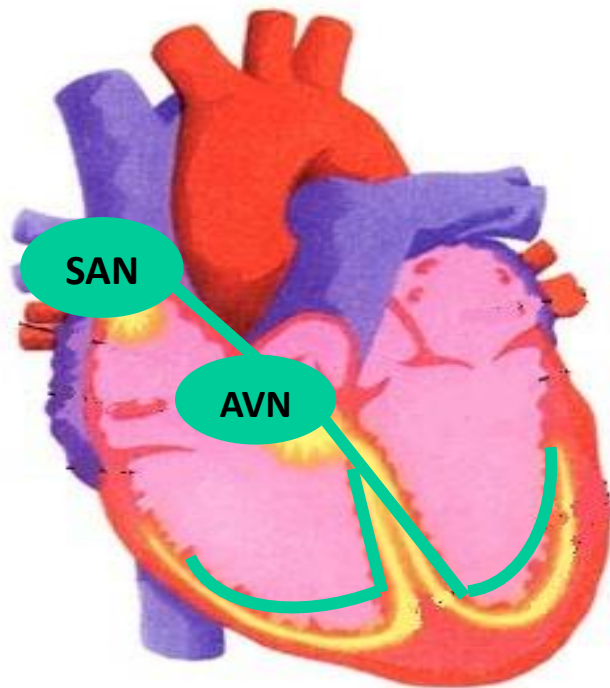
Diagnosis

Based on the association of signs and symptoms of right heart failure and impaired diastolic filling due to pericardial constriction by one or more imaging methods, including echocardiography, chest X- ray, CT, CMR, and cardiac catheterization. The main differential diagnosis is with restrictive cardiomyopathy



Treatment

The mainstay of treatment of chronic permanent cases is surgery (pericardial resection).



Impulse conduction

Impulses originate regularly at a frequency of 60-100 beat/ min

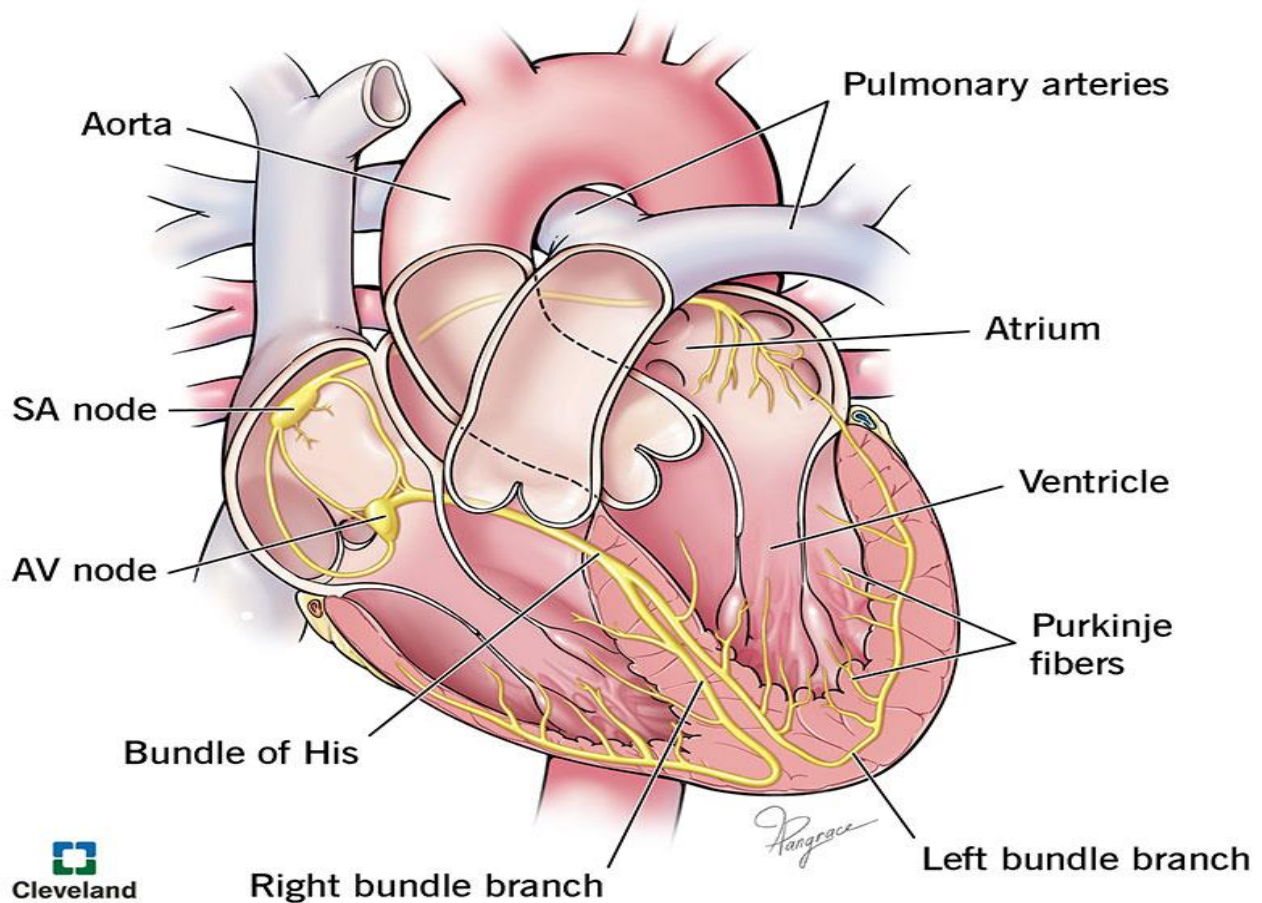


What are the parts of the cardiac conduction system?

Your cardiac conduction system contains specialized cells and nodes that control your heart beat.

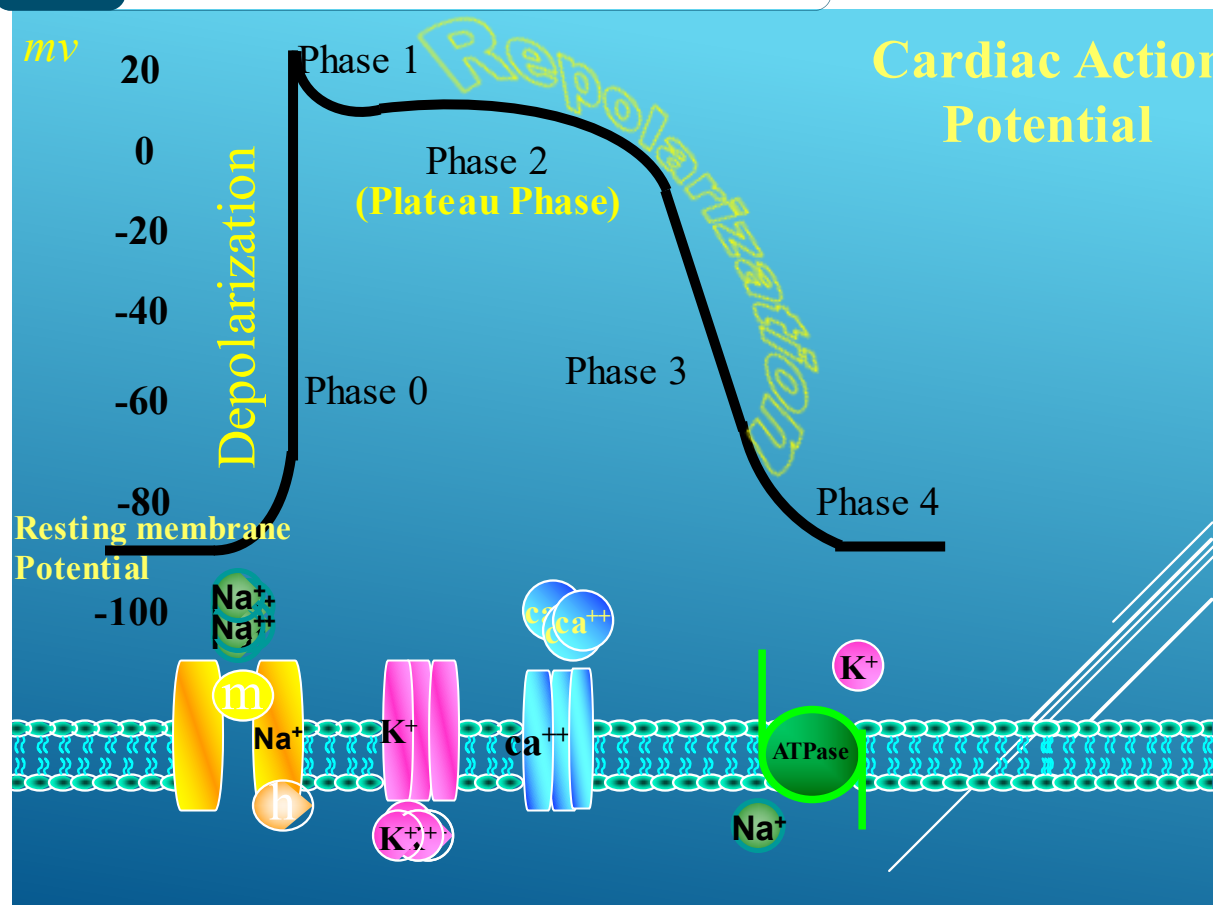
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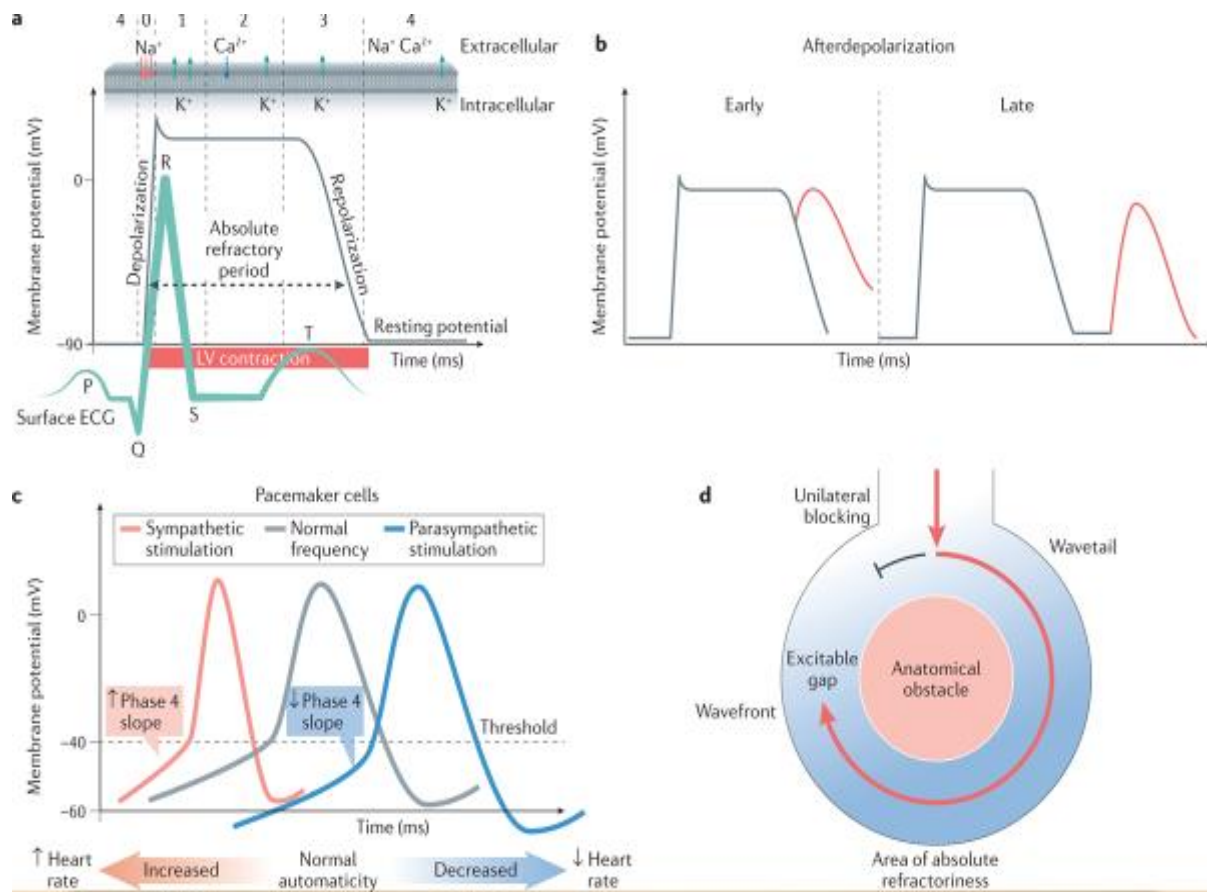
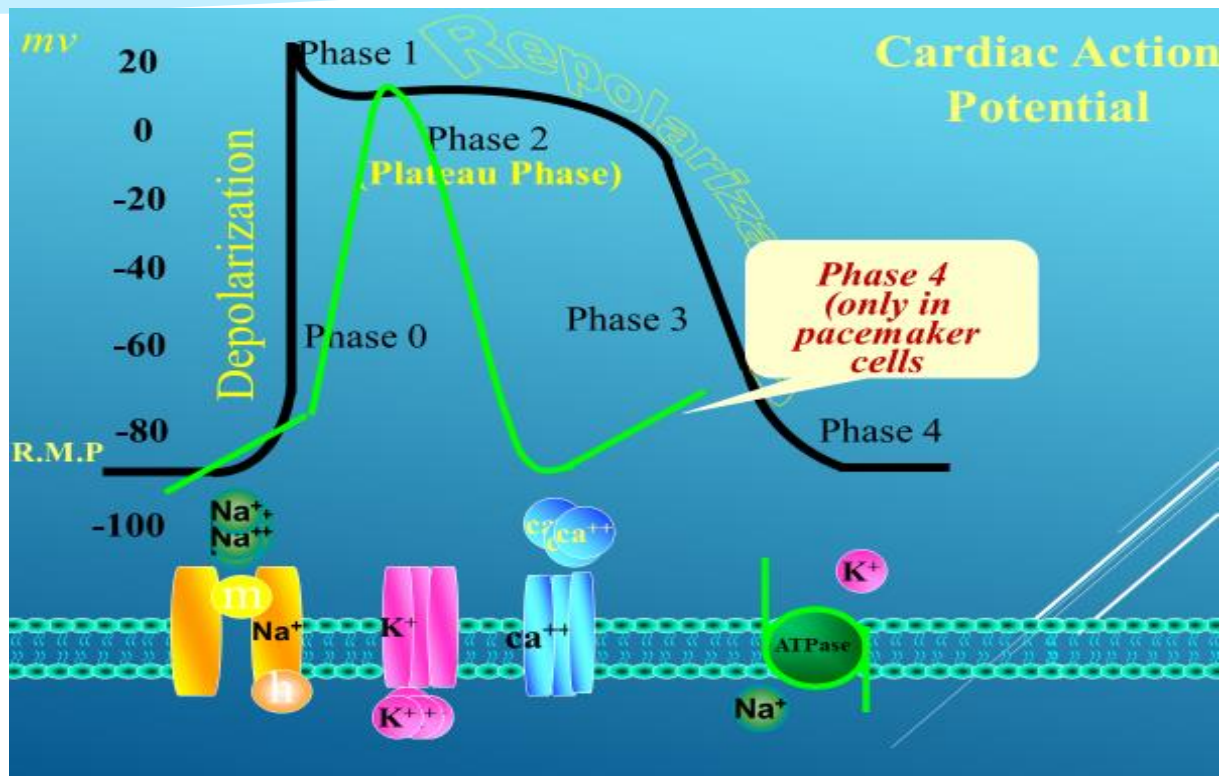
- Sinoatrial node.
- Atrioventricular node.
- Bundle of His (atrioventricular bundle).
- Purkinje fibers.




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Cardiac Action Potential





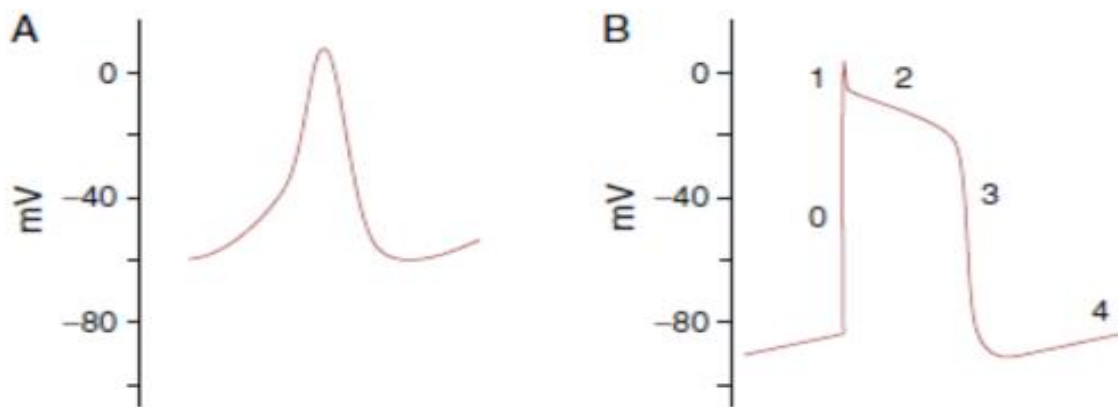
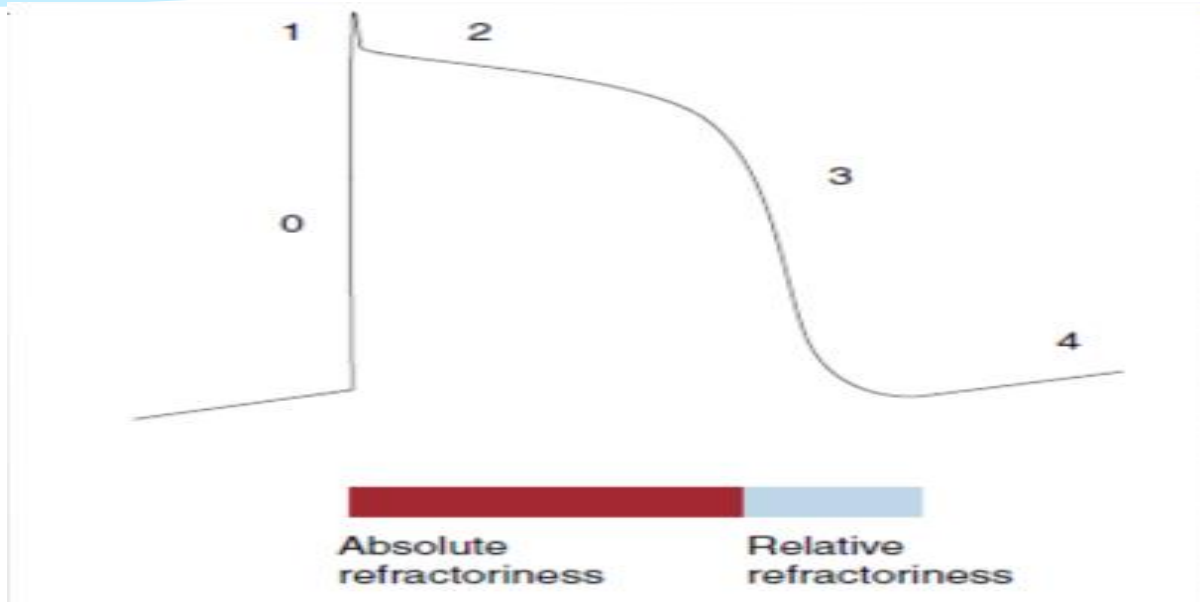


Figure 1. The cardiac action potential. A: sinus node action potential. B: muscle cell action potential.

CARDIAC ARRHYTHMIAS

- An abnormality of the cardiac rhythm is called a cardiac arrhythmia.
- Arrhythmias may cause sudden death, syncope, heart failure, dizziness, palpitations or no symptoms at all.
- There are two main types of arrhythmias:
 - Bradycardia: the heart rate is slow (< 60 b.p.m).
 - Tachycardia: the heart rate is fast (> 100 b.p.m).

Mechanisms of Cardiac Arrhythmias

Disorders of impulse formation	Disorders of impulse conduction
<i>Automaticity</i>	<i>Reentry</i>
Altered normal automaticity	Anatomic reentry
Abnormal automaticity	Functional reentry
<i>Triggered activity</i>	Block
Delayed afterdepolarization	
Early afterdepolarization	

Altered Normal

- Some specialized heart cells, such as sinoatrial nodal cells, the atrioventricular (AV) node, and the His- Purkinje system, as well as some cells in both atria, possess the property of pacemaker activity or automaticity.
- Suppression or enhancement of this activity may lead to clinical arrhythmias.

Abnormal Automaticity

- Atrial and ventricular non pacemaker myocardial cells, which in the normal heart typically do not exhibit spontaneous activity, may exhibit automaticity properties.
- This can happen under conditions that drive the maximum diastolic potential towards the threshold potential, which is explained by the interplay of numerous currents that together result in a net inward depolarizing current associated with a decrease in potassium conductance.

Triggered Activity

- Triggered activity (TA) is defined by impulse initiation caused by afterdepolarizations (membrane potential oscillations that occur during or immediately following a preceding action potential {AP}).
- Afterdepolarizations occur only in the presence of a previous AP (the trigger), and when they reach the threshold potential, a new AP is generated. This may be the source of a new triggered response, leading to self-sustaining TA.

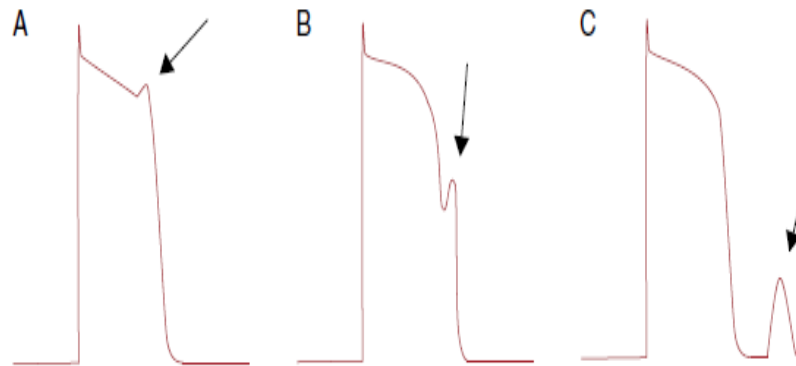
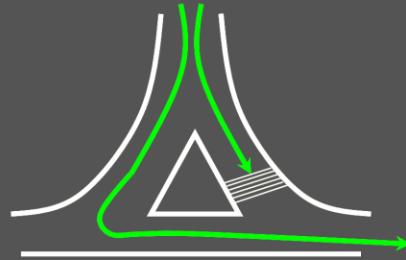


Figure 6. Representation of triggered activity. A: phase 2 early afterdepolarization. B: phase 3 early afterdepolarization. C: delayed afterdepolarization.

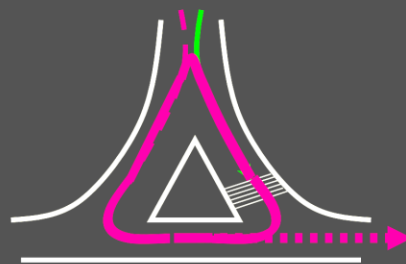
Re-entry

- During normal electrical activity, the cardiac cycle begins in the sinoatrial node and continues to propagate until the entire heart is activated.
- This impulse dies out when all fibers have been depolarized and are completely refractory.
- However, if a group of isolated fibers is not activated during the initial wave of depolarization, they can recover excitability in time to be depolarized before the impulse dies out.
- They may then serve as a link to re-excite areas that were previously depolarized but have already recovered from the initial depolarization.
- Such a process is commonly denoted as reentry, reentrant excitation, circus movement, reciprocal or echo beats, or reciprocating tachycardia (RT), referring to a repetitive propagation of the wave of activation, returning to its site of origin to reactivate that site.

Reentry Arrhythmias



Normal



Re-entrant
Tachycardia

Mechanisms of bradycardias:

Sinus bradycardia is a result of abnormally slow automaticity while bradycardia due to AV block is caused by abnormal conduction within the AV node or the distal AV conduction system.

Atrial

Sinus arrhythmia:

- A condition in which the heart rate varies with breathing.
- This is usually a benign condition

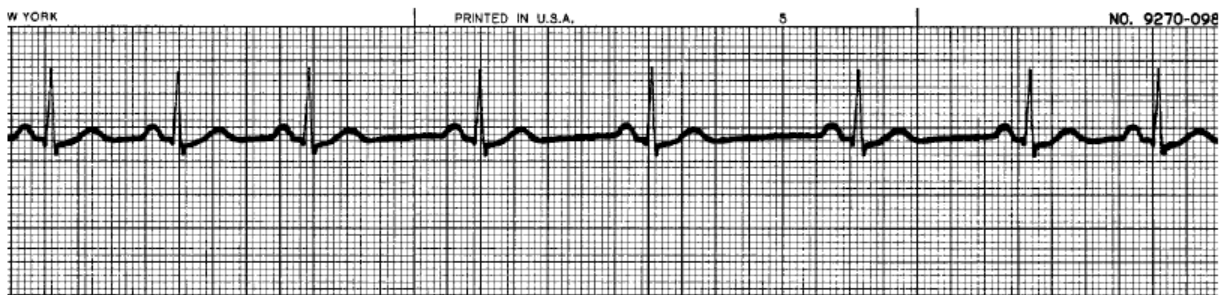


FIGURE 27-8 Sinus arrhythmia in lead II.



SUPRAVENTRICULAR TACHYCARDIAS

- Supraventricular tachycardias (SVTs) arise from the atrium or the atrioventricular junction.
- Conduction is via the His-Purkinje system; therefore, the QRS shape during tachycardia is usually similar to that seen in the same patient during baseline rhythm.

Causes of SVT

Tachycardia	ECG features	Comment
Sinus tachycardia	P wave morphology similar to sinus rhythm	Need to determine underlying cause
AV nodal reentry tachycardia (AVNRT)	No visible P wave, or inverted P wave immediately before or after QRS complex	Commonest cause of palpitations in patients with normal hearts
AV reciprocating tachycardia (AVRT)	P wave visible between QRS and T wave complexes	Due to an accessory pathway. If pathway conducts in both directions, ECG during sinus rhythm may be pre excited
Atrialfibrillation	Irregularly irregular RR intervals and absence of organized atrial activity	Commonest tachycardia in patients over 65 years
Atrialflutter	Visible flutter waves at 300/min (saw-tooth appearance) usually with 2 : 1 AV conduction	Suspect in any patient with regular SVT at 150/min
Atrialtachycardia	Organized atrial activity with P wave morphology different from sinus rhythm	Usually occurs in patients with structural heart disease
Multifocaktrialtachycardia	Multiple P wave morphologies (≥ 3) and irregular RR intervals	Rare arrhythmia; most commonly associated with significant chronic lung disease
Accelerated junctional tachycardia	ECG similar to AVNRT	Rare in adults

Sinus tachycardia

- A condition in which the heart rate is 100-160/min
- Symptoms may occur with rapid heart rates including; weakness, fatigue, dizziness, or palpitations.
- Sinus tachycardia is often temporary, occurring under stresses from exercise, strong emotions, fever, dehydration, thyrotoxicosis, anemia and heart failure.

- If necessary, beta-blockers may be used to slow the sinus rate, e.g. in hyperthyroidism

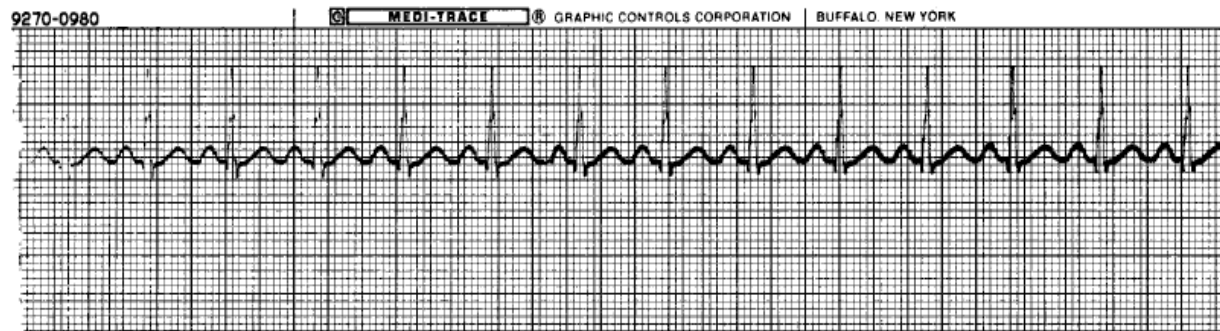
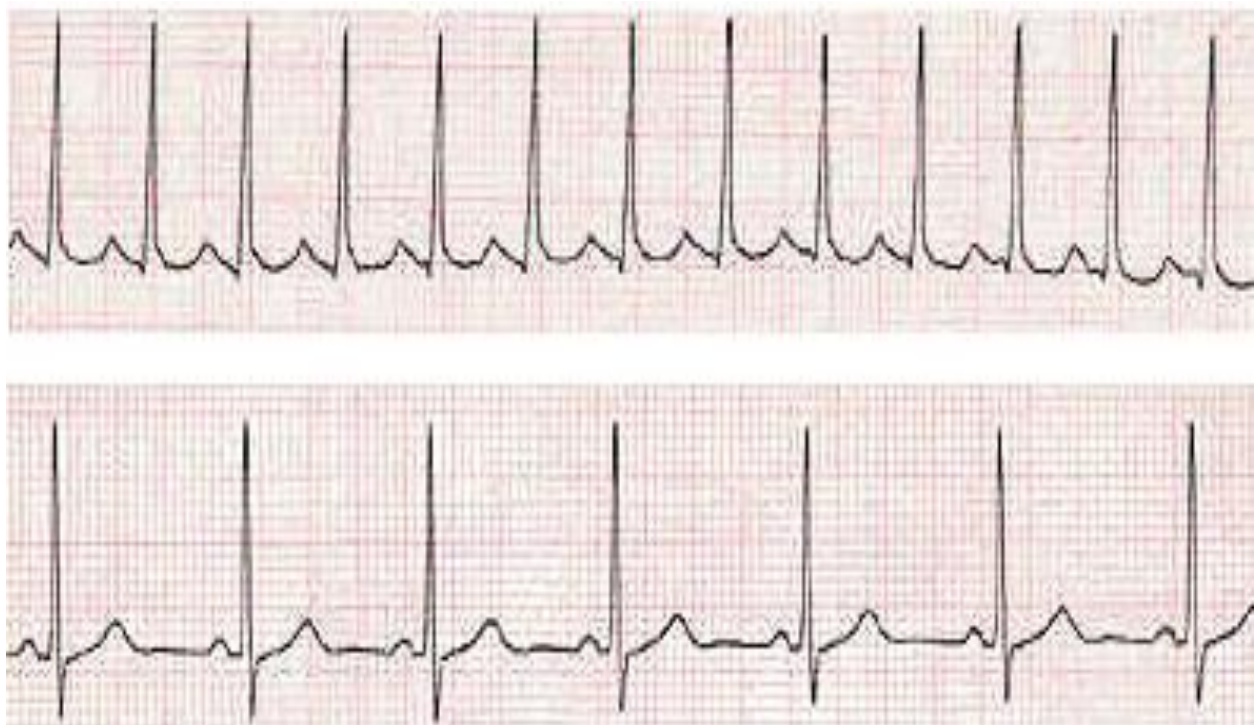


FIGURE 27-7 Sinus tachycardia in lead II.



Premature supraventricular contractions or premature atrial contractions (PAC)

- A condition in which an atrial pacemaker site above the ventricles sends out an electrical signal early. The ventricles are usually able to respond to this signal, but the result is an irregular heart rhythm.
- PACs are common and may occur as the result of stimulants such as coffee, tea, alcohol, cigarettes, or medications.
- Treatment is rarely necessary.

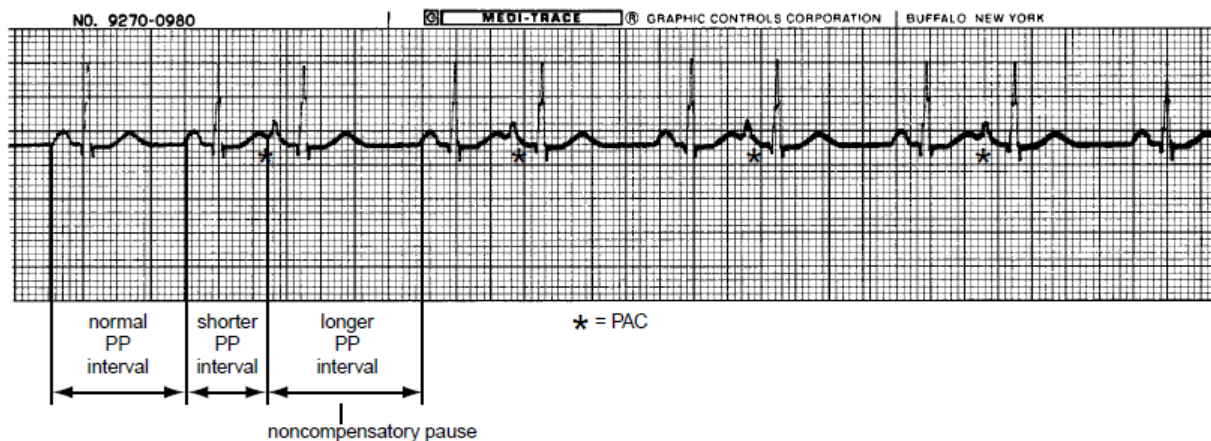


FIGURE 27-9 Premature atrial complexes (PACs) in lead II. Note pause following PAC that is longer than the normal PP interval, but shorter than twice the normal PP interval.

Paroxysmal Supraventricular tachycardia

Atrioventricular nodal re-entry tachycardia (AVNRT)

- It usually begins and ends rapidly, occurring in repeated periods. This condition can cause symptoms such as weakness, fatigue, dizziness, fainting, or palpitations if the heart rate becomes too fast.

- In AVNRT, there are two functionally and anatomically different pathways within the AV node: one is characterized by a short effective refractory period and slow conduction, and the other has a longer effective refractory period and conducts faster.
- In sinus rhythm, the atrial impulse that depolarizes the ventricles usually conducts through the fast pathway.
- If the atrial impulse (e.g. an atrial premature beat) occurs early when the fast pathway is still refractory, the slow pathway takes over in propagating the atrial impulse to the ventricles. It then travels back through the fast pathway which has already recovered its excitability, thus initiating the most common 'slow-fast', or typical, AVNRT.
- The rhythm is recognized on ECG by normal regular QRS complexes, usually at a rate of 140-240 per minute. Sometimes the QRS complexes will show typical bundle branch block. P waves are either not visible or are seen immediately before or after the QRS complex because of simultaneous atrial and ventricular activation.

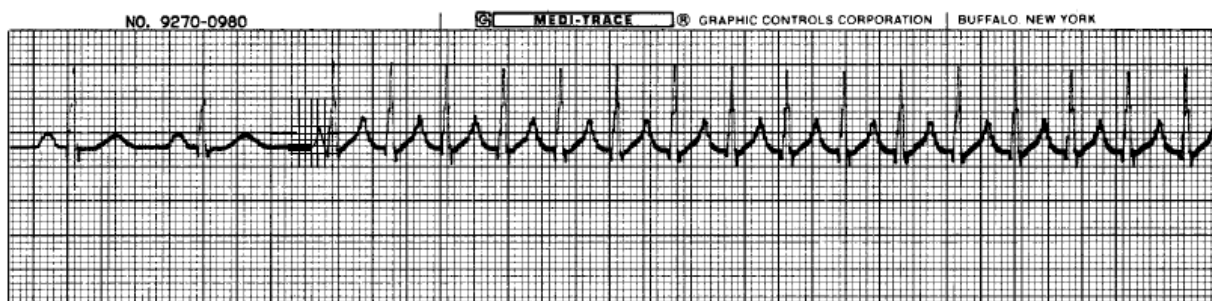


FIGURE 27-13 AV nodal reentry tachycardia in lead II.



Atrioventricular reciprocating

- In AVRT there is a large circuit comprising the AV node, the His bundle, the ventricle and an abnormal connection from the ventricle back to the atrium. This abnormal connection is called an accessory pathway or bypass tract.
- Bypass tracts result from incomplete separation of the atria and the ventricles during fetal development.
- Atrial activation occurs after ventricular activation and the P wave is usually clearly seen between the QRS and T complexes

PSVT

Acute Management

- Patients presenting with SVTs and hemodynamic instability require emergency cardioversion.
- If the patient is hemodynamically stable, vagal maneuvers, including right carotid massage, Valsalva maneuver and facial immersion in cold water can be successfully employed.
- If not successful, intravenous adenosine (up to 0.25 mg/kg) , verapamil 5-10 mg i.v. over 5-10 minutes, i.v. diltiazem, or beta-blockers should be tried.

Long-term management

- It includes ablation of an accessory pathway. Also, verapamil, diltiazem & β -blockers; are effective in 60-80% of patients.

Wolf Parkinson White Syndrome (WPW)

An abnormal band of atrial tissue connects the atria and ventricles and can electrically bypass the normal pathways of conduction; a re-entry circuit can develop causing paroxysms of tachycardia.

ECG shows:

- Short PR interval
- Delta wave on the upstroke of the QRS complex

► **Drug treatment:** includes flecainamide, amiodarone or disopyramide.

► Digoxin and verapamil are contraindicated.

► Transvenous catheter radiofrequency ablation is the treatment of choice.

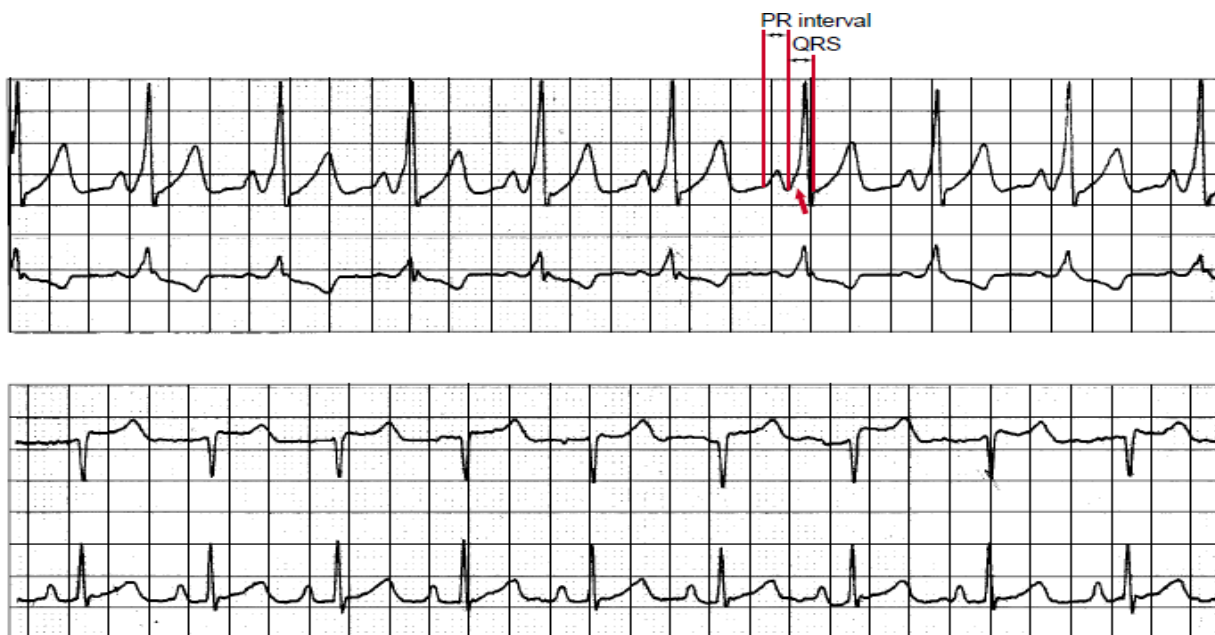


FIGURE 27-33 Wolff-Parkinson-White syndrome. (A) Sinus rhythm. Note the short PR interval, slurred initial upstroke of the QRS complex (delta wave, at the arrow), and prolonged QRS duration, upper lead II, lower lead V₁. (B) Rhythm strip of same patient following ablation, upper lead V₁, lower lead II. ECG strips courtesy of Linda Ardini and Catherine Berkmeier, Inova Fairfax Hospital, Falls Church, VA.

Atrial flutter (HR200–350/min)

- A condition in which the electrical signals come from the atria at a fast but even rate, often causing the ventricles to contract faster and increase the heart rate.
- When the signals from the atria are coming at a faster rate than the ventricles can respond to, the ECG pattern develops a signature "sawtooth" pattern, showing two or more flutter waves between each QRS complex.

Atrial flutter (TREATMENT)

- Treatment of the symptomatic acute paroxysm is electrical *cardioversion*.
- Patients who have been in atrial flutter more than 1-2 days should be treated in a similar manner to patients with atrial fibrillation and *anticoagulated* for 4 weeks prior to cardioversion.
- Recurrent paroxysms may be prevented by *class Ic* and *class III* agents
- The treatment of choice for patients with recurrent atrial flutter is radiofrequency catheter *ablation*

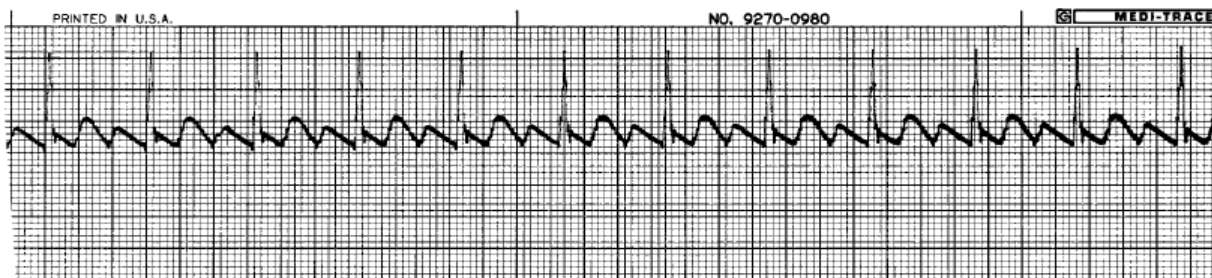


FIGURE 27-10 Atrial flutter in lead II.

Atrial fibrillation (AF)

- A condition in which the electrical signals come from the atria at a very fast and erratic rate. The ventricles contract in an irregular manner because of the erratic signals coming from the atria.
- The ECG shows normal but irregular QRS complexes, fine oscillations of the baseline (so-called fibrillation or f waves) and no P waves.
- Common causes include CAD, valvular heart disease, hypertension, hyperthyroidism and others. In some patients no cause can be found 'lone' atrial fibrillation.



FIGURE 27-11 Atrial fibrillation in lead II.

Management

- When atrial fibrillation is due to an acute precipitating event such as alcohol toxicity, chest infection or hyperthyroidism, the provoking cause should be treated.
- Strategies for the acute management of AF are ventricular rate control or cardioversion ($\pm$ anticoagulation).
 - Ventricular rate control is achieved by drugs which block the AV node



- Cardioversion is achieved electrically by DC shock or medically either by IV infusion of an anti-arrhythmic drug such as a class Ic or a class III agent

The choice depends upon:

- How well the arrhythmia is tolerated (is cardioversion urgent?)
- Whether anticoagulation is required before considering elective cardioversion
- Whether spontaneous cardioversion is likely (previous history? reversible cause?).
- Patients are anticoagulated with warfarin for 4 weeks before cardioversion.
- Anticoagulants are used to minimize the risk of thromboembolism associated with cardioversion unless atrial fibrillation is of less than 1-2 days' duration.
- Trans-esophageal echocardiography is being used to document the presence or absence of atrial thrombus as a guide to the necessity for long-term anticoagulation.

The choice depends upon:

- Long-term management of atrial fibrillation include two strategies:
 - **Rhythm control**: antiarrhythmic drugs plus DC cardioversion plus warfarin
 - **Rate control**: AV nodal slowing agents plus warfarin
- Recurrent paroxysms may be prevented by oral medication; class Ic agents are employed in patients with no significant heart disease and class III agents are preferred in patients with structural heart disease.
- Rate control is usually achieved by a combination of digoxin beta-blockers or calcium channel blockers (diltiazem or verapamil).



- **Anticoagulation** (target INR 2.0-3.0) This is indicated in patients with atrial fibrillation and one of the following major or two of the moderate risk factors:
- Major risk factors: Prosthetic heart valve, Rheumatic mitral valve disease, Prior history of CVA/TIA, Age > 75 years, Hypertension, diabetes mellitus, heart failure.
- Moderate risk factors: Age 65-75 years, coronary artery disease but normal LV function.

VENTRICULAR TACHYARRHYTHMIAS

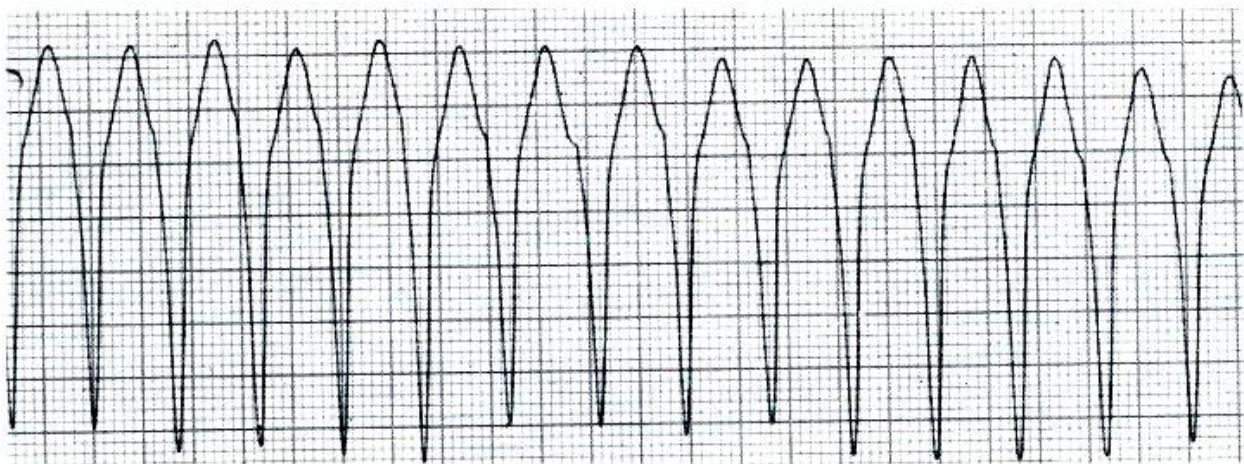
Ventricular tachyarrhythmias can be considered under the following headings:

- life-threatening ventricular tachyarrhythmias (Sustained ventricular tachycardia and ventricular fibrillation)
- torsades de pointes
- normal heart ventricular tachycardia
- non-sustained ventricular tachycardia
- ventricular premature beats

Ventricular

- A condition in which an electrical signal is sent from the ventricles at a very fast but often regular rate.

- The ECG shows a rapid ventricular rhythm with broad (often 0.14 s or more), abnormal QRS complexes. AV dissociation may result in visible P waves
- **Treatment:** in hemodynamically compromised patients, emergency DC cardioversion may be required. If the blood pressure and cardiac output are well maintained, intravenous therapy with class I drugs or amiodarone is usually used. First-line drug treatment consists of lidocaine (50-100 mg i.v. over 5 minutes) followed by a lidocaine infusion (2-4 mg i.v. per minute). DC cardioversion is necessary if medical therapy is unsuccessful.



Ventricular

- A condition in which many electrical signals are sent from the ventricles at a very fast and erratic rate. As a result, the ventricles are unable to fill with blood and pump.
- This rhythm is life-threatening because there is no pulse and complete loss of consciousness.

- The ECG shows shapeless, rapid oscillations and there is no hint of organized complexes
- A person in VF requires prompt defibrillation to restore the normal rhythm and function of the heart. It may cause sudden cardiac death. Basic and advanced cardiac life support is needed
- Survivors of these ventricular tachyarrhythmias are, in the absence of an identifiable reversible cause (e.g. acute myocardial infarction, severe metabolic disturbance), at high risk of sudden death. Implantable cardioverter-defibrillators (ICDs) are first-line therapy in the management of these patients

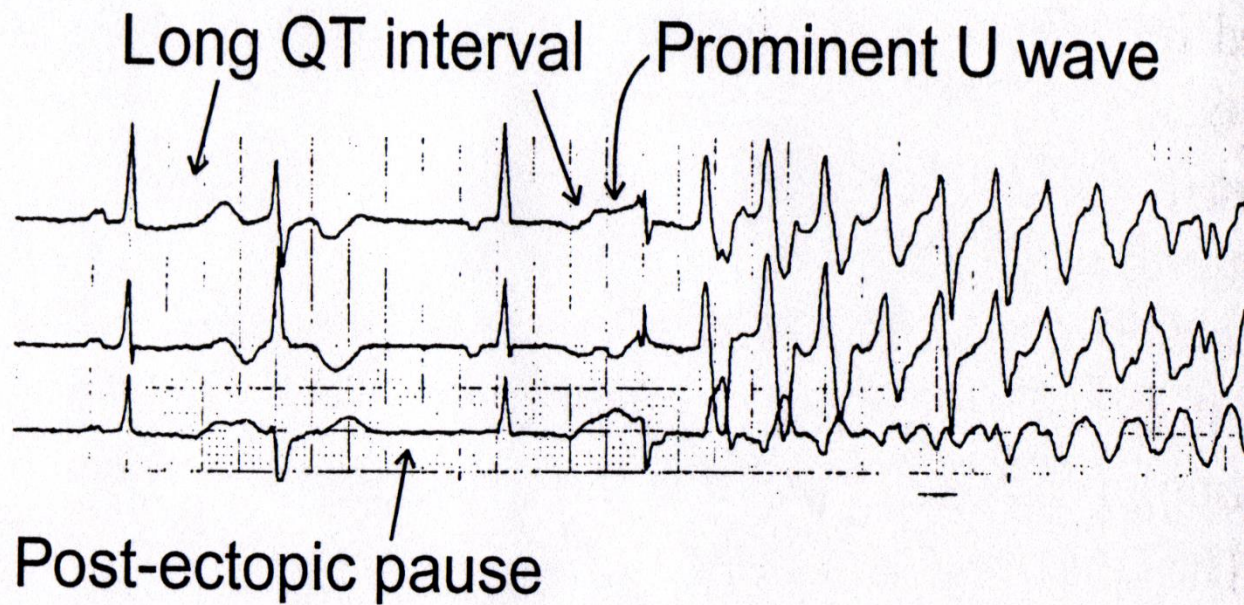


Torsade's De Pointes:

- This is a type of short duration tachycardia that reverts to sinus rhythm spontaneously.
- It may be due to:
 - Congenital
 - Electrolyte disorders e.g. hypokalemia, hypomagnesemia, hypocalcemia.
 - Drugs e.g. tricyclic antidepressant, class IA and III antiarrhythmics.



- It may present with syncopal attacks and occasionally ventricular fibrillation.
- QRS complexes are irregular and rapid that twist around the baseline. In between the spells of tachycardia, the ECG show prolonged QT interval.
- Treatment includes; correction of any electrolyte disturbances, stopping of causative drug, atrial or ventricular pacing, Magnesium sulphate 8 mmol (mg^{2+}) over 10-15 min for acquired long QT, IV isoprenaline in acquired cases and B blockers in congenital types
- Long-term management of acquired long QT syndrome involves avoidance of all drugs known to prolong the QT interval. Congenital long QT syndrome is generally treated by **beta-blockade**, left cardiac sympathetic denervation, and pacemaker therapy. Patients who remain symptomatic despite conventional therapy and those with a strong family history of sudden death usually need ICD therapy.



Premature ventricular contractions (PVCs)

- A condition in which an electrical signal originates in the ventricles and causes the ventricles to contract before receiving the electrical signal from the atria.
- ECG shows wide and bizarre QRS complex
- Early 'R-on-T' ventricular premature beats may induce ventricular fibrillation
- PVCs are not uncommon and often do not cause symptoms or problems.

Treated only if symptomatic with beta-blockers.





BRADYCARDIAS

Sinus Bradycardia

- Physiological variant due to strong vagal tone or atheletic training.
- Rate as low as 50 at rest and 40 during sleep.
- Common causes of sinus bradycardia include:
- Extrinsic causes; Hypothermia, hypothyroidism, cholestatic jaundice and raised intracranial pressure. Drug therapy with beta-blockers, digitalis and other antiarrhythmic drugs.
- Intrinsic causes; Acute ischaemia and infarction of the sinus node (as a complication of acute myocardial infarction). Chronic degenerative changes such as fibrosis of the atrium and sinus node (sick sinus syndrome).

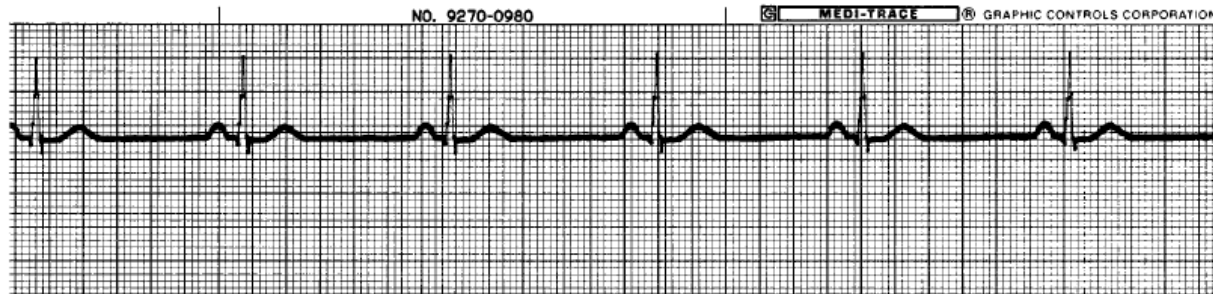


FIGURE 27-6 Sinus bradycardia in lead II.

Sick sinus

- A condition in which the sinus node sends out electrical signals either too slowly or too fast. There may be alternation between too-fast and too-slow rates.
- This condition may cause symptoms if the rate becomes too slow or too fast for the body to tolerate.
- Chronic symptomatic sick sinus syndrome requires permanent pacing (AAI), with additional antiarrhythmic drugs (or ablation therapy) to manage any tachycardia element.
- Thromboembolism is common in tachy-brady syndrome and patients should be anticoagulated unless there is a contraindication.

Atrioventricular (AV) Block

- Seldom of clinical significance, and unlikely to progress.
- ECG shows prolonged PR interval.

- May be associated with acute rheumatic fever, diphtheria, myocardial infarction or drugs as digoxin

Mobitz type I (Wenckebach phenomenon):

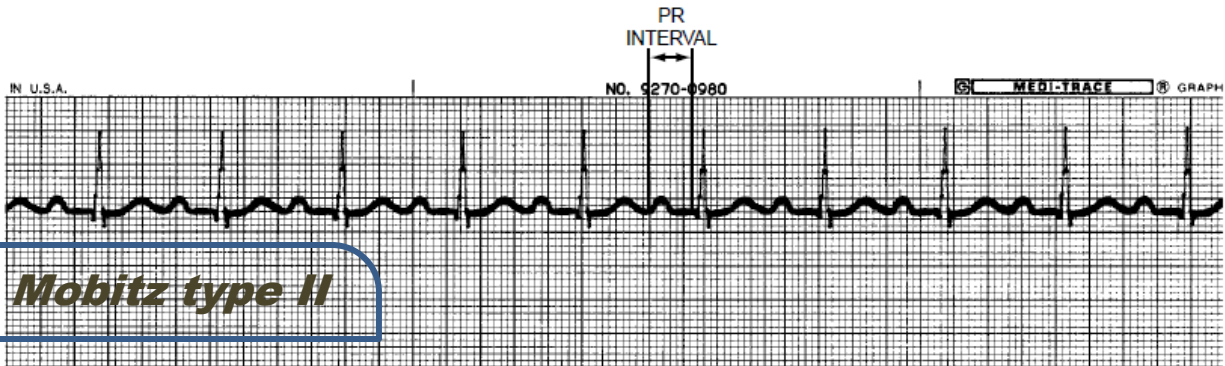


FIGURE 27-20 Sinus rhythm with first-degree AV block in lead II.

Second degree A-V Block

Acute myocardial infarction may produce second-degree heart block. In inferior myocardial infarction, close monitoring and transcutaneous temporary back-up pacing are all that is required.

In anterior myocardial infarction, second-degree heart block is associated with a high risk of progression to complete heart block, and temporary pacing followed by permanent pacemaker implantation is usually indicated.

Two types:

- Gradually increasing P-R intervals culminating in an omission.
- When isolated, usually physiological and due to increased vagal tone and abolished by exercise and atropine.

- The P wave is sporadically not conducted. Occurs when a dropped QRS complex is not preceded by progressive PR interval prolongation.
- Pacing is usually indicated in Mobitz II block, whereas patients with Wenckebach AV block are usually monitored.

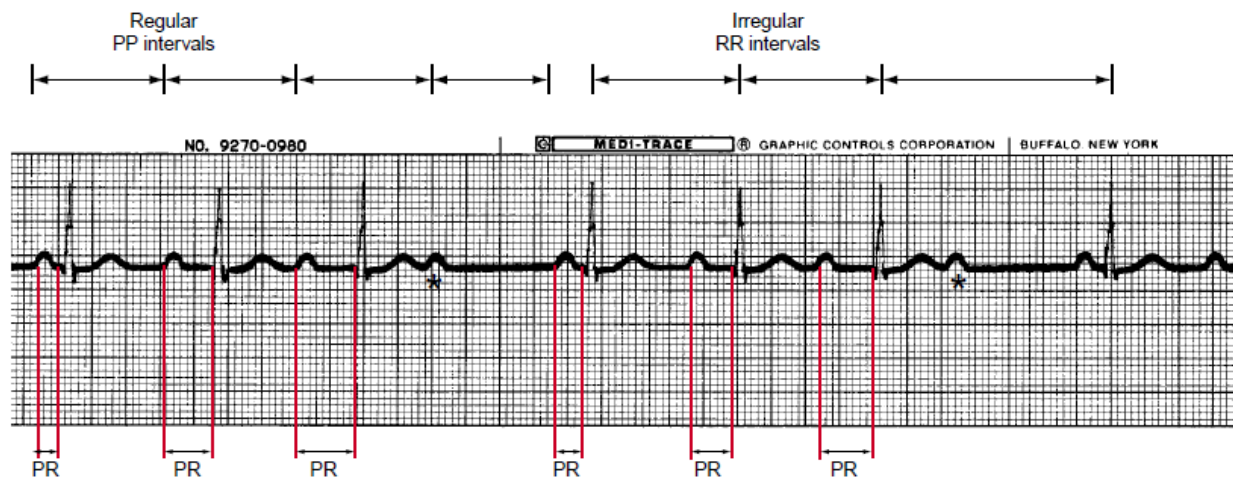


FIGURE 27-21 Sinus rhythm with second-degree AV block, type I in lead II. Note progressively longer PR durations until there is a nonconducted P wave, indicated by the asterisk (*).

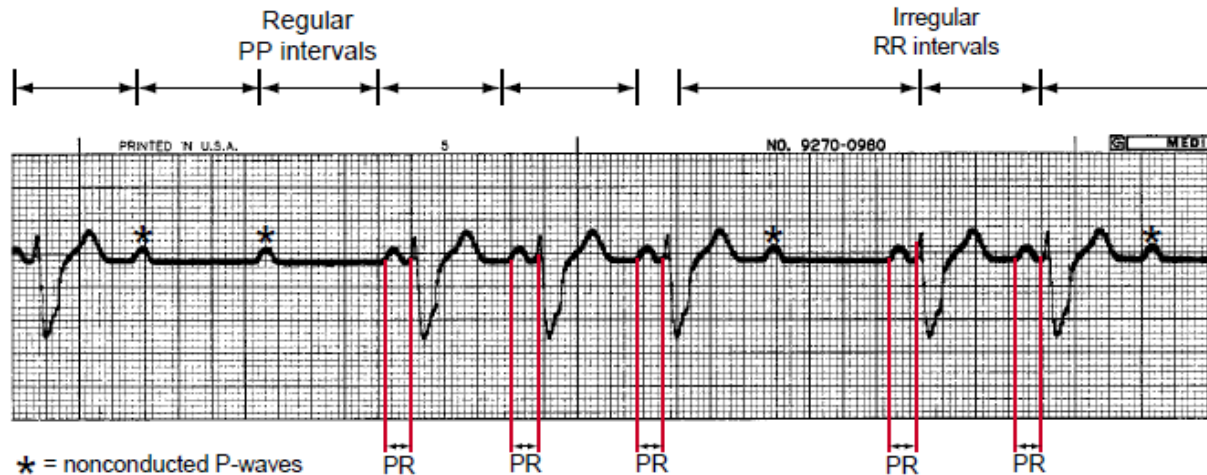


FIGURE 27-22 Sinus rhythm with second-degree AV block, type II in lead V_1 ; note constant PR interval.

Third degree A-V Block

- Common in elderly age groups due to idiopathic bundle branch fibrosis.
- Other causes include coronary heart disease, calcification from aortic valve, sarcoidosis or congenital.
- ECG shows bradycardia, P wave continue, unrelated to regular slow idioventricular rhythm.
- Treatment is permanent pacing.

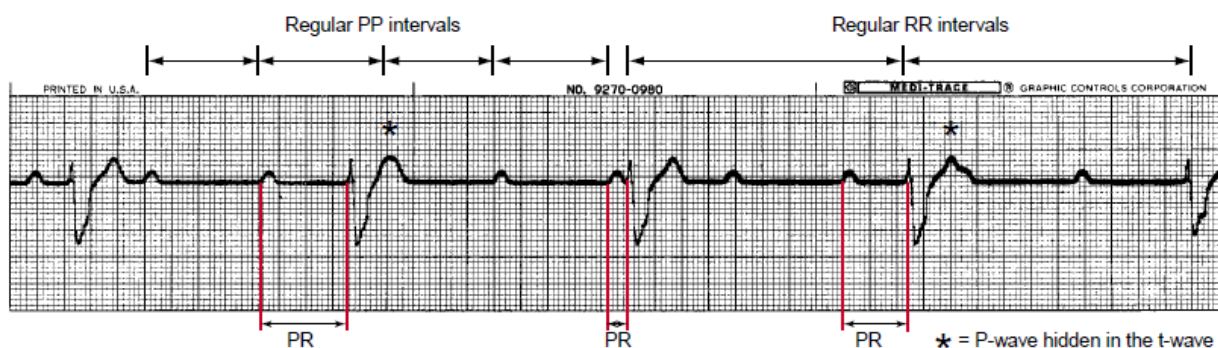


FIGURE 27-23 Sinus rhythm with third-degree AV block and idioventricular rhythm in lead V_1 ; note irregular PR intervals.



Bundle Branch Block (BBB):

- Interruption of the right or left branch of the bundle of His delays activation of the corresponding ventricle leading to broadening of the QRS complex
- Unlike right BBB, left BBB is always associated with an underlying heart disease.
- Both RT and LT BBB show wide deformed QRS complex. In RBBB there is rSR pattern in lead V1, while in LBBB there is a broad monophasic (or notched) R wave in leads V5 and V6.

Hemiblock

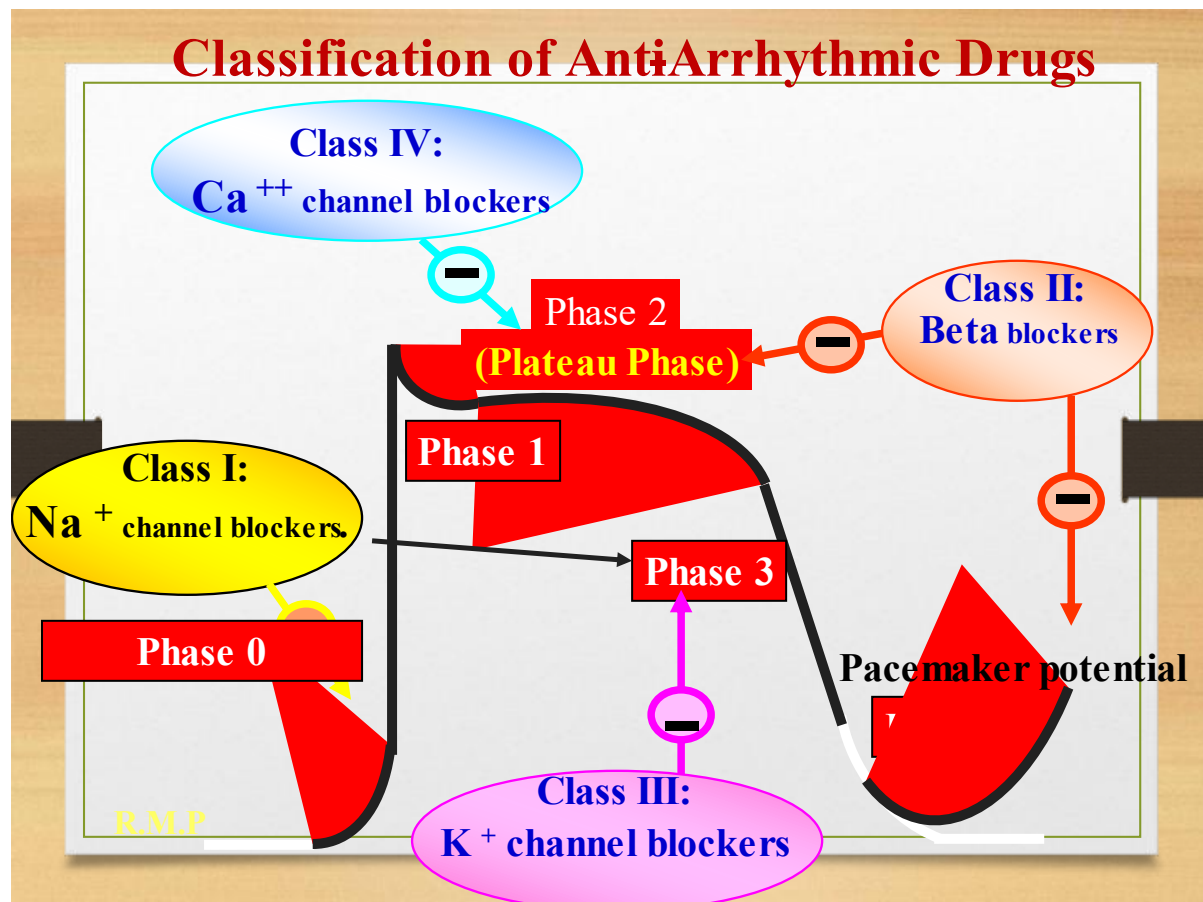
Delay or block in the divisions of the left bundle branch produces a swing in the direction of depolarization (electrical axis) of the heart. When the anterior division is blocked (left anterior hemiblock), there is left axis deviation. Delay or block in the postero-inferior division causes (right axis deviation).

Bifascicular block

This is a combination of a block of any two of the following: the right bundle branch, the left antero-superior division and the left postero-inferior division. Block of the remaining fascicle will result in complete AV block.

MANAGEMENT OF ARRHYTHMIAS

- Pharmacological therapy.
- Cardioversion.
- Pacemaker therapy.
- Surgical therapy e.g. aneurysmal excision.
- Interventional therapy “ablation”.



Classification of Antiarrhythmic Drugs based on Drug Action

CLASS	ACTION	DRUGS
I.	Sodium Channel Blockers	
IA.	Moderate phase 0 depression and slowed conduction (2+); prolong repolarization	Quinidine, Procainamide, Disopyramide
IB.	Minimal phase 0 depression and slow conduction (0-1+); shorten repolarization	Lidocaine
IC.	Marked phase 0 depression and slow conduction (4+); little effect on repolarization	Flecainide
II.	Beta-Adrenergic Blockers	Propranolol, esmolol
III.	K ⁺ Channel Blockers (prolong repolarization)	Amiodarone, Sotalol, Ibutilide
IV.	Calcium Channel Blockade	Verapamil, Diltiazem



HIGH BLOOD PRESSURE (HYPERTENSION)

What is the blood pressure (BP)?

Blood pressure is the force exerted by the blood against the walls of blood vessels. The magnitude of this force depends on the cardiac output and the resistance of the blood vessels. High blood pressure is a common condition that affects the body's arteries. It's also called hypertension. In patients with hypertension, the force of the blood pushing against the artery walls is consistently too high.

Types of hypertension

1. **Primary (Essential):** The most common type of hypertension >85%
2. **Secondary:** rare

Definition of hypertension:

Hypertension is defined as having a blood pressure equal or higher than 140 over 90 mmHg, with repeated measurements

Table 3 Definitions and classification of office blood pressure levels (mmHg)^a

Category	Systolic		Diastolic
Optimal	<120	and	<80
Normal	120–129	and/or	80–84
High normal	130–139	and/or	85–89
Grade 1 hypertension	140–159	and/or	90–99
Grade 2 hypertension	160–179	and/or	100–109
Grade 3 hypertension	≥180	and/or	≥110
Isolated systolic hypertension	≥140	and	<90

Risk factors

High blood pressure has many risk factors, including:

1. **Age.** The risk of high blood pressure increases with age. Until about age 64, high blood pressure is more common in men. Women are more likely to develop high blood pressure after age 65.
2. **Race.** High blood pressure is particularly common among Black people.
3. **Family history.**
4. **Obesity** or being overweight. Excess weight causes changes in the blood vessels, the kidneys and other parts of the body. These changes often increase blood pressure.
5. **Lack of exercise.** Not exercising can cause weight gain. Increased weight raises the risk of high blood pressure.
6. **Smoking**
7. **Too much salt.**



8. **Low potassium levels.** Potassium helps balance the amount of salt in the body's cells. A proper balance of potassium is important for good heart health.
9. **Drinking alcohol.** Alcohol use has been linked with increased blood pressure, particularly in men.
10. **Stress.** High levels of stress can lead to a temporary increase in blood pressure.
11. **Certain chronic conditions.** Kidney disease, diabetes and sleep apnea
12. **Some medications** such as anti-inflammatories and nonsteroidal anti-inflammatories, certain types of birth control pills, and stimulant medications used for weight loss.
13. **Pregnancy.**

Blood Pressure measurement.

1-Office BP measurements (at clinics) are the most commonly used method for BP measurement.

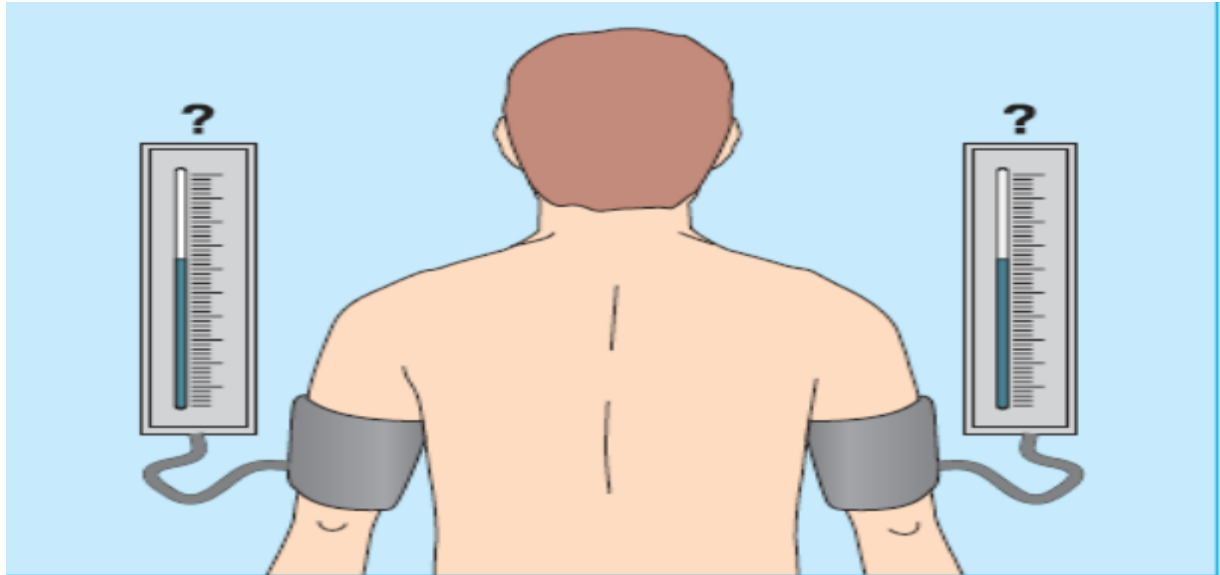


▪ *Patient preparation*

1. No acute anxiety, stress or pain.
2. No caffeine, smoking or nicotine in the preceding 30 minutes.
3. No use of substances containing adrenergic stimulants such as phenylephrine or pseudoephedrine.
4. Bladder and bowel comfortable.
5. No tight clothing on arm or forearm.
6. Quiet room with comfortable temperature.
7. Rest for at least 5 minutes before measurement.

- **Which arm?**

For the first time measurements should be done from both arms, taking the higher reading from either one of them, would more accurately capture the real BP.



- **Causes of difference in blood pressure in the two arms**

1. Arrhythmia
2. Anomalies of the aorta
3. Polyarthritis nodosa
4. Mediastinal syndrome
5. Hemiplegia

- **The cuff size does matter**

- Too small cuff/bladder can overestimate the blood pressure
- Too large cuff/bladder can underestimate the blood pressure

Mismatching of bladder and arm

Bladder too small (undercuffing)

- Overestimation of blood pressure
- Range of error:
 - 3/3 to 12/8 mm Hg
 - As much as 30 mm Hg in patients who are obese

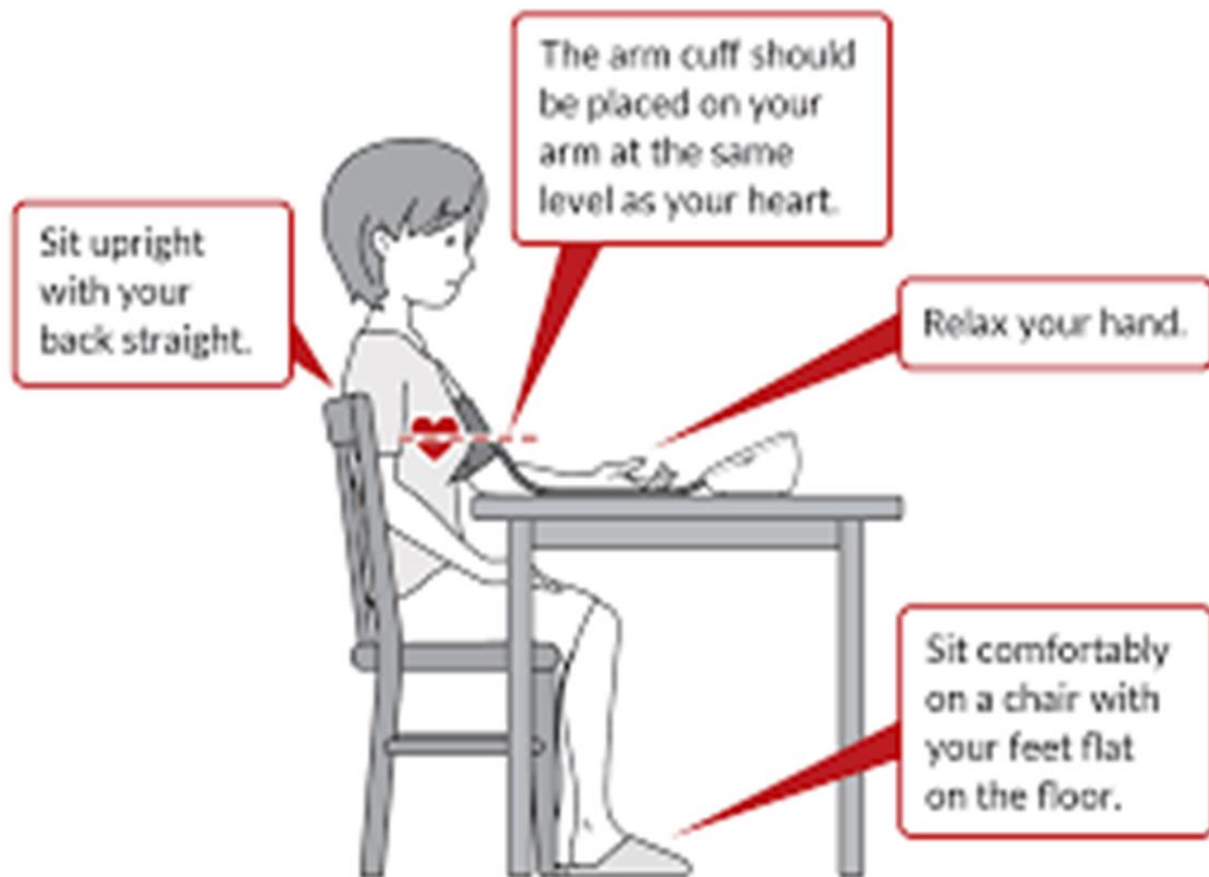
Bladder too large

- (overcuffing)
- Underestimation of blood pressure
- Range of error:
 - 10–30 mm Hg

Undercuffing is more common than overcuffing

▪ Position of the patient

The seated BP measurement is the standard position to determine diagnostic and therapeutic treatment decisions as this figure illustrates.



The standing blood pressure is used to test for postural hypotension.

Postural hypotension (orthostatic hypotension)

Is defined as a fall in systolic blood pressure of at least 20 mmHg and/or in the diastolic blood pressure of at least 10 mmHg between the supine reading and the upright reading.

Confirmed by measuring a person's blood pressure after lying flat for 5 minutes, then 1 - 3 minutes after standing.

Causes of postural hypotension

1. Dehydration
2. Anemia
3. Medications
4. Stroke
5. Arrhythmias
6. Valve abnormalities
7. DM and other causes of A. neuropathy
8. Parkinson disease
9. Electrolytes abnormalities
10. Adrenal gland malfunction

2-Other methods for BP measurements

A. Home BP measurements (at home)



B. Ambulatory BP measurements (outside clinics): This method measures blood pressure on a continuous basis for 24 hours. The blood pressure is measured even during daily activity or during sleep.



These devices helped in the classification of the blood pressure into 4 categories:

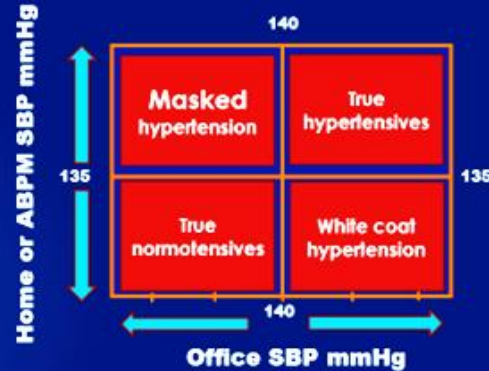
1-True Normotension: Blood pressure which is normal at clinic (office) and non-office measurements

2-True hypertension: Blood pressure which is elevated at clinic (office) and non-office measurements

3-White coat hypertension: a high blood pressure reading in the clinic (office) and a normal reading at home (by ambulatory or home BP measuring).

4-Masked hypertension: a normal blood pressure in the clinic (office) but an elevated BP out of the clinic (by ambulatory or home BP measuring)

BP categories



Derived from Pickering et al. Hypertension 2002; 40: 795-796.

Potential indications for the use of ambulatory blood pressure monitoring

- Unusual variability
- Possible white coat or masked hypertension
- Evaluation of nocturnal hypertension
- Evaluation of drug-resistant hypertension
- Determining the efficacy of drug treatment over 24 hours
- Diagnoses and treatment of hypertension in the elderly and in pregnant women
- Evaluation of symptomatic hypotension

Complications of hypertension:

The excessive pressure on the artery walls caused by high blood pressure can damage blood vessels and body organs. The higher the blood pressure and the longer it goes uncontrolled, the greater the damage.

Complications include:

1. **Atherosclerosis** of the arteries due to high blood pressure or other factors can lead to a heart attack, stroke or other complications.
2. **Aortic Aneurysm**. Increased blood pressure can cause a blood vessel to weaken and bulge, forming an aneurysm. If an aneurysm ruptures, it can be life-threatening.
3. **Heart failure**. When you have high blood pressure, the heart has to work harder to pump blood. The strain causes the walls of the heart's pumping chamber to thicken. This condition is called left ventricular hypertrophy. Eventually, the heart can't pump enough blood to meet the body's needs, causing heart failure
4. **Kidney problems**. High blood pressure can cause the blood vessels in the kidneys to become narrow or weak. This can lead to kidney damage.
5. **Eye problems**. Increased blood pressure can cause thickened, narrowed or torn blood vessels in the eyes (retinopathy). This can result in vision loss.
6. **Dementia**. Narrowed or blocked arteries can limit blood flow to the brain. This can cause a certain type of dementia called vascular dementia.

Management of hypertension

1-Assessment of the overall CV risk factors

Presence of other Risk Factors

- A. Smoking
- B. Family history of premature cardiovascular disease Dyslipidemia
- C. Lifestyle
- D. Abdominal obesity
- E. Drugs

Target organ damage

- A. Proteinuria
- B. Left ventricular hypertrophy
- C. Chronic kidney disease (glomerular filtration rate < 60 ml/min/1.73 m²)

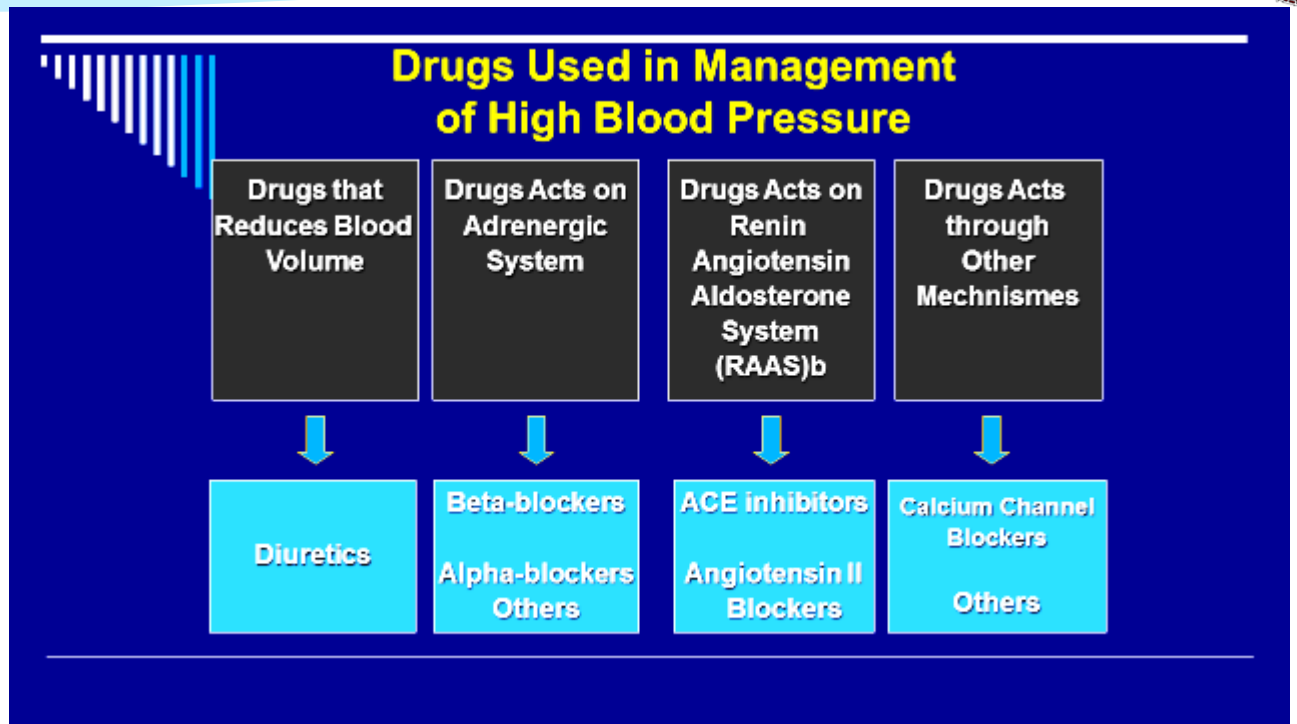
Presence of atherosclerotic vascular disease

- A. Previous stroke or TIA
- B. CHD or PAD

2-Routine laboratory tests

3-Non pharmacological therapy as weight loss ,exercise ,salt restriction and healthy diet

4-Pharmacotherapy



Any drug of the listed drugs in the above table can be used as an initial therapy for hypertension treatment, however specific agents are preferred in selected cases of hypertension (compelling indications)

COMPELLING INDICATIONS FOR INDIVIDUAL ANTIHYPERTENSIVE DRUG CLASSES

Compelling Indication*	Recommended Drugs					
	DIURETIC	BB	ACEI	ARB	CCB	Aldo ANT
Heart failure	•	•	•	•		•
Post-MI		•	•			•
High coronary disease risk	•	•	•		•	
Diabetes	•	•	•	•	•	
Chronic kidney disease			•	•		
Recurrent stroke prevention	•		•			

Treatment of HT in pregnancy

- Methyldopa ,labetalol or nifedipine are preferable antihypertensive medications in pregnancy, with magnesium sulfate for women with preeclampsia who are at high risk for seizures .
- All major antihypertensive agents with the exception of (ACE) inhibitors and (ARBs(are recommended .
- Diuretics do not cause fetal malformations but are generally avoided in pregnancy, as they prevent the physiologic volume expansion seen in normal pregnancy.

Blood pressure targets

Reduction of the BP (<140/90 mm Hg) in all patients.

Reduction of the BP(<130/80 mm Hg) in high risk patients, if the patient can tolerate.

Levels <120/70 mmHg are not allowed as the mortality rate increases with these levels.

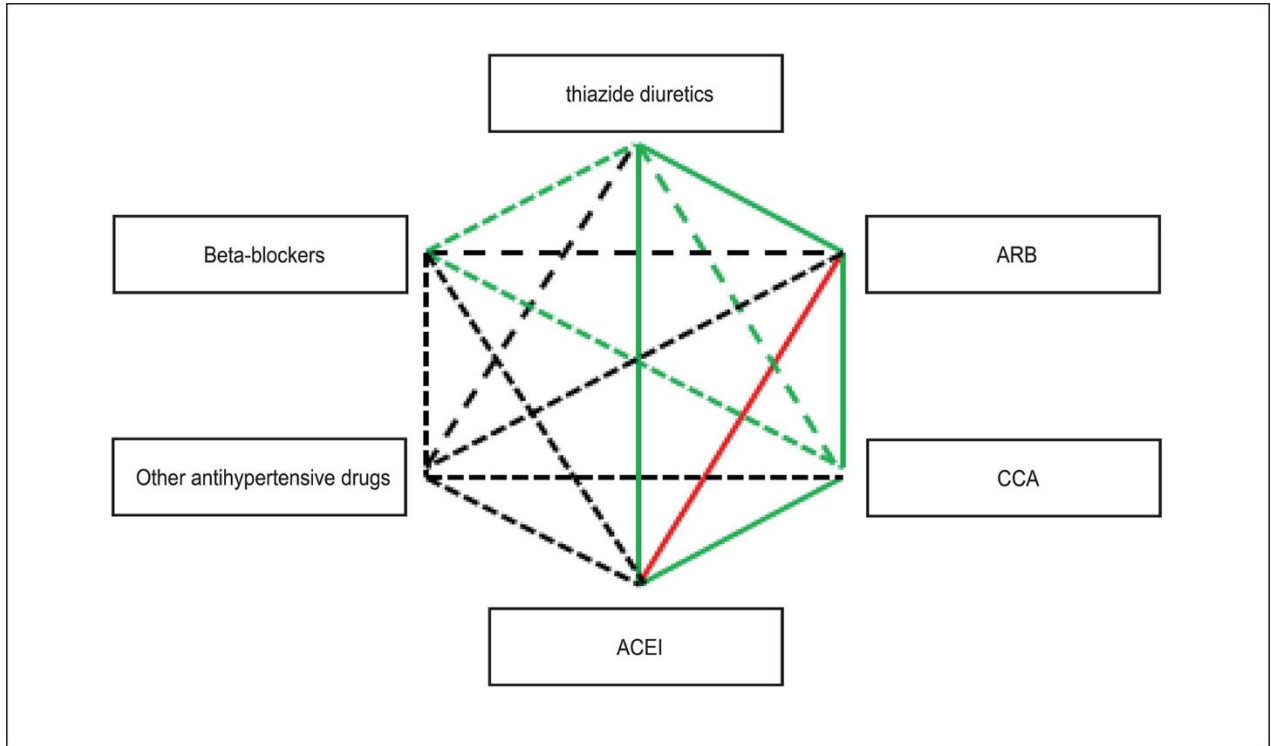
Combination Therapy:

Most hypertensive patients will require two or more antihypertensive medications to achieve the BP goal

Possible combination of antihypertensive drugs are shown in the following graph :

Continuous green line (preferential combinations); dotted green line (acceptable combinations); dotted black line (less usual combinations); red

line (unusual combinations) ... (ARB: angiotensin receptor blocker; CCA: calcium channel antagonist; ACEI: angiotensin-converting-enzyme inhibitor).



Resistant Hypertension

Blood pressure remaining above goal in spite of concurrent use of 3 antihypertensive agents of different classes.

Ideally, 1 of the 3 agents should be a diuretic & all agents should be prescribed at optimal dose amounts.

Causes of resistant hypertension

Pseudo resistance

- A. Patient resistance
- B. Physician resistance
- C. White coat hypertension
- D. Masked hypertension

True resistance

- A. Secondary causes
- B. True drug resistance

Treatment of Resistant HT

1. Non-Pharmacologic therapy.
2. Withdrawal of interfering medications e.g. NSAID.
3. Diuretic therapy (lack or under use).
4. Combination therapy.
5. Mineralcorticoid receptor antagonists.
6. New interventional therapy.

Secondary Causes of HT

Common

1. Obstructive sleep apnea
2. Renal parenchymal disease
3. Primary aldosteronism
4. Renal artery stenosis

Uncommon

1. Pheochromocytoma
2. Cushing's disease
3. Hyperparathyroidism
4. Aortic Coarctation
5. Intracranial tumor

Secondary Hypertension is Suggested By:

1. Severe or refractory hypertension.
2. An acute rise in blood pressure.
3. Age of onset before puberty.
4. Age less than 30 years or above 60.
5. Negative family history of hypertension.
6. Signs suggesting secondary cause.

Hypertensive Crisis

- BP elevation to 180/120 mmHg or more.
- **Classified into** hypertensive urgency or emergency according to the presence or absence of target organ damage (TOD):

TOD

Hypertensive emergency

- ☐ Immediate treatment
- ☐ Hospital admission
- ☐ IV therapy

No TOD

Hypertensive urgency

- ☐ Less rapid BP reduction
- ☐ Oral therapy
- ☐ Outpatient management

Who are at risk of hypertensive crisis?

1. Non adherence to anti HTN medications
2. Secondary hypertension
3. Reno vascular hypertension
4. Pheochromocytoma
5. Hyperaldosteronism
6. Severe preeclampsia
7. Perioperative setting
8. Patients with associated comorbidities
9. Cocaine use

Parenteral Drugs For Treatment of Hypertensive Emergencies

VASODILATORS

- Nitroprusside
- Fenoldopam
- Nitroglycerine
- Enalaprilat
- Nicardipine
- Hydralazine

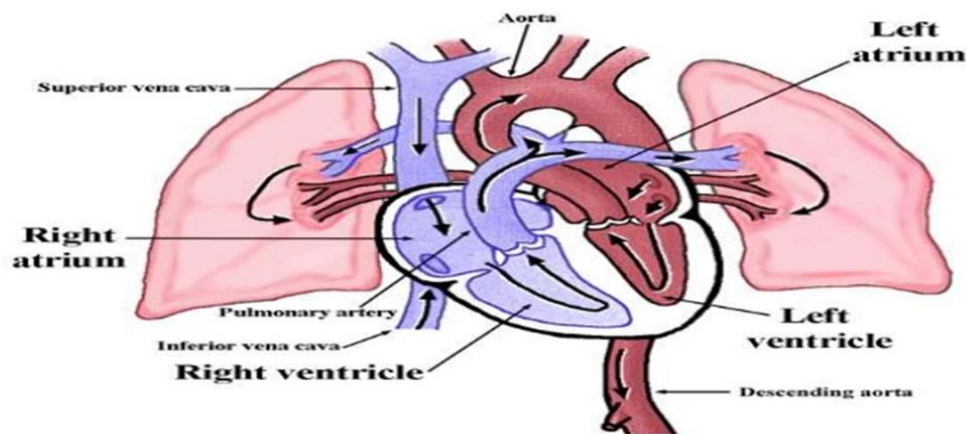
ADRENERGIC INHIBITORS

- Labetalol
- Esmolol
- Phentolamine

PULMONARY EMBOLISM

Pulmonary embolism is a blood clot that stops blood flow to an artery in the lung. In most cases, the blood clot starts in a deep vein in the leg and travels to the lung. Rarely, the clot originates from the right side of the heart or from another vein in the body.

Pulmonary circulation



Lifestyle

- Pulmonary embolism occurs when a clump of material, most often a blood clot, gets stuck in an artery in the lungs, blocking the flow of blood. Blood clots most commonly come from the deep veins of legs, a condition known as deep vein thrombosis (DVT).



- Occasionally, blockages in the blood vessels are caused by substances other than blood, such as:
 1. Fat from a broken bone
 2. Part of a tumor
 3. Air bubbles

Risk factors

Although anyone can develop blood clots that result in a pulmonary embolism, certain factors can increase the risk:

1. History of blood clots
2. Family history of thromboembolism
3. Medical conditions such as heart disease, renal disorders, diabetes
4. Cancer
5. Chemotherapy
6. Surgery.
7. Inherited disorders affecting blood coagulation
8. Coronavirus disease. People who have severe symptoms of COVID-19 have an increased risk of pulmonary embolism.
9. Extended periods of inactivity
10. Smoking.
11. Overweight.
12. Supplemental estrogen.
13. Pregnancy.

Symptoms:

Symptoms can vary greatly, depending on how much of lung involvement, the size of the clots, and whether you have underlying lung or heart disease.

Common symptoms

1. Shortness of breath.
2. Chest pain.
3. Lightheadedness or dizziness
4. Leg pain or swelling, or both, usually in the back of the lower leg

Other symptoms

1. Cough that may include bloody or blood-streaked mucus
2. Rapid or irregular heartbeat
3. Excessive sweating
4. Fever

Complications

- Pulmonary embolism can be life-threatening. About one-third of people with an undiagnosed and untreated pulmonary embolism don't survive.
- Pulmonary embolisms can lead to pulmonary hypertension, a condition in which the blood pressure in the lungs and in the right side of the heart is too high.
- In rare cases, small clots called emboli remain in the lungs and scarring develops in the pulmonary arteries over time. This results in chronic pulmonary hypertension.

Diagnosis

1. Blood tests

D dimer. High levels may suggest an increased likelihood of blood clots, although many other factors can cause high D dimer levels. D dimer is a good negative test

2. **Chest X-ray:** This noninvasive test shows images of the heart and lungs. Although X-rays can't diagnose a pulmonary embolism, they can rule out other conditions with similar symptoms.

Chest X-Ray

- Usually abnormal, but non-specific
 - ☐ Normal
 - ☐ Cardiac enlargement
 - ☐ Pleural effusion
 - ☐ Elevated hemidiaphragm
 - ☐ Pulmonary artery enlargement
 - ☐ Atelectasis : wedge-shaped configuration at lung periphery
 - ☐ Parenchymal pulmonary infiltrates
 - ☐ Lung oligemia

3. ECG:

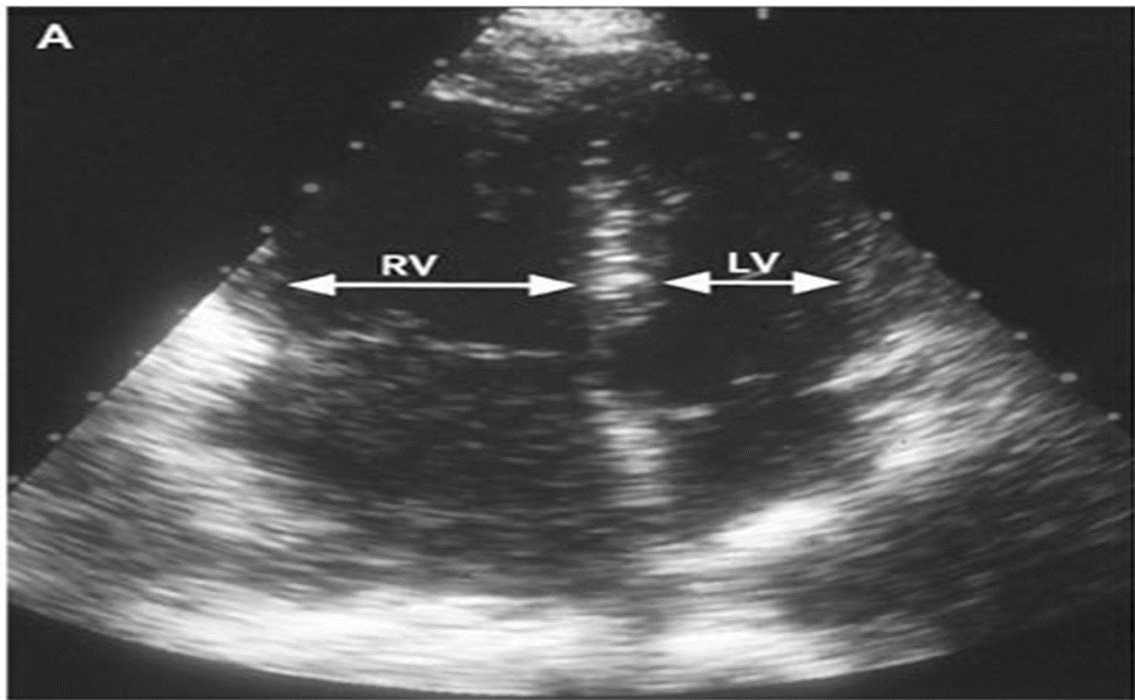
Also, non-specific and their values have been reduced in the recent guidelines:

Findings may include:

- a. Tachycardia
- b. ST/T waves changes
- c. Right ventricular (RV) strain patterns

4. Echocardiography:

May shows RV dilation, pulmonary artery dilation or pulmonary hypertension



5. Ultrasound on the venous system:

A noninvasive test known as duplex ultrasonography, sometimes called a duplex scan uses sound waves to scan veins to check for deep vein blood clots.

6. CT pulmonary angiography:

Creates 3D images that can find changes such as a pulmonary embolism within the arteries in the lungs. In some cases, contrast material is given through a vein in the hand or arm during the CT scan to outline the pulmonary arteries.

7. MRI:

MRI is a medical imaging technique that uses a magnetic field and computer-generated radio waves to create detailed images of the organs and tissues in the body. MRI is usually only done in those who are pregnant to avoid radiation to the baby and in people whose kidneys may be harmed by dyes used in other tests.

8. Pulmonary angiography: The gold standard test

Provides a clear picture of the blood flow in the arteries of the lungs. It's **the most accurate way to diagnose a pulmonary embolism.**

Treatment:

1-Emergency management and resuscitation in the CCU:

Oxygen therapy

IV access

Analgesics

Management of cardiogenic shock

2-Pharmacotherapy

- a. **Heparin** is a frequently used anticoagulant that can be given through a vein or injected under the skin. It acts quickly and is given along with an **oral anticoagulant (warfarin)** until the oral medicine becomes effective. This can take several days.
- b. **Newer direct oral anticoagulants (DOAC)** work more quickly and have fewer interactions with other medicines. They have the



advantage of being given by mouth until they're effective, without the need for heparin.

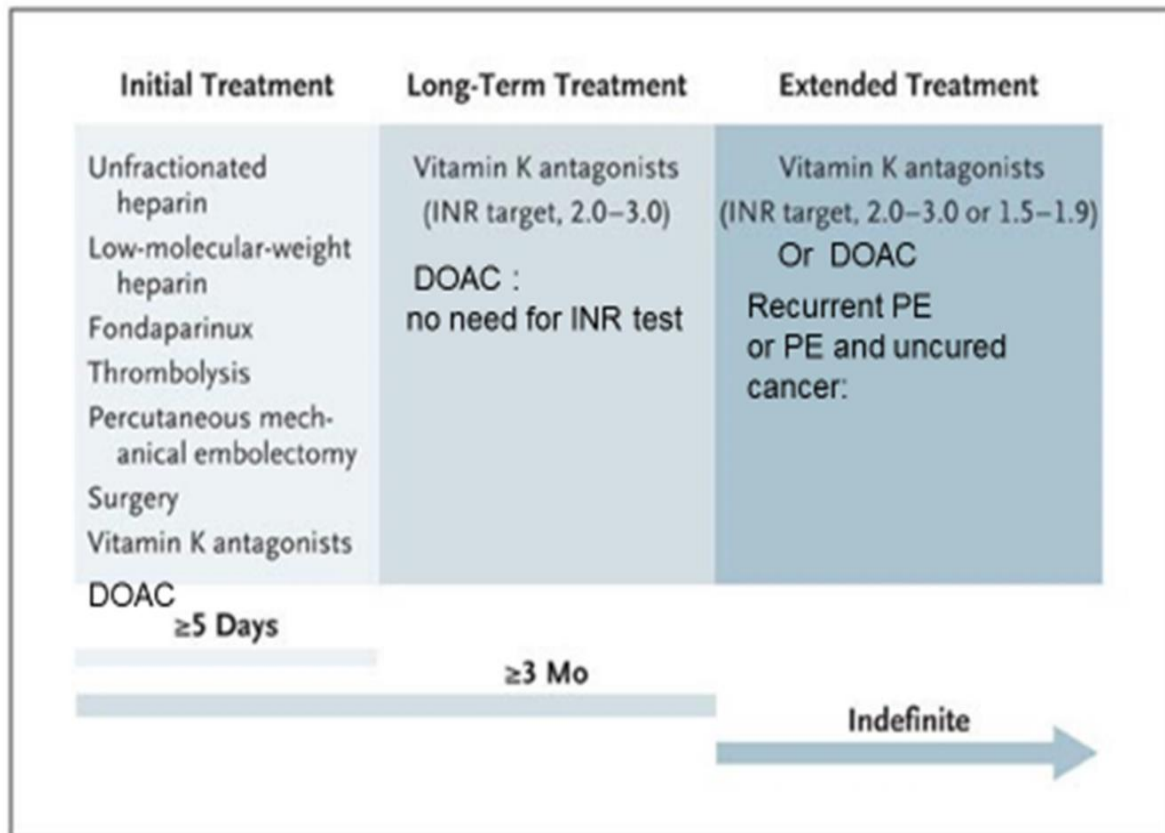
- c. **Clot dissolvers.** While clots usually dissolve on their own, thrombolytics medicines that dissolve clots given through a vein can dissolve clots quickly. Because these clot-busting medicines can cause sudden and severe bleeding, **they usually are reserved for life-threatening situations.**

3-Surgical and other procedures

- a. Clot removal. In large, life-threatening clot in the lung using a thin, flexible catheter threaded through blood vessels.
- b. Vein filter. A catheter also can be used to position a filter in the body's main vein, the inferior vena cava that leads to the right side of your heart. The filter can help keep clots from going to the lungs. This procedure is usually only used for people who can't take anticoagulant drugs or those who get blood clots even with the use of anticoagulants.

4-Post discharge anticoagulation therapy

is shown in the following table



Shock

Definition

A state of decreased tissue perfusion due to failure of circulatory system and resulting in decreased supply of oxygen & nutrients to tissues.

Pathophysiology of shock:

Failure of Circulatory System → Decreased Tissue Perfusion→

Decreased Supply of Oxygen & Nutrients to Tissues → Organ Dysfunction→
Cell Death

Etiology

1. Hypovolemic shock
2. Cardiogenic shock
3. Obstructive shock
4. Distributive shock

A) Hypovolemic shock:

1. Severe hemorrhage.
2. Severe vomiting & diarrhea.
3. Severe polyuria & sweating.
4. Severe trauma.

5. Extensive burns.

Clinical manifestations include:

1. Acute myocardial infarction.
2. Severe myocarditis & cardiomyopathy.
3. Acute mitral regurgitation & aortic regurgitation.

B) Cardiogenic shock:

1. Cardiac tamponade.
2. Massive pulmonary embolism.

C) Obstructive shock:

1. Septic shock due to severe infections.
2. Anaphylactic shock.
3. Neurogenic shock due to severe spinal cord injury.

D) Distributive shock:

Clinical picture:

- 1- Features of compensatory mechanisms mainly due to sympathetic stimulation.
- 2- Features of organ failure.
- 3- Features of the cause.
- 4- Irreversible shock & death.

Clinical manifestations are explained by:

- 1- Hypotension: decreased mean arterial blood pressure < 60 mmHg
(Normal level: 70-105 mmHg).
- 2- Tachycardia.
- 3- Cold sweats.
- 4- Dyspnea & tachypnea.
- 5- Irritability, disturbed consciousness.
- 6- Oliguria.
- 7- Mottled skin.
- 8- Features of organ dysfunction & failure:
 - a) Myocardial depression
 - b) ARDS
 - c) Pre-renal failure, acute tubular necrosis & renal failure
 - d) Jaundice & liver failure
 - e) Stress ulcers
 - f) Decreased coagulation factors
 - g) DIC
 - h) Metabolic acidosis due to increased production of lactic acid
 - i) Pancreatitis
- 9- Features of the cause: e.g.
 - a. In myocardial infarction: acute chest pain & heart failure especially left-sided.
 - b. In septic shock: fever & manifestations of infection e.g. cholecystitis.

c. In anaphylactic shock: urticaria, pruritus, stridor, wheezing, dyspnea.

2) Investigations:

Shock is an emergency state & rapid resuscitation is required.

Management of Shock

1. Initial resuscitation.
2. Investigations.
3. Monitoring.
4. Treatment.

Lines of management include:

1. IV fluids.
 2. Vasopressor agents.
 3. Inotropic agents.
- IV fluids are usually the first line of treatment in shock.
 - If there is no adequate response:
 - Vasopressor agents are added to increase BP.
 - Inotropic agents are given to increase CO.

1) Initial

- Initial investigations:

- Chest X-ray.
- ECG.
- CBC.
- Arterial blood gases (ABGs).
- Electrolytes: Na, K, Ca, Mg, P.
- Renal & liver function tests.
- * Investigations for the cause: e.g.
 - Cardiac enzymes for myocardial infarction.
 - Blood culture for sepsis.

3) Monitoring:

- ECG monitoring...
- Central venous catheter.
- Arterial catheter.
- May be Swan-Ganz catheter.
- Mental Status.
- Urine output (urinary catheter should be placed).
- CBC.
- Electrolytes: Na, K, Ca, Mg. P.
- Arterial blood gases (ABGs)
- Renal & liver function tests.

-Coagulation tests.

- Blood lactate.

4) Treatment:

***Goals of treatment:**

1. Mean ABP>60 mmHg.
2. Cardiac index>2.2 L/min/m²
3. Oxygen saturation > 92%.

*** Lines of treatment:**

1. Hospitalization & admission to ICU.
2. Bed rest & proper positioning.
 - Patient with shock is placed in a flat position with elevation of lower limbs.
 - Patient with cardiogenic shock is placed in a sitting or semi sitting position.

3. Care of respiration:

- Maintenance of a patent airway.
- Suction of secretions.
- Oxygen inhalation.
- Mechanical ventilation may be indicated.

4. Sedation:

- Small doses of morphine (2-5 mg) or pethidine (10-25 mg) IV may be needed. Caution should be given to patients with impairment of respiratory functions.

5. **IV fluids** (See below).

6. **Vasopressor agents** (See below).

7. **Inotropic agents** (See below).

8. **Treatment of the cause** (See below).

IV Fluids

Amount of fluids:

- Usually starting with 500-1000 ml.
- Additional amounts depend on:
 - Type of shock e.g. large volumes in hypovolemic shock.
 - BP.
 - CVP.
 - Urine output.
 - CO.

Type of fluids:

- 1- Crystalloids e.g. Normal saline, Ringers lactate.
- 2- Colloids e.g. Albumin if plasma level is <2 gm/dl.
- 3- Blood transfusion in hemorrhagic shock.
- 4- Packed red cells transfusion in anemia.

Vasopressor Agents

- They are used to elevate BP & improve tissue perfusion.
- They include:
- Norepinephrine: 2-20 $\mu\text{g}/\text{min}$.
- Epinephrine: 1-20 $\mu\text{g}/\text{min}$.
- Vasopressin.

Inotropic Agents

- They are used to elevate CO & improve tissue perfusion.
- They include:
- Dopamine: 2-20 $\mu\text{g}/\text{kg}/\text{min}$.
- Dobutamine: 2.5-15 $\mu\text{g}/\text{kg}/\text{min}$.
- Amrinone, Milrinone.

Treatment of the Cause

In myocardial infarction.

In massive pulmonary embolism.

- Surgical drainage if needed.

In anaphylactic shock:

- Epinephrine: 0.5 ml of 1:1000 solution SC & could be repeated if necessary.
- Inhalation of B2- agonist.
- Antihistaminics.
- Hydrocortisone 100 IV/6 h.



Acute Arterial Ischemia

Definition

- Acute occlusion of a peripheral artery, usually without a history of claudication.
- Urgent management required
- Skeletal muscle can tolerate 6 h of ischemia before irreversible damage and myonecrosis; exception is in acute-on-chronic occlusion, where previously developed collaterals allow more time
- Tends to be lower in lower extremity > upper extremity; femoropopliteal > aortoiliac

Etiology and Risk

- Embolism vs. thrombosis
 - examples of conditions that predispose to embolism are: arrhythmias, endocarditis, and arterial aneurysms
 - existing atherosclerotic plaques (i.e. chronic PAD) can rupture causing thrombosis
 - previous vascular grafts/reconstructions can fail and thrombose leading to acute presentation
 - hypercoagulable states can contribute to arterial thrombosis

Clinical features

Symptoms of Acute Limb Ischemia



6 Ps – all may not be present:

1-Pain: absent in 20% of cases.

2-Pallor: within a few hours becomes mottled cyanosis.

3-Paresthesia: light touch lost first then sensory modalities.

4-Paralysis/Power loss: most important, heralds impending gangrene.

5-Polar/Poikilothermia/'Perishing cold' .

6-Pulselessness: not reliable.

Arterial Embolism VS Thrombus

Presentation	Embolus	Thrombus
Onset	Acute	Progressive, acute-on-chronic
Loss of Function/Sensation	Prominent	Less profound (due to underlying collaterals)
Hx of Claudication	No	Maybe
Atrophic Changes	No	Maybe
Contralateral Limb Pulses	Classically normal	Decreased or absent

Investigations

- History and physical exam are essential:

depending on degree of ischemia one may have to forego investigations and go straight to the operating room.

- ABI: extension of physical exam, easily performed at bedside.
- ECG and troponin: rule out recent MI or arrhythmia.
- CBC:



Rule out leukocytosis, thrombocytosis or recent drop in platelets in patients receiving heparin (may suggest heparin induced thrombocytopenia syndrome or HITS).

- PT/INR, PTT: Patient anticoagulated / sub-therapeutic INR.
- Echo:

Identify wall motion abnormalities, intracardiac thrombus, valvular disease, aortic dissection (Type A).

- CT angiogram:

Underlying atherosclerosis, aneurysm, aortic dissection.

- Conventional catheter-based angiography: can be obtained.

Treatment

- Immediate heparinization with 5000IU bolus(80Units/kg) and continuous infusion to titrate PTT to 70-90s.
- If impaired neurovascular status: emergent revascularization
- If intact neurovascular status: time for workup (including angiogram-CTA)
- Definitive treatment

Embolus: embolectomy

Thrombus: thrombectomy $\pm$ bypass graft $\pm$ endovascular therapy

- Irreversible ischemia (complete loss of power or sensation, absent venous and arterial dopplers, rigor): primary amputation
- Identify and treat underlying cause



- Continue heparin post-operatively, start oral anticoagulant post operative day 1x3 mo depending on underlying etiology

Complications

- compartment syndrome with prolonged ischemia; requires 4 compartment (anterior/lateral/superficial and deep posterior) fasciotomy.
- Risk of arrhythmia and death with reperfusion injury.
- Renal failure and multi-organ failure due to toxic metabolites from ischemic muscle.

Prognosis

- 12-15% mortality rate.
- 5-40% morbidity rate (amputation).

Chronic Arterial Occlusion/Insufficiency

Definition

Chronic ischemia due to inadequate arterial supply to meet cellular metabolic demands during walking (claudication) or at rest (limb threat/critical limb ischemia).

Etiology and Risk

Predominantly due to atherosclerosis (usually occurs in the lower extremities).

- Major risk factors: smoking, DM, older age, HTN, hyperlipidemia, obesity, sedentary lifestyle.



Clinical features

-Claudication: must differentiate vascular from neurogenic claudication or Musculo-skeletal.

-Symptoms

1. pain with exertion: usually in calves or any exercising muscle group.
2. relieved by short rest: 2-5 min, and no postural changes necessary.
3. reproducible: same distance or time to elicit pain, same location of pain, same amount of rest to relieve pain.

-Signs

- ◆ pulses may be absent at some locations, bruits may be present.
- ◆ signs of poor perfusion: hair loss, hypertrophic nails, atrophic muscle, ulcerations and infections, slow capillary refill, prolonged pallor with elevation and rubor on dependency, venous troughing (collapse of superficial veins of foot).

Differential Diagnosis of Claudication

Vascular

- Atherosclerotic disease.
- Vasculitis (e.g. Buerger's disease, Takayasu's arteritis).
- Diabetic neuropathy.
- Popliteal entrapment syndrome .

Neurogenic

- Neurospinal disease (e.g. spinal stenosis) .



- Complex regional pain syndrome.

Skeletal

- Rheumatoid arthritis/connective tissue disease.
- Remote trauma.

Investigations

- Non-invasive:
 - routine blood work, fasting metabolic profile.
 - ABI: take highest brachial and highest ankle (dorsalis pedis artery or posterior tibial artery) pressures for each side generally (may be falsely normal or elevated in those with calcified vessels e.g. diabetics).

- CTA and MRA

- Invasive arteriography:

superior resolution to CTA/MRA, better for tibial arteries, can be done intraoperatively or as part of endovascular intervention.

Treatment

- Conservative:

- Risk factor modification (smoking cessation, HbA1c control, treatment of HTN, hyperlipidemia (statin), antiplatelet therapy.
- Exercise program (30 min 3x/wk): improves collateral circulation and oxygen extraction at the muscle level
- Foot care (especially in DM): keep wounds clean/dry, avoid trauma and pressure on wounds



- Pharmacotherapy:

- Antiplatelet agents (e.g. Aspirin, clopidogrel).
- Cilostazol (cAMP-phosphodiesterase inhibitor with antiplatelet and vasodilatory effects): improves walking distance for some patients with claudication.

- Surgical/endovascular:

- Indications: severe lifestyle impairment, vocational impairment, critical ischemia

- ◆ endovascular (angioplasty ± stenting).

- ◆ endarterectomy: removal of plaque and repair with patch (usually distal aorta or common/ profunda femoral).

- ◆ bypass graft sites: aortofemoral, axillofemoral, femoropopliteal).

- ◆ amputation: if not suitable for revascularization, persistent serious infections/gangrene, unremitting rest pain poorly controlled with analgesics.

Prognosis

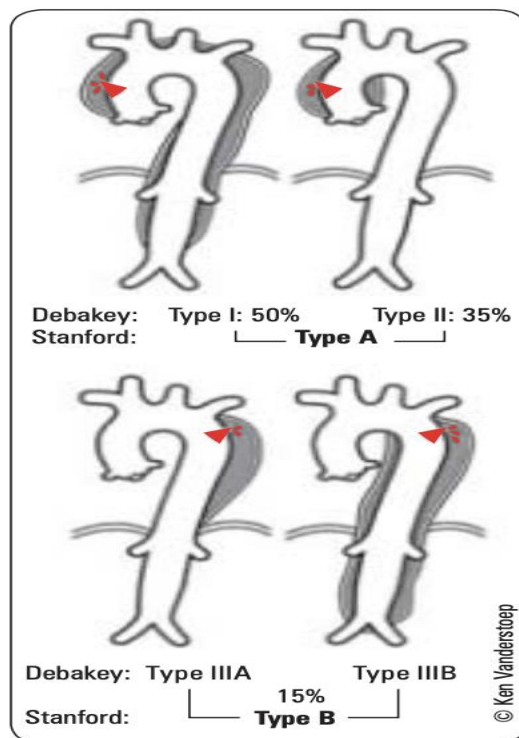
Conservative therapy: 60-80% improve, 20-30% stay the same, 5-10% deteriorate, 5% will require intervention within 5 yr, <4% will require amputation.

Aortic Diseases

Aortic Dissection

Definition

- Tear in aortic intima allowing blood to dissect into the media
- Stanford classification: Type A (involve the ascending aorta) vs. Type B (do not)
- Acute <2 wk • chronic >2 wk



Etiology

- Most common: HTN → degenerative/cystic changes → damage to aortic media.



- Other: connective tissue disease (e.g. Marfan's, Ehlers-Danlos), cystic medial necrosis, atherosclerosis, congenital conditions (e.g. coarctation of aorta, bicuspid aortic valves, patent ductus arteriosus), infection (e.g. syphilis), trauma, arteritis (e.g. Takayasu's).

Epidemiology

- M: F = 3.2:1
- Small increased incidence in African-Canadians (related to higher incidence of HTN); lowest incidence in Asians.
- Peak incidence 50-65 yr old; 20-40 yr old with connective tissue diseases.

Clinical Features:

- Sudden onset tearing chest pain that radiates to back with:
 - HTN (75-85% of patients).
 - Asymmetric BPs and pulses between arms (>30 mmHg difference indicates poor prognosis).
 - Ischemic syndromes due to occlusion of aortic branches: coronary (MI), carotids (ischemic stroke, Horner's syndrome), splanchnic (mesenteric ischemia), renal (AKI), peripheral (ischemic leg), intercostal vessels (spinal cord ischemia).
 - "unseating" of aortic valve cusps (new diastolic murmur in 20-30%) in Type A dissection.
 - rupture into pleura (dyspnea, hemoptysis) or retroperitoneum (hypotension, shock) or pericardium (cardiac tamponade).

- syncope.
- Aortic dissection is ‘the great imitator’ thus increased risk to patient and always be considered.

Investigations

- CXR: pleural cap (pleural effusion in lung apices), widened mediastinum, left pleural effusion with extravasation of blood.
- TEE: can visualize aortic valve and thoracic aorta but not abdominal aorta.
- ECG: LVH ± ischemic changes, pericarditis, heart block.
- CTA (gold standard), aortography, MRA: 100% sensitive and specific.
- Blood work: lactate (elevated in ischemic gut, shock), amylase (rule out pancreatitis), troponin (rule out MI).

Treatment

- Type A dissection needs referral to cardiac surgeon for urgent repair.
- Type B dissection is managed medically with selective intervention for complications or refractory symptoms/progression despite medical therapy.

Pharmacologic;

- Acute therapy is typically with intravenous antihypertensives, BP measured by arterial line in critical care setting.
- May transition to oral medications after initial control.
- β -blocker to lower BP and decrease cardiac contractility.

- Use non-dihydropyridine CCB if there is a clear contraindication to β -blockers.
- Targets BP of 110 mmHg and HR <60 bpm- need to manage hypertension and pain-failure to do so is a relative indication for surgical intervention.
- ACEI and/or other vasodilators if insufficient BP or HR control.

• **Surgical:**

- Urgent surgical consult if thoracic aortic dissection diagnosed or highly suspected.
- Type A to cardiac surgery, Type B to vascular surgery.
- 2/3 of patient's die of operative or post-operative complications.
- Type A: requires emergent surgery with cardiopulmonary bypass;
 - ◆ initial mortality rate without surgery is 3% per hour for first 24 h, 30% 1 wk, 80% 2 wk.
- Type B: managed medically in absence of spinal/mesenteric/limb malperfusion syndrome.
 - ◆ Treatment can be surgical or endovascular (now more often endovascular)
 - ◆ require follow-up over time to monitor for aneurysmal degeneration
 - ◆ role for early endovascular intervention controversial.

Aortic Aneurysm

Definition of Aneurysm

- Localized dilatation of an artery having a diameter at least 1.5x that of the expected normal diameter.
- True aneurysm: involving all vessel wall layers (intima, media, adventitia).
- False aneurysm (also known as pseudo-aneurysm): disruption of the aortic wall or the anastomotic site between vessel and graft with containment of blood by a fibrous capsule made of surrounding tissue.
- Aneurysms can rupture, thrombose, embolize, erode, and fistulize.

Classification

- Thoracic aortic aneurysm (TAA): ascending, transverse arch, descending.
- Thoracoabdominal.
- Abdominal aortic aneurysm (AAA):
 - 90-98% are infrarenal

Common Locations of Aneurysms; Abdominal Aorta Alone: 65% Thoracic Aorta Alone: 19% Abdominal + Iliac: 13% Thoraco-Abdominal: 2% Isolated Iliac: 1% Popliteal: 2.5% Visceral (splenic, SMA, renal, celiac): 2%

Etiology and Risk Factors

- Risk factors: smoking, HTN, DM, PVD, age >70, family history.
- Degenerative.



- Traumatic.
- Mycotic (Salmonella, Staphylococcus, usually suprarenal aneurysms).
- Connective tissue disorder (Marfan syndrome, Loeys-Dietz Syndrome, Ehlers-Danlos syndrome).
- vasculitis.
- Infectious (syphilis, fungal).
- Ascending thoracic aneurysms are associated with bicuspid aortic valve.

Clinical Features

- 75% asymptomatic (discovered incidentally).
- Most commonly in the abdominal aorta.
- Common presentation: due to acute expansion or rupture: -
 - Syncope.
 - Pain (chest, abdominal, flank, back).
 - Hypotension.
 - Palpable pulsatile mass above the umbilicus.
 - Airway or esophageal obstruction, hoarseness (left recurrent laryngeal nerve paralysis), hemoptysis, or hematemesis (indicates thoracic or thoracoabdominal aortic aneurysm).
 - Distal pulses may be intact.
- Uncommon presentations:

- Ureteric obstruction and hydronephrosis (often with inflammatory aortic aneurysm)
- Gastrointestinal bleed (duodenal mucosal hemorrhage, aortoduodenal fistula-most commonly a result of previous aortic surgery), aortocaval fistula.
- Distal embolization (blue toe syndrome) occurs in <1% of AAA's.

Investigations

- Blood work: CBC, electrolytes, urea, creatinine, PTT, INR.
- Abdominal U/S (approaching 100% sensitivity, up to ± 0.6 cm accuracy in size determination) – useful for screening and surveillance.
- CT with contrast (accurate visualization, size determination, EVAR planning)
- Peripheral arterial doppler/duplex (rule out aneurysms elsewhere, e.g. popliteal)

Treatment

- **Conservative** (for asymptomatic aneurysms that do not meet the size threshold for repair).
- cardiovascular risk factor reduction: smoking cessation; control of HTN, DM, hyperlipidemia, regular exercise.

• Surgical;

■ indications

- ◆ -ruptured.
- ◆ - symptomatic.

◆ -prophylaxis when risk of rupture is greater than risk of surgery (size >5.5 cm for AAA).

■ risk of rupture depends on: size, family history of rupture, rate of enlargement (>1cm/yr in diameter), symptoms, and comorbidities (HTN, COPD, dissection), smoking

■ surgical options;

- open surgery (laparotomy or retroperitoneal) with graft replacement.
- endovascular aneurysm repair (EVAR).

Coarctation of the Aorta

Definition

Narrowing of aorta.

Epidemiology

Commonly associated with bicuspid aortic valve (50%); Turner syndrome (35%).

Clinical presentation

Clinical presentation

■ History:

often asymptomatic.

■ Physical exam;

- blood pressure discrepancy between upper and lower extremities (increased suspicion/severity if >20 mmHg difference)
- ◆ diminished or delayed femoral pulses relative to brachial (i.e. brachial-femoral delay)
- ◆ possible systolic murmur at back

Investigations

ECG shows Left Ventricular Hypertrophy later on.

Echo or CTA or MRI for diagnosis.

Management

Balloon arterioplasty.

Surgical correction.